

**MAEASE: DEVELOPING A MOBILE APPLICATION TO EMPOWER
CORN FARMERS WITH COMPREHENSIVE SERVICE SOLUTIONS**

**SUBMITTED TO THE FACULTY OF THE
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DEDICATION

This study is lovingly dedicated to our families, whose unconditional love, sacrifices, and constant encouragement gave us the strength and motivation to pursue and complete this research. Their patience and understanding during the long hours of study and development have been our greatest source of inspiration.

We also dedicate this work to our friends and classmates who offered support, shared knowledge, and provided encouragement throughout the course of this project. Their presence made the journey more meaningful and fulfilling.

Above all, this research is dedicated to the local corn farmers and sellers of Bobontugan, Jasaan, Misamis Oriental. It is our hope that this study and the developed system may contribute to improving their livelihood and supporting their efforts through accessible digital solutions.

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ABSTRACT

Small-scale corn sellers in rural communities such as Bobontugan, Jasaan, Misamis Oriental often rely on manual methods for managing product sales and customer transactions. These traditional practices result in challenges such as unorganized order handling, delayed updates, and lack of proper transaction records, which limit business growth and efficiency. The absence of a digital platform further restricts market reach and operational coordination among sellers and buyers.

This study aimed to design and develop MaEase, a mobile-based application intended to support corn sellers and buyers by providing an organized digital platform for product listing, order management, transaction tracking, and communication. The study utilized a Design and Development Research approach, following the Waterfall Model in system development. Data were gathered through interviews and usability testing using the System Usability Scale (SUS) and ISO/IEC 25010-based evaluation to assess system quality and user satisfaction.

Results of the evaluation indicate that the MaEase system effectively improves transaction organization, communication efficiency, and order tracking for users. The application was found to be user-friendly, functional, and suitable for small-scale corn business operations. The study concludes that MaEase has the potential to support digital transformation in rural corn businesses by enhancing operational efficiency and income opportunities. Recommendations include system enhancement, wider user adoption, and future studies focusing on scalability and integration of additional features.

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Chapter I

1.1 Background and Rationale

In rural communities like Bobontugan, Jasaan, Misamis Oriental, corn farming remains one of the primary sources of livelihood for many families. These families often operate small-scale corn businesses, facing recurring challenges in organizing their product sales and managing customer transactions. Many still rely on traditional methods such as phone calls or handwritten notes to receive and confirm orders, which can lead to errors, delays, and a lack of transaction records. These inefficiencies limit their potential to grow and sustain their businesses, especially when trying to reach more customers or fulfill orders on time.

Recent studies reinforce the importance of digital transformation in agriculture. Serrano et al. (2024) introduced the Harvester mobile e-commerce platform, which improved both user experience and productivity among Filipino farmers. Briones, Galang, and Latigar (2023) highlighted how digital tools enhance market access and sustainability for smallholder farms. Additionally, Taylor and Broløs (2024) discussed how mobile platforms have contributed to the growth and stability of farming businesses in the Philippines by making transactions more organized and traceable. The Asian Development Bank (2023) further emphasized that digital innovations can boost agricultural efficiency by reducing waste and increasing income through better connectivity and online access.

To directly address these issues, this study proposes MaEase, a mobile application specifically designed for corn sellers. The system will allow users to post available corn products, receive and confirm customer orders, update the status of transactions, and monitor deliveries in real time. This digital solution aims to simplify the buy-and-sell process and promote a more organized and efficient workflow, ultimately reducing manual errors and improving income opportunities for corn-based families. MaEase also introduces basic inventory tracking and notification features, helping local sellers stay updated on low stock, new orders, and customer inquiries tools often missing in rural business setups.

This study aligns with Sustainable Development Goals (SDGs) 8, 9, and 12 Decent Work and Economic Growth, Industry, Innovation, and Infrastructure, and Responsible Consumption and Production. By digitizing core business tasks, improving market connectivity, and enhancing inventory control, MaEase seeks to empower local entrepreneurs and help rural communities thrive through sustainable digital practices.

1.2 Conceptual Framework

This study is anchored on the Input-Process-Output (IPO) Model, providing a structured and systematic approach to guide the design, development, and implementation of MaEase, a mobile and web-based application that supports handlers, and sellers in Bobontugan, Jasaan, Misamis Oriental.

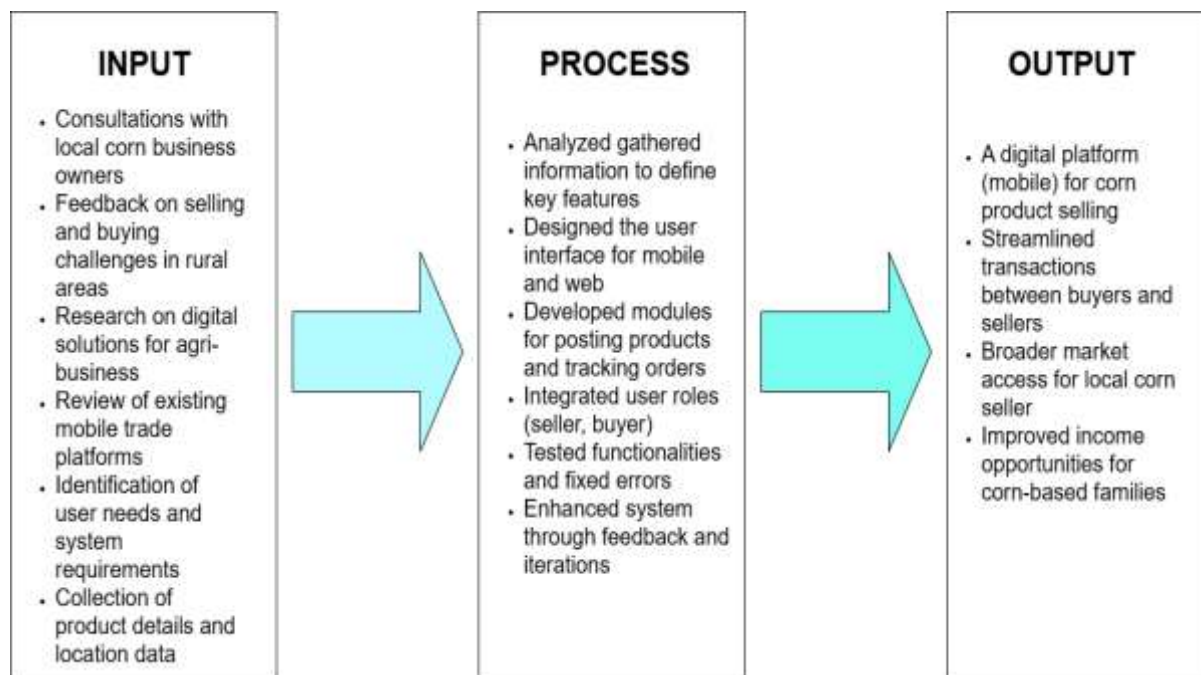


Figure 1.1 - Input-Process-Output (IPO) Model

INPUT

- Consultations with local corn business owners:
The researchers talked with people who are already selling or handling corn to understand their current processes and problems.
- Feedback on selling and buying challenges in rural areas:
Buyers and sellers shared what makes transactions hard in areas with poor internet or limited digital tools.
- Research on digital solutions for agri-business:
The team studied existing apps and systems that help with farming and selling to get ideas for useful features.
- Review of existing mobile trade platforms:
Other online marketplaces were reviewed to check what features are commonly used and how they can be improved for rural users.
- Identification of user needs and system requirements:
The team listed what users need most in an app—such as simple navigation, low data use, and fast posting of corn products.
- Collection of product details and location data:

Details like product name, price, available stock, and seller location were gathered to help buyers make better decisions.

PROCESS

- Analyzed gathered information to define key features:
The data from users was studied to decide which app features are most needed, like posting products or tracking orders.
- Designed the user interface for mobile and web:
The app layout was made simple and clean so that users can easily use it even on basic smartphones.
- Developed modules for posting products and tracking orders:
Functions like adding a product, viewing a buyer's order, and updating its status were coded and tested.
- Integrated user roles (seller, buyer):
Different user types were created with specific access: sellers can post, buyers can order, and both can track updates.
- Tested functionalities and fixed errors:
After building the app, it was tested on different devices. Bugs and issues were fixed based on the results.
- Enhanced system through feedback and iterations:
After testing, users gave feedback. The team used their suggestions to improve the app design and flow.

OUTPUT

- A digital platform (mobile) for corn product selling:
The final output is a working mobile application made specifically for buying and selling corn products in rural areas.
- Streamlined transactions between buyers and sellers:
With the app, sellers can post quickly and buyers can place orders directly, making transactions faster and easier.
- Broader market access for local corn seller:
Sellers can now reach more buyers even outside their area, helping them grow their small business.
- Improved income opportunities for corn-based families:
By using the app, families who rely on corn can earn more because they connect with more customers and manage orders better.

1.3 Statement of the Problem

1.3.1 General Problem

In Bobontugan, Jasaan, Misamis Oriental, small-scale corn business owners continue to rely on manual methods such as in-person communication or text messaging to manage sales. Without a digital system in place, they face difficulty in finding buyers, confirming orders,

and tracking transactions. This often leads to disorganized operations, missed selling opportunities, delayed deliveries, and a lack of proper records that could help improve business decisions. As a result, the business struggles to grow and maintain consistent income, especially in areas with limited access to digital tools.

1.3.2 Specific Problem

1. There is no existing digital platform that allows sellers to easily post their corn products and reach potential buyers.
2. Transactions are handled manually through word-of-mouth or text, making the order process confusing and unorganized.
3. Sellers cannot properly monitor the status or progress of customer orders, resulting in delays, repeated mistakes, or missed deliveries.
4. The lack of proper transaction records limits the ability to improve business strategies and increase income.

1.4 Objectives of the Problem

1.4.1 General Objectives

The main objective of this study is to develop a mobile-based application called MaEase that helps small-scale corn business owners in Bobontugan, Jasaan, Misamis Oriental manage their sales by providing a digital platform for posting products, receiving customer orders, tracking order status, and recording transactions. The goal is to reduce disorganized sales processes, minimize errors, and allow users to operate more efficiently. Through MaEase, the study aims to support the digital transformation of local corn businesses and improve their income opportunities.

1.4.2 Specific Objectives

1. To develop a digital platform through MaEase that allows sellers to post their corn products and connect with potential buyers.
2. To replace manual communication methods with a structured and organized order system within the app.
3. To provide a real-time order tracking feature that lets sellers monitor the progress and status of customer orders.
4. To include an automated sales recording system that helps users review transaction history and plan for business improvements.

1.5 Significance of the Study

This study aims to provide practical benefits to individuals and groups involved in corn-related businesses in Bobontugan, Jasaan, Misamis Oriental. By introducing the MaEase mobile application, the study may support the digital shift of local corn enterprises, potentially improving how they sell, manage, and monitor their operations.

This study may benefit the following:

- Corn Sellers – They may benefit from an easier and faster way to manage their business. With MaEase, they may be able to post their products, track orders, and get notified about new purchases or low stock, reducing manual errors and saving time.
- Corn Handlers – The system may help handlers coordinate with sellers in real time, making it easier to know what to handle, when to deliver, and where to go – ensuring timely and organized product movement.
- Buyers – Buyers may benefit from convenient browsing of corn products, easier order placement, and real-time updates about their orders. This may promote a more transparent and reliable buying experience.
- Admin – Acting as the system supervisor, the admin may ensure that the platform is being used correctly by monitoring product listings and orders. This role may give families better control over business operations.
- Corn-Based Families and Entrepreneurs – Families involved in corn businesses may increase their income potential by reaching more buyers, reducing mistakes, and offering better services through technology.
- Future Researchers and System Developers – They may use this study as a reference to build similar systems or explore more ways to improve rural agri-business through mobile and digital innovations.

Through MaEase, this study may support the digital empowerment of rural corn businesses and may contribute to sustainable economic growth through innovation and improved operations.

1.6 Scope and Delimitation of the Study

This study focuses on the design and development of MaEase, a mobile-based application intended to support small-scale corn business owners in Bobontugan, Jasaan, Misamis Oriental. The main purpose of the application is to assist in managing buy-and-sell activities such as posting products, receiving customer orders, tracking order status, and recording transactions. It aims to provide a convenient and organized platform for sellers to manage sales and monitor their business operations more efficiently.

The scope of this study is limited to the features related to product listing, order management, transaction monitoring, and basic sales recording. The application does not include features for payment processing, logistics

services, or customer reviews. MaEase is not designed for large-scale agricultural operations or for crops other than corn. It is intended to support small business owners who are engaged in selling corn directly to local buyers.

Geographically, this study is limited to the area of Bobontugan, Jasaan, Misamis Oriental, and the system is developed based on the common practices, challenges, and needs of corn sellers within this rural community. The findings, implementation, and testing of the system will only cover this specific location. The results may not be fully generalizable to other agricultural sectors or regions, but they aim to provide a practical solution that reflects real-life situations and supports digital transformation in local corn businesses.

1.7 Definition of Terms

API (Application Programming Interface) – A set of rules and protocols that allow different software applications to communicate with each other. In MaEase, the API server facilitates communication between the mobile app and the database server.

Cloud-Based System – A system that stores and processes data over the internet instead of on local servers or devices. In MaEase, the backend services may be hosted on the cloud for scalability and accessibility.

Data Synchronization – Ensures that information in the mobile app is consistent and updated across all devices and user accounts, preventing data mismatches or confusion.

Database – A structured collection of data that is stored and accessed electronically. The MaEase system uses a database to store information about products, orders, and users.

Digital Platform – An online or mobile-based system used to perform tasks such as selling products or managing transactions. MaEase is a digital platform developed for local corn businesses.

Information System – A combination of hardware, software, data, and processes designed to collect, process, and distribute information. MaEase functions as an information system for corn businesses.

Iteration – The process of making continuous improvements to the system based on feedback or testing results. Each version improves system performance or usability.

Low-Stock Alert – A notification feature that informs sellers when the quantity of a listed product is running low, helping them restock in time.

MaEase – The name of the mobile app in this study. It combines “Maize” (corn) and “Ease” (to make easier). The app is made to help make corn-related services like planting, harvesting, and processing faster and more convenient.

Manual Transactions – Sales processes done without the help of digital tools, often through text, calls, or in-person communication.

Mobile Application – A type of software that runs on smartphones or tablets. In this study, it refers to the digital platform used by corn sellers to manage their products and orders.

Multi-Role Access – The ability to assign different roles to users within a system, each with specific permissions. In MaEase, users may have different access rights based on whether they are sellers, buyers, or administrators.

Order Tracking – A system feature that allows sellers to check and update the status of customer orders, from placement to delivery.

Real-Time Updates – Information that is delivered instantly as events happen. In MaEase, sellers and buyers receive immediate updates about new orders or changes in order status.

Scalability – The ability of a system to handle increasing amounts of work or to accommodate growth. MaEase is designed to be scalable, allowing it to serve more users or expand to additional regions as needed.

Seller Roles – The specific functions assigned to sellers in MaEase, including the ability to manage products, update inventory, process orders, and track sales.

System Dashboard – A visual interface that displays key information, such as product status, sales, or pending orders, allowing users to monitor business activity at a glance.

System Module – A specific part or feature within the application designed to perform a particular function, such as product posting, order management, or notification alerts.

Sustainability – The practice of conducting business in a way that minimizes negative environmental impact while promoting economic growth. By digitizing processes, MaEase contributes to more efficient resource use, reducing paper waste and improving transaction tracking.

Transaction History – A record of all completed transactions within the system, including details such as product purchased, buyer information, order status, and date of transaction. This feature helps users review past transactions and make informed business decisions.

User Authentication – The process of verifying the identity of a user, typically involving the use of a username and password. In MaEase, this ensures that only authorized users can access the system.

User-Friendly Interface – A design approach that makes the app easy to understand and use, especially for users with little to no technical background.

User Roles – The specific functions assigned to users in a system. In MaEase, users have roles such as seller, buyer, or admin, each with different access and permissions.

Wi-Fi – A wireless networking technology that allows devices like smartphones and computers to connect to the internet without physical cables. In MaEase, users may need Wi-Fi or mobile data to access the app.

Chapter II

Review of Related Literature

This chapter reviews local and foreign literature on digital buy-and-sell platforms, sales management, and e-commerce in agriculture. These studies highlight how technology supports farmers in improving market access, transaction tracking, and overall business operations. The findings serve as a foundation for developing *MaEase*, a mobile application aimed at addressing the challenges faced by corn farmers and sellers in managing their transactions.

2.1 Digital Buy-and-Sell Platforms in Agriculture

Digital Buy-and-Sell Platforms have reshaped the agricultural sector by transforming how producers and consumers connect, transact, and build long-term relationships. These platforms operate through mobile applications and websites, allowing agricultural sellers to post product listings, monitor inventory, receive orders, and communicate with buyers in real time. They eliminate the need for in-person negotiations and bring significant time and cost savings, especially for rural communities.

Taylor and Broløs (2024) conducted an in-depth study on how smallholder farmers in developing regions, particularly in Asia and Africa, are adjusting to these technological shifts. Their findings reveal that increased digital literacy enhances farmers' trust in using mobile applications, which in turn results in higher adoption and transaction rates. They stress that digital inclusion must go hand-in-hand with education. In a related study, Chang (2023) investigated how AI-enhanced mobile tools can revolutionize agricultural operations, particularly by optimizing logistical systems, such as crop tracking and delivery coordination. These AI tools not only support production but also play a growing role in enabling farmers to engage in digital marketplaces with confidence.

Further emphasizing systemic support, Context.ph (2024) noted the role of public-private partnerships in accelerating digital integration in agriculture. Their report highlights that when government agencies and private tech firms work together, platforms become more scalable and trustworthy to rural users. They observed that farmers were more likely to engage with a platform when it was backed by known institutions or cooperatives. In parallel, Farmonaut (2023) emphasized how mobile-based agri-tech applications can boost market reach by up to 40% by providing real-time product visibility and pricing transparency. Their research underlined the role of digital alerts and GPS-based market intelligence in helping sellers make strategic sales decisions. Together, these studies emphasize that both institutional support and robust app functionality are essential for effective platform adoption.

Sun, Wang, and Li (2024) focused on the transformation of the corn industry in China through e-commerce solutions. Their research found that mobile platforms helped farmers bypass traditional middlemen, allowing them to directly reach urban markets. This disintermediation resulted in higher profit margins and faster sales turnover. Similarly, Lin, Zheng, and Yu (2023) examined coconut farmers using digital tools and reported significant improvements in market access, transparency in pricing, and production planning efficiency. Both studies suggest that regardless of crop type, mobile buy-and-sell platforms can provide scalable benefits that directly empower producers to participate more actively in the marketplace.

Another compelling innovation was presented by Hinojosa et al. (2023) through AgroTIC, a mobile platform enhanced with machine learning features. AgroTIC connects farmers to agronomists and merchants while analyzing data to support better crop management and market predictions. The integration of such intelligent systems has shown promising improvements in productivity, customer targeting, and service access. This development aligns well with the findings of Briones, Galang, and Latigar (2023), who studied the Philippine agri-food system and emphasized the transformative potential of mobile tools in enabling access to wider markets and improving overall farm productivity. They argued that for mobile platforms to be truly inclusive, they must be designed with rural realities and challenges in mind.

Localized innovation has also played a critical role in platform usability. Serrano et al. (2024) created "Harvester," a mobile application developed specifically for Filipino farmers. The platform focused on creating a user-friendly interface, fast transaction processing, and secure payment methods. Their study reported increased incomes among users and a stronger sense of digital confidence. Supporting this, Guay, Reyes, and Villanueva (2025) explored the case of digital extension tools in Bukidnon and found that user readiness—determined by age, digital experience, and perceived tool usefulness—directly influenced whether farmers engaged with digital services. This points to the importance of designing systems that are not only technically robust but also tailored to the end user's learning curve and digital habits.

However, despite the promise of these tools, barriers remain. Francisco et al. (2024) documented challenges such as limited smartphone access, poor internet connectivity, and low levels of digital literacy among many Filipino farmers. These limitations slow the effectiveness of e-commerce and reduce the reach of otherwise helpful platforms. To address this, they recommend the development of more affordable, offline-capable tools with vernacular-language support. Echoing this need for resilient systems, MDPI (2022) examined how small-scale producers in Naic, Cavite, used mobile e-commerce during the COVID-19 lockdowns. Even with mobility restrictions, farmers were able to reach new customers and stabilize their incomes through digital tools.

This highlights how mobile platforms can serve not only as tools of growth but also of crisis resilience.

Other researchers have explored mobile platforms built for specific regions and communities. Aspuria (2023) developed AlayPalay.PH, a mobile app aimed at promoting the online trade of rice and other goods. The app offered better visibility for sellers and encouraged local sourcing. The findings indicated that digital commerce can support sustainable practices and promote economic stability in rural trade. In a similar context, Gupta and Mallari (2024) assessed a mobile agriculture app in Silang, Cavite. They found that user-friendliness and word-of-mouth within the farming community were the two strongest drivers of adoption. Their study emphasized that even the best technology must be supported by community trust and local influence to succeed.

Finally, Mun, Crispin, and de Guzman (2020) designed a mobile platform for sustainable agriculture that connects local farmers directly with consumers and community businesses. Their tool emphasized streamlined transactions, local branding, and farmer empowerment. Their findings demonstrate how mobile commerce can be not only a commercial tool but also a community-building mechanism that supports the values of local sustainability and self-sufficiency.

In summary, the literature strongly supports the effectiveness of digital buy-and-sell platforms in agriculture. From AI-based innovations and government-supported systems to community-driven applications and localized tools, the research demonstrates that mobile platforms can improve efficiency, expand markets, and empower producers—especially in rural and underserved areas. However, for these tools to succeed long-term, issues of affordability, accessibility, and user design must continue to be addressed through inclusive, collaborative development.

2.2 Order and Sales Management in Agriculture

Order and Sales Management in Agriculture refers to the systematic handling of agricultural product transactions through digital platforms, allowing for real-time status tracking, improved order accuracy, and well-organized sales records. This digital shift reduces errors associated with manual processes and improves operational efficiency.

Basir et al. (2024) emphasized the importance of transitioning from manual to digital order management in farming, stating that automated recordkeeping significantly improves the reliability of sales tracking and supports better business decisions. Complementing this, Chen and Li (2023) studied an e-commerce platform that integrates order management into the supply chain, highlighting that digital systems improve transparency,

coordination among stakeholders, and overall transaction speed in agricultural commerce.

Patil and Tripathi (2021) introduced a machine learning model designed to predict commodity prices and guide sales planning for agricultural businesses. Their forecasting system empowers sellers to make informed decisions and manage orders based on price trends and seasonal changes. Meanwhile, Ding and Wang (2022) explored how point-of-sale (POS) data can enhance supply chain forecasting, arguing that accurate sales information strengthens inventory control and refines order planning. Together, these studies confirm that predictive tools and real-time sales data significantly boost order accuracy and reduce waste in agri-sales.

To support consistent performance in agribusiness, Boehlje and Gray (2021) proposed a comprehensive checklist that sellers can follow to manage sales and coordinate orders. Their work acts as a structured guide for tracking customer transactions and maintaining workflow balance. Similarly, Datacalculus (2023) focused on the role of smart technologies and analytics in optimizing order processing. Their research showed that digital systems not only reduce human error but also enhance order speed and data reliability, suggesting a practical link between structured tools and technological integration in agriculture.

HQSoft (2024) introduced a distribution system that digitizes order and sales management across multiple layers of the agricultural value chain. It offers features like flexible tracking, delivery coordination, and order verification to enhance operational clarity. Supporting this perspective, TraceX (2023) emphasized the benefits of a platform that combines procurement, sales, and inventory functions—stressing that comprehensive systems promote quality control and help align orders with supply capability. These findings reinforce the importance of multi-functional platforms in managing complex agricultural operations.

Unleashed Software (2022) developed an agricultural inventory solution that simplifies sales processing and stock monitoring for both large-scale agribusinesses and smallholder farmers. Their system promotes real-time visibility into product movement, reducing overstocking and missed sales opportunities. Alongside this, Harvest Harmonies (2023) offered digital sales strategies focused on order organization and expanding customer outreach. Their content advocates for structured workflows supported by digital platforms to help sellers enter new markets and increase customer satisfaction.

Terms of transaction visibility and seller convenience, Yadav and Sharma (2022) developed a web-based tracking system that allows agricultural vendors to monitor order status, update records, and manage sales activities efficiently. The dashboard feature improves traceability and supports

performance reviews. EasyBA (2022) echoed this approach, sharing a set of sales management strategies that include seasonal demand planning, order documentation, and automated updates. Their recommendations enable sellers to respond to market fluctuations while maintaining an orderly sales process.

Rizki (2021) applied the ABC inventory classification method to a digital agricultural order control system. By prioritizing product groups based on their contribution to overall sales, the system improved supply-demand balance and optimized resource use. Likewise, Alhassan and Kumah (2020) utilized the Economic Order Quantity (EOQ) model to reduce understocking and overstocking in agricultural product management. Their findings show how mathematical inventory tools can be integrated into digital platforms to guide sellers in making cost-effective order decisions.

Finally, Raybiztech (2023) presented an SMS-based agricultural order system with dashboard features for order tracking and delivery management. This case study revealed how simple mobile technologies can dramatically improve communication between sellers and buyers, especially in areas with limited internet access. The platform not only enhances response times but also ensures better recordkeeping, showcasing how even low-bandwidth solutions can support digital transformation in agricultural sales.

2.3 E-Commerce in Agricultural Systems

E-Commerce in Agricultural Systems plays a critical role in reshaping how agricultural products are marketed, distributed, and sold. Through online platforms, sellers gain direct access to buyers, bypassing middlemen and unlocking wider markets.

Gao (2022) studied key factors that influence farmers' adoption of e-commerce platforms, such as internet infrastructure, platform trust, and digital literacy. The study highlighted that farmers who were equipped with digital skills and had access to reliable networks were more likely to engage in online sales, experiencing increased income predictability and broader market access. Building on this, Tajammul et al. (2025) examined several digital marketplaces that promote direct farm-to-consumer sales, finding that transaction speed and reduced postharvest waste made these platforms more appealing. Their findings reinforce Gao's conclusion that user-friendly and inclusive platforms foster digital participation and rural development.

Zhang, Dong, and Zhang (2024) conducted an empirical analysis of how rural e-commerce participation impacts farmer income, noting that digital platforms enabled sellers to reach urban markets while avoiding high overhead costs. Government subsidies were also cited as an essential driver for platform onboarding. In relation to this, Xiong, Li, and Chen (2024) explored how e-commerce usage influenced green farming behavior, revealing that farmers using digital inputs became more environmentally aware and reduced fertilizer

overuse. Their structural modeling connected platform usability with sustainable farming practices, complementing Zhang et al.'s view that digital adoption improves both income and farming behavior.

Huang (2023) focused on livestream e-commerce for agricultural goods in China and found that interactive, visually appealing live selling sessions resulted in increased brand engagement and minimized unsold inventory. Farmers benefited from real-time feedback and stronger consumer trust. Similarly, Chen, Zhang, and Chen (2024) applied the Technology Acceptance Model to the same trend, confirming that ease of use and perceived usefulness drove higher adoption of livestream platforms. Their research adds that government support plays a crucial role in boosting farmer participation. Together, both studies suggest that blending entertainment and commerce could be a powerful strategy for increasing farm product sales.

Nurhayati et al. (2025) explored the role of consumer trust and credibility in shaping e-commerce decisions, showing that product reviews, verified sellers, and transparent processes significantly influenced purchase behavior. Buyers were more likely to return to trustworthy vendors. Schwering, Meyer, and Stein (2024) added to this discussion by comparing pricing strategies in digital agriculture, concluding that presale models offered income stability and better customer commitment for smallholder farmers. Both sources show that trust and pricing structure are essential components in sustaining e-commerce adoption in agriculture.

Penn State Extension (2023) provided a practical overview of the pros and cons of adopting e-commerce in agriculture. While online selling reduces marketing costs and expands market reach, the brief also pointed out challenges such as delivery logistics and product perishability. In parallel, Zhang and Kraemer (2024) looked at how traditional wholesale agriculture is transforming in China, where wholesalers are adapting to digital fulfillment roles. Their research emphasized that hybrid systems—combining online ordering with offline logistics—maximize delivery speed and reduce spoilage. These insights confirm that strategic adaptation is necessary to meet e-commerce demands in the perishable agri-product market.

Briones, Galang, and Latigar (2023) assessed digital adoption in Philippine agri-food systems and identified uneven infrastructure and digital literacy as major limitations, especially in remote areas. However, they also reported that producers with digital access enjoyed wider sales reach and better income opportunities. This is echoed by Valencia-Aquino and Co (2024), who developed the “Harvester” app that lets Filipino farmers list and sell products directly to consumers. The app demonstrated excellent usability and enabled real-time transactions, suggesting that digital tools must be tailored to local contexts for maximum impact.

Cruz and Palencia (2023) investigated e-commerce adoption behavior among farmers in Naic, Cavite using an extended Technology Acceptance Model. Their findings showed that perceived usefulness and peer influence were significant motivators, while internet cost served as a barrier. Rivera (2024) supports this by introducing “FarmBill,” an application designed for listing crops and managing orders. The system enabled faster transactions, improved payment flexibility, and reduced postharvest losses. These findings suggest that digital literacy and peer encouragement, combined with low-cost, feature-rich tools, can boost farmer participation in e-commerce.

Finally, Mendoza and Reyes (2024) presented a case study from Quirino Province where smallholder farmers were successfully linked to buyers through an online platform. Within one planting season, farmers experienced over 20% revenue growth. Feedback highlighted better delivery reliability and communication. The authors advocate for more support from local governments and cooperatives to sustain and scale these digital efforts. Their study reflects the broader theme across the literature: that rural farmers can thrive in the digital economy when supported by infrastructure, training, and accessible e-commerce models.

2.4 Research Gap Analysis

Area of Study	Existing Studies / Findings	Identified Gaps	How <i>MaEase</i> Addresses the Gap
E-commerce platforms for agriculture	Many platforms focus on large-scale farming or export supply chains, such as rice, coffee, and livestock.	Lack of tailored e-commerce solutions for small-scale corn farmers and their direct customers.	MaEase provides a localized mobile solution specifically designed for corn farmers to sell and manage their products.
Order & Inventory Management	Complex inventory systems exist	Small farmers struggle to track stock and	MaEase features a simplified inventory and order management

	but are not suitable for low-resource farmers with limited digital skills.	orders efficiently using existing platforms.	module, tailored to their needs and local context.
Farmer-Buyer Communication	Limited or no real-time communication tools in most farming marketplaces.	Communication gaps often lead to missed or delayed transactions.	MaEase includes a messaging feature to bridge communication between buyers and sellers directly within the app.
Sales Tracking and Reporting	Most systems lack accessible financial reporting tools for small agricultural sellers.	Farmers lack insights on sales performance and profitability.	MaEase includes easy-to-read sales summaries and reports to help farmers manage and grow their business.
Product Visibility and Updates	Product information on many agriculture apps is often outdated or not customizable by the farmer.	Customers face confusion due to lack of live updates on product availability.	MaEase allows real-time product editing and availability status, improving transparency and buyer satisfaction.

Digital Inclusion for Rural Farmers	Studies show many rural farmers are left out due to low tech literacy and poor app accessibility.	Existing apps are not designed for offline use, non-English speakers, or basic phones.	MaEase is designed to be intuitive, light on data usage, and includes support for local language and possibly offline use.
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Table 2.1 - Research Gap Analysis

Despite the growing number of digital platforms in the agricultural sector, existing solutions often cater to large-scale operations or export markets, leaving small-scale corn producers underserved—especially those in rural areas. The identified gaps include the lack of tailored tools for simple inventory management, real-time communication, localized e-commerce features, and digital accessibility for low-tech users. MaEase addresses these gaps by offering a user-friendly, mobile-based solution specifically built for corn farmers and merchants, with features that promote inclusivity, transparency, and operational efficiency at the grassroots level.

Chapter III

Methodology

This chapter explains how the study was conducted, including the methods used to gather and analyze data. It outlines the research design, respondents, tools, and procedures that guided the development of the MaEase system.

3.1 Research Design

This study uses a Design and Development Research Design, which is appropriate for creating and testing technology-based solutions like mobile applications. The goal is to develop MaEase, a mobile-based system that supports corn transactions and operations of a family-run corn business in Bobontugan, Jasaan, Misamis Oriental. The research includes key phases: analyzing user needs, designing features, developing the system, testing its functions, and evaluating it based on user feedback. This design allows continuous improvements through direct input from users, especially in rural areas. Each phase helps ensure that the final system is user-friendly, functional, and useful for corn sellers, handlers, and buyers.

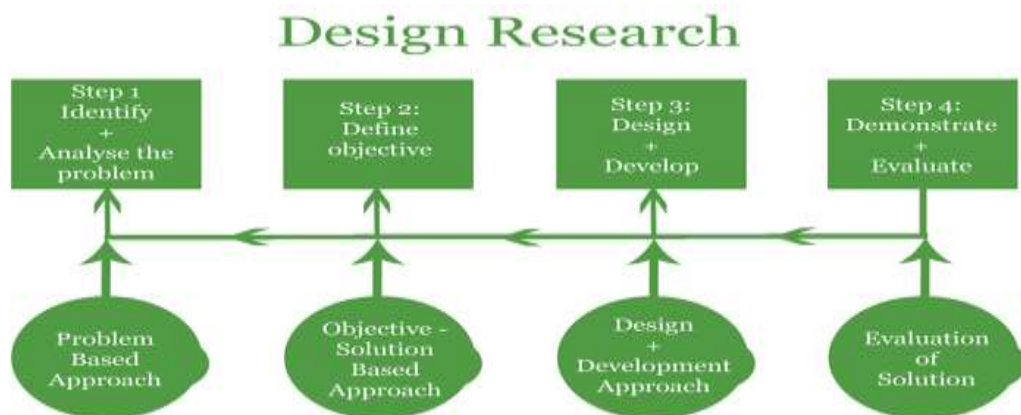


Figure 3.1 - *Design and Development Research*

3.2 System Development Life Cycle

To the proponents, the researchers used the Waterfall System Development Life Cycle (SDLC) because it provides a clear, structured approach suited for the nature of the MaEase project. The Waterfall Model is a traditional SDLC method that follows a linear and step-by-step sequence, where each phase—requirements gathering, design, development, testing, deployment, and maintenance—must be completed before moving to the next. This model is appropriate for the study because the system's scope, features, and user roles were already well-defined at the beginning of the project, based on interviews with corn product handlers and community consultations. With minimal expected changes during development, the Waterfall model allowed the team to follow a predictable and low-risk workflow, ensuring that each deliverable was fully completed and documented before advancing to the next phase.



Figure 3.2 - Waterfall Model

Requirements Gathering

In this phase, data for the MaEase application was collected through surveys, interviews, and focus group discussions with corn farmers and related stakeholders. This helped identify the users' specific needs and challenges, such as market price updates, financial tracking, and easier access to information. This step guarantees that the app's development is based on real user requirements.

System Design

Using the gathered requirements, the system design phase created the user interface (UI) and the app's overall structure. The design focused on making the app user- friendly and accessible, especially for farmers in Bobontugan, Jasaan, Misamis Oriental. Considerations included simplicity and features suited to their daily activities.

Implementation(Development)

During this phase, the MaEase app was built according to the design plans. The development team coded the app's functions, such as transaction tracking, market price updates, and user management for farmers, sellers, and admins. The focus was on creating an easy-to-use tool that meets farmers' needs.

Testing

Once developed, the app underwent testing to check its functions, performance, security, and user-friendliness. Testing ensured the app works correctly, is free from bugs, and offers a smooth experience. Feedback from users during this phase was used to improve the app's features.

Deployment

After successful testing, the MaEase app was launched for use by corn farmers in Bobontugan, Jasaan, Misamis Oriental. This stage involved making the app available for download and helping users get started with its tools and features.

Maintenance and Support

Following deployment, ongoing maintenance and support are necessary to keep the app updated and working well. This phase handles fixing problems, adding new features, and adapting the app to users' changing needs based on their feedback and changes in the corn farming environment.

3.3 Research Setting

This study was conducted in Bobontugan, Jasaan, Misamis Oriental, a rural barangay where corn farming and selling serve as a major livelihood. The business involved is a family-owned enterprise that handles all aspects of corn production and sales internally. The chosen setting reflects an authentic agricultural environment where manual order tracking and poor communication often result in delayed processing and income loss.

The application to be developed, MaEase, aims to digitize their current manual operations by providing a mobile-based solution for tracking customer orders, monitoring the status of corn production (e.g., number of sacks to be prepared), updating order progress (e.g., preparing, packed, out for delivery), and handling buyer transactions. This setting was selected because of its relevance to the purpose of the system and the feasibility of working closely with the users throughout the development process. Its real-world challenges offer meaningful input in making the system more practical and user-adaptive.

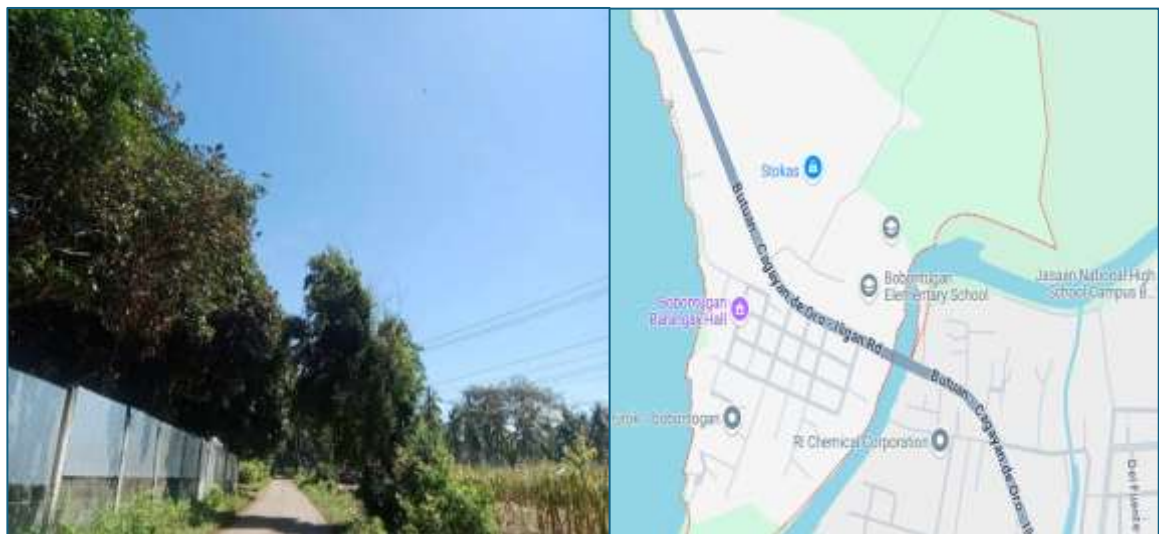


Figure 3.3 - *Bobontugan, Jasaan Mis. Or*

3.4 Unit of Analysis, Respondents and Sampling

3.4.1 Unit of Analysis

The unit of analysis in this study is the group of individuals involved in the corn processing and selling operations of the family-run business. These include:

- **Corn Product Handlers** – Those responsible for preparing and organizing the corn based on the volume of buyer orders.

- Corn Sellers – Those managing order status, communication with buyers, and final transaction or delivery updates.
- Buyers– Individuals who place orders, receive updates regarding their purchases, and coordinate with sellers for the fulfillment of transactions.

3.4.2 Respondents of the Study

The respondents of this study will be individuals directly involved in corn-related services in Bobontugan, Jasaan, Misamis Oriental. They will be selected based on their specific roles and hands-on experience in the areas of corn selling and buying. Their responses will serve as valuable input to assess how the proposed MaEase application can address the service needs and challenges within the local corn business sector.

A total of 12 respondents will be purposively selected, distributed across the following categories:

- 2 Corn Sellers – These individuals are responsible for managing product listings, handling orders, and communicating with buyers. Their feedback will help identify challenges in the transaction process and communication gaps that MaEase aims to improve.
- 10 Corn Buyers – These buyers include various local merchants or traders who regularly purchase corn products from sellers. The diversity among the buyers will provide insights into how the system can improve product visibility, ordering convenience, and order tracking.
-

While the sample size is relatively small, it will provide focused and relevant data for each user type. In quantitative studies, a clearly defined and homogenous sample, even with fewer respondents, can yield statistically useful insights, especially when the study's scope is narrow and exploratory. As noted by Saunders, Lewis, and Thornhill (2019), pilot studies commonly involve between 5 and 30 participants to test research instruments and collect initial quantitative data.

Therefore, the selected 12 respondents will be sufficient for generating structured data and identifying service gaps to guide the development of MaEase.

3.4.3 Sampling Technique

This study employed purposive sampling, a non-probability sampling method that allows the deliberate selection of individuals based on their involvement in the operational processes of the corn business. The chosen respondents are considered information-rich cases because of their experience in corn production and transactions.

According to Ahmad & Wilkins (2024), purposive sampling is suitable for qualitative studies that aim to extract detailed insights from a specific group of people who have firsthand experience related to the topic being studied. This sampling technique ensures that the information collected is directly aligned with the study's goals and offers meaningful contributions to the design of the system.

3.5 Data Source, Data Gathering Instruments and procedure

3.5.1 Data Source

The study will utilize both primary and secondary sources of data. The primary data will be gathered from selected corn product handlers, sellers, and buyers who are directly involved in corn-related operations in Bobontugan, Jasaan, Misamis Oriental. Their firsthand experiences will offer valuable insights into the challenges of corn processing, selling, and purchasing—informing the system's design and intended features. Secondary data will include relevant literature such as academic journals, research articles, and case studies on mobile technology, agricultural practices, and rural e-commerce. These materials will support the conceptual framework of the MaEase application and guide its technical development. All primary data will be collected with informed consent, ensuring that participation is voluntary and data confidentiality is strictly observed in accordance with ethical research standards.

3.5.2 Data Gathering Instruments

This study utilized the System Usability Scale (SUS), a standardized tool based on ISO standards, as the main instrument for assessing the usability of the MaEase application. The SUS questionnaire was administered to selected sellers and buyers after they interacted with the system prototype. It measured usability factors such as ease of use, satisfaction, efficiency, and overall system quality. To ensure a realistic testing environment, the prototype was developed using the Plyker Skin interface, a mobile-responsive design that enhanced user interaction and allowed for more accurate feedback on usability and visual appeal.

In addition, the researchers used a structured questionnaire aligned with the ISO/IEC 25010 Software Product Quality Model to evaluate the system across eight quality attributes: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. A 5-point Likert scale (from 1 – Strongly Disagree to 5 – Strongly Agree) was used to capture participant responses. All respondents were properly briefed about the study and gave informed consent. Their answers were kept

confidential and used strictly for academic and evaluation purposes in line with ethical research standards.

3.5.3 Data Gathering Procedure

Before starting the data gathering, the researchers will first seek permission and obtain approval from their research adviser to conduct the study. Once approved, purposive sampling will be used to select respondents who are directly involved in corn-related work such as handling, selling, and buying in Bobontugan, Jasaan, Misamis Oriental.

Before the actual interviews, the researchers will explain the purpose of the study to each participant and request their consent to participate. They will be told that their responses will be kept confidential and that they may withdraw from the interview at any time if they feel uncomfortable. This will ensure that the data collection process is conducted in an ethical and respectful manner.

Face-to-face interviews will then be scheduled based on the most convenient time for the respondents. During the interviews, the researchers will actively listen and record the participants' responses. After data collection, the gathered information will be reviewed, categorized according to common themes, and analyzed. The results will help the researchers identify the shared needs and challenges of the respondents. These findings will serve as the basis for designing the MaEase application, ensuring that it addresses real issues experienced by the community.

3.6 Mode of Analysis

This study uses qualitative data analysis to understand the information collected from the interviews. The responses were carefully read and grouped to find common ideas, patterns, and suggestions related to corn product handling, selling, and mobile application needs. This method helps the researchers understand the actual experiences and challenges of the respondents. Instead of using numbers, the focus is on the meaning behind the participants' answers. The results of the analysis were used to design system features that match what the users really need.

3.7 Statistical Instruments Procedure

The main method used for data analysis in this study is quantitative. After collecting the survey responses, the researchers organized the data and encoded the answers using Microsoft Excel and Google Sheets. Basic statistical tools such as frequency, percentage, and weighted mean were used to analyze

the results. These tools helped identify the most common needs, preferences, and problems experienced by the corn handlers, sellers, and buyers. The analyzed data provided a clear basis for designing the features and functions of the MaEase system to ensure that it responds to the actual needs of the users.

Chapter IV

Result and Discussion

This chapter presents the results of the MaEase system's development and testing, including an analysis of its effectiveness in addressing the needs of corn sellers, buyers, and administrators. It highlights the key findings, evaluates system performance, and discusses how the platform meets its objectives in improving business operations.

4.1. Requirements Specifications

The MaEase system was developed with both functional and non-functional requirements to address the needs of corn sellers, buyers, and admins. Functional requirements include product management, order tracking, user management, notifications, and reporting. Non-functional requirements focus on ensuring usability, performance, security, scalability, and reliability. These specifications were key to the system's design, ensuring that **MaEase** provides a user-friendly and efficient platform.

4.1.1 Schedule Feasibility

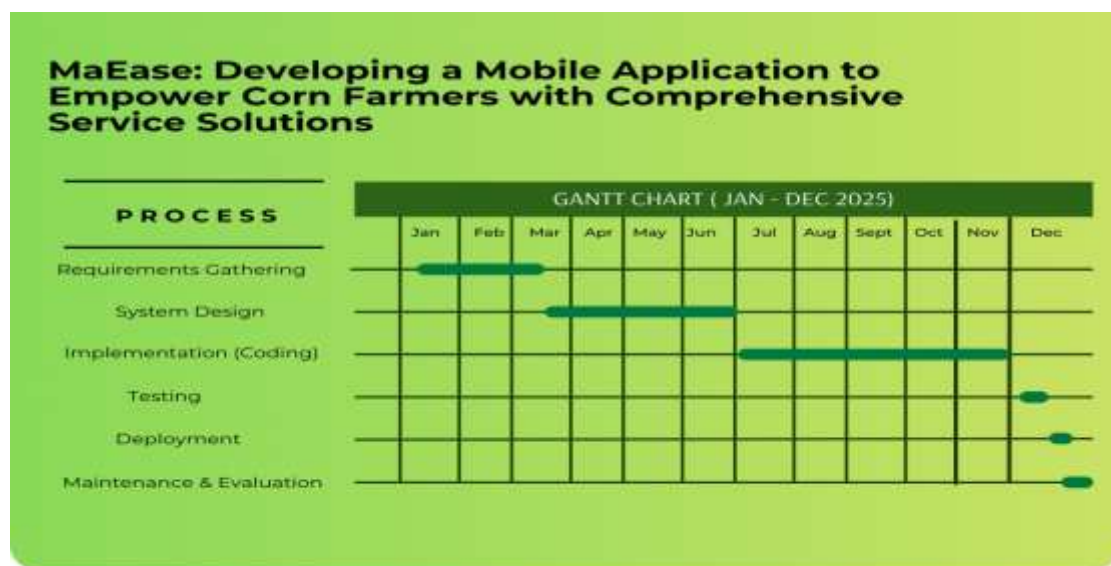


Figure 4.1 - Gantt Chart

The Gantt chart outlines the timeline for the development of the MaEase mobile application, from January to December 2025. The project begins with the Requirements Gathering phase in January and February, where the team collects data and defines system features based on user needs. This is followed by the System Design phase in March and April, which focuses on planning the

system architecture and creating design prototypes. From May through July, the Implementation (Coding) phase takes place, where the actual coding and development of the application occur. Testing is conducted in December to ensure functionality and resolve any bugs. The system is then Deployed in October, making it available for use. Finally, the Maintenance & Evaluation phase also takes place in December, ensuring that the system remains operational and continues to meet user needs through feedback and updates.

4.2 Requirements Modeling

4.2.1 User Definition

The **MaEase** system is designed for three primary user roles, each with distinct functionalities. These roles include **Sellers**, **Buyers**, and **Admin**, all of whom interact with different system modules based on their specific needs. Each role has access to a tailored set of features to ensure the system operates efficiently for all users.

- **Sellers (Farmers)**

Sellers can manage their product listings, including adding, editing, and deleting products from their inventory. They also track incoming orders and update their order statuses, such as marking orders as "Preparing" or "Delivered." Additionally, sellers can communicate directly with buyers to clarify order details or address inquiries.

- **Buyers**

Buyers can browse the available corn products, search by category or price, and add items to their cart. Once an order is placed, buyers can track its status and communicate with the seller to inquire about delivery times or product details. They also have the option to cancel orders or adjust quantities before the order is processed.

- **Admin**

Admins have the ability to manage all users within the system, including adding, editing, or removing sellers and buyers. They also have full control over product listings, ensuring that all products are up-to-date and accurate. Admins can generate detailed reports on system performance, sales trends, and user activity, which are crucial for evaluating system success and making informed decisions.

4.2.2 System Module Specification

Authentication Module

- Purpose:
To securely authenticate users and provide role-based access to system functions.
- Inputs:
Username, password, assigned user role.
- Processes:
 - Validates user credentials.
 - Determines role (Sellers, Buyers, Admin).
 - Redirects user to their appropriate dashboard.
 - Restricts access to unauthorized modules.
- Outputs:
Successful login session, access to permitted features, login status notifications.
- Users:
Sellers, Buyers, Admin.
- Limitations:
Does not manage or create user accounts; only validates them.

Product Management Module (for Sellers)

- Purpose:
To allow sellers to add, update, and manage their corn products.
- Inputs:
Product name, description, price, quantity, product image.
- Processes:
 - Validates input fields.
 - Saves product data to the database.
 - Allows editing or deleting existing products.
- Outputs:
Confirmation of product added/updated, error messages for invalid inputs.
- Users:
Sellers.
- Limitations:
Does not allow buyers to interact with products unless they are published by the seller.

Order Management Module (for Sellers and Buyers)

- Purpose:
To manage orders placed by buyers and update their statuses for sellers.
- Inputs:
Order details (product, quantity, buyer details), order status update.
- Processes:
 - Tracks order details and status.
 - Sends notifications to both sellers and buyers.

- Updates the order status (Preparing, Shipped, Delivered).
- Outputs:
Order status updates, notifications to users.
- Users:
Sellers, Buyers.
- Limitations:
Only sellers can update order statuses; buyers can view and track but not update.

Admin Dashboard Module

- Purpose:
To provide admins with an overview of system activity, including user and order management.
- Inputs:
User activity data, sales data, order data.
- Processes:
 - Displays key metrics such as total users, total orders, revenue trends.
 - Provides data visualizations (charts, graphs).
 - Allows admin to manage users and products.
- Outputs:
System statistics, user management actions (add/edit/delete users).
- Users:
Admin.
- Limitations:
Admin can only manage data within the scope of the platform, not external systems.

Product Listing and Order Placement Module (for Buyers)

- Purpose:
To allow buyers to browse available products, place orders, and manage their shopping cart.
- Inputs:
Product selection, quantity, payment method (COD or online), delivery address.
- Processes:
 - Allows buyers to search and browse products.
 - Enables adding products to the shopping cart.
 - Handles order placement, including payment and delivery details.
 - Sends order confirmation and status updates to the buyer.
- Outputs:
Order confirmation, order tracking details, payment receipt.
- Users:
Buyers.

- **Limitations:**
Does not allow buyers to manage or edit product details; only provides the interface for browsing and purchasing.

Messaging Module (for Sellers and Buyers)

- **Purpose:**
To enable direct communication between buyers and sellers within the app.
- **Inputs:**
Message content, sender and receiver details.
- **Processes:**
 - Sends and receives messages between buyers and sellers.
 - Notifies users of new messages.
 - Stores message history for reference.
- **Outputs:**
New message notifications, message history.
- **Users:**
Sellers, Buyers.
- **Limitations:**
Limited to text-based communication only; no support for attachments or media files.

Notifications Module

- **Purpose:**
To notify users (buyers, sellers, and admins) about important system events such as new orders, low stock, and system updates.
- **Inputs:**
Event triggers (new order, low stock, user actions).
- **Processes:**
 - Sends notifications to users based on predefined triggers.
 - Allows users to view and manage their notifications.
 - Filters notifications by priority or type (e.g., order status updates, promotional alerts).
- **Outputs:**
Notification messages, alert sounds, visual cues on the dashboard.
- **Users:**
Sellers, Buyers, Admin.
- **Limitations:**
Does not support email notifications; notifications are limited to the mobile app.

Analytics and Reporting Module (for Admin)

- **Purpose:**
To provide detailed reports and visual analytics on system performance, sales trends, and user activity.
- **Inputs:**
Sales data, user activity logs, order data.

- Processes:
 - Generates reports on revenue, order volumes, top products, and user roles.
 - Provides visualizations such as pie charts, bar graphs, and line graphs.
 - Allows admins to filter and customize reports.
- Outputs:
Generated reports (PDF/Excel), graphical visualizations.
- Users:
Admin.
- Limitations:
Reports are limited to data within the platform and may not integrate with external systems or databases.

4.3 Data and Process Modeling

4.3.1 Context Diagram

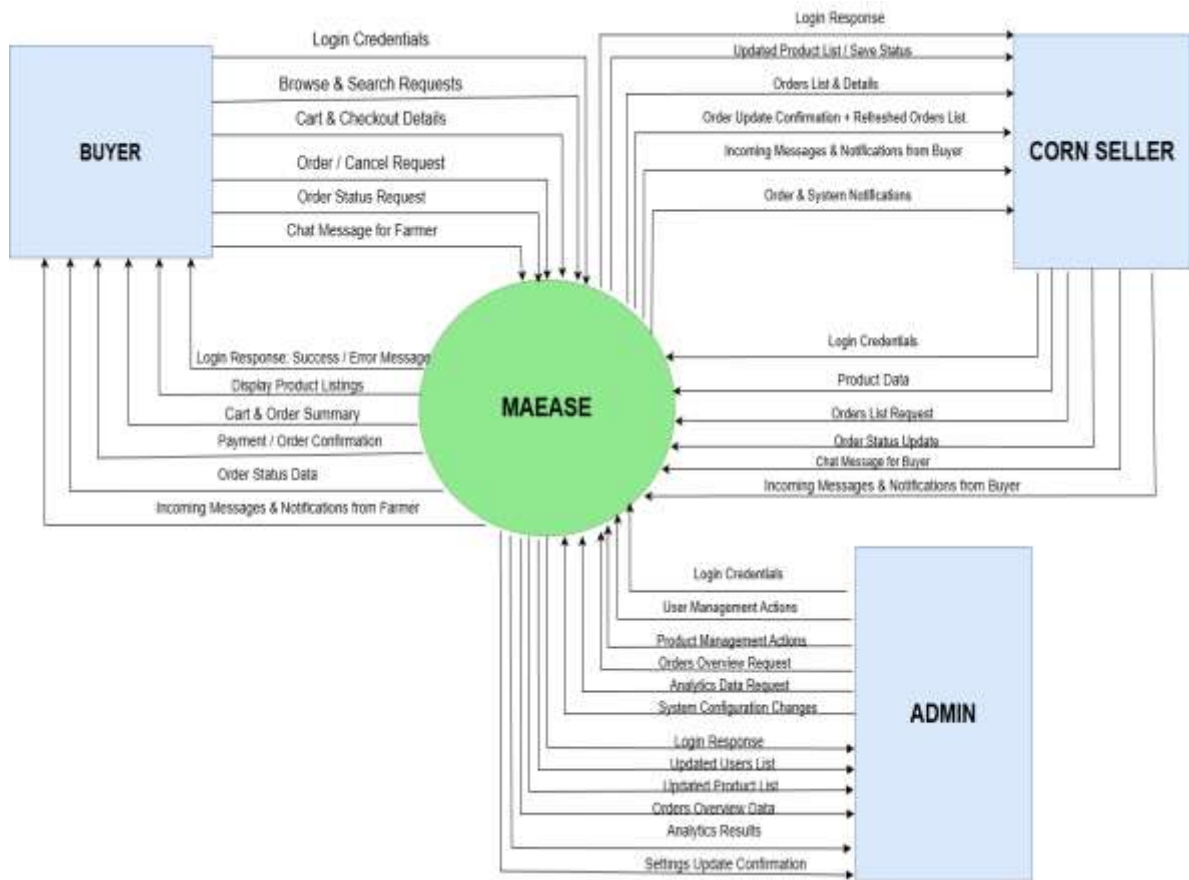


Figure 4.2 - Context Diagram

The context diagram of the MaEase system shows the interaction between the BUYER, SELLER, and ADMIN with the central MaEase System. The BUYER places orders, views product listings, and makes payments, while the SELLER manages products and order statuses. The ADMIN oversees user management, generates reports, and manages system settings, with all data stored in the Database.

4.3.2 Data Flow Diagram

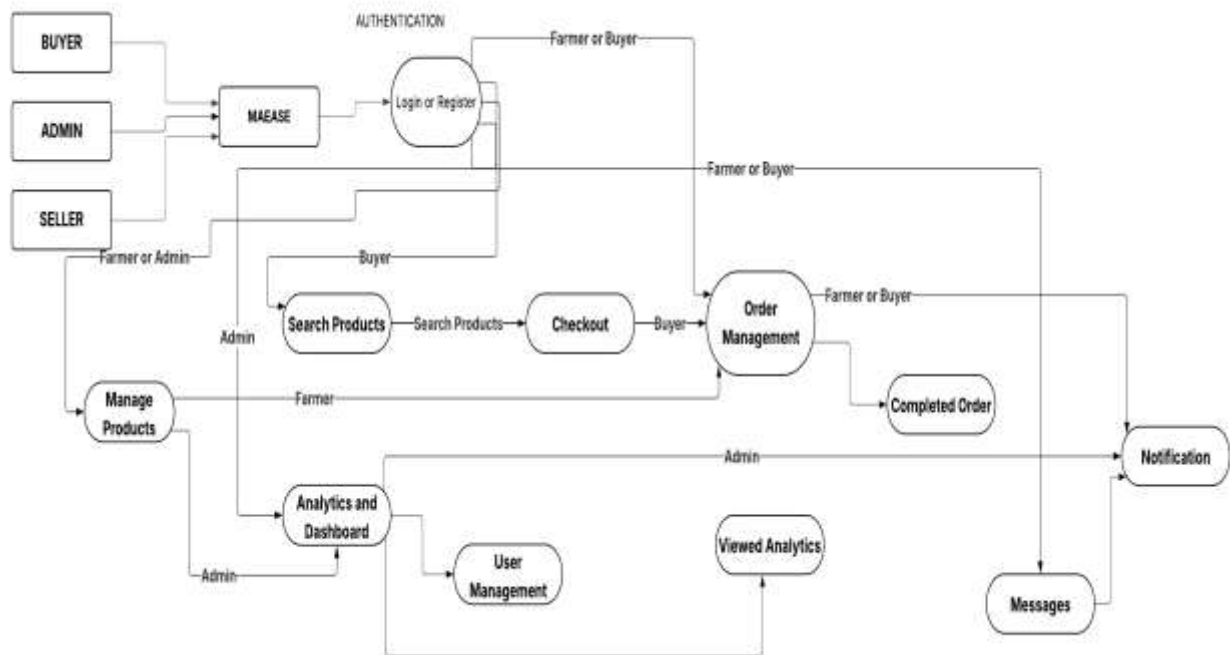


Figure 4.3 - Data Flow Diagram

The Data Flow Diagram (DFD) for the MaEase System shows the flow of data between buyer, seller, admin, and the MaEase System. The buyer searches for products, places orders, and receives notifications, while the seller manages products and updates orders. The admin oversees user management, analytics, and order completion, with the system processing and storing all related data.

4.3.4 System Flowchart

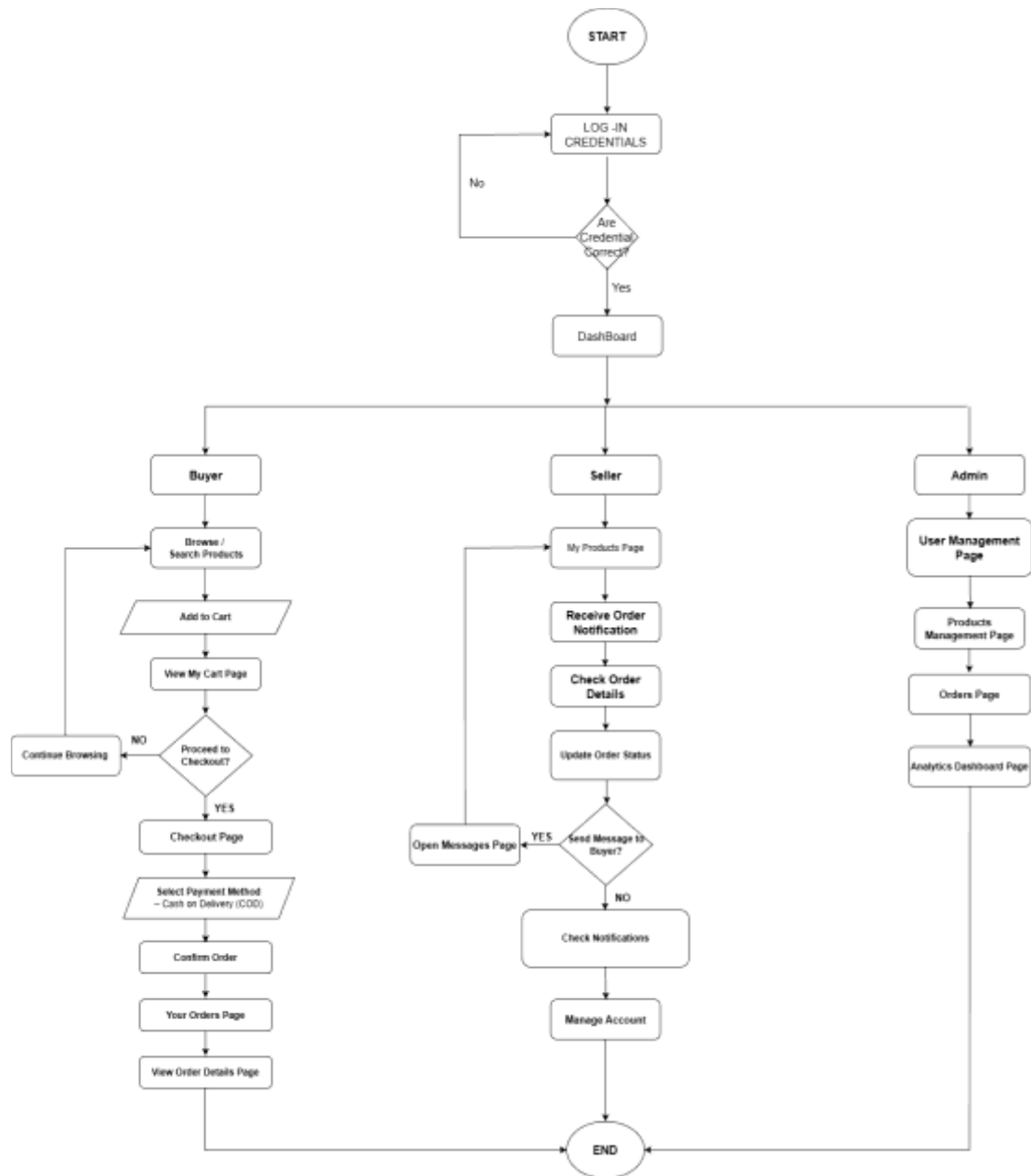


Figure 4.4 – System Flowchart

The system flowchart for the maease system illustrates the login process for buyer, seller, and admin. after successful authentication, the buyer can browse products, place orders, and manage their account, while the seller receives notifications and updates order statuses. the admin oversees user management, product management, and analytics, ensuring system functionality and operations.

Program Flow

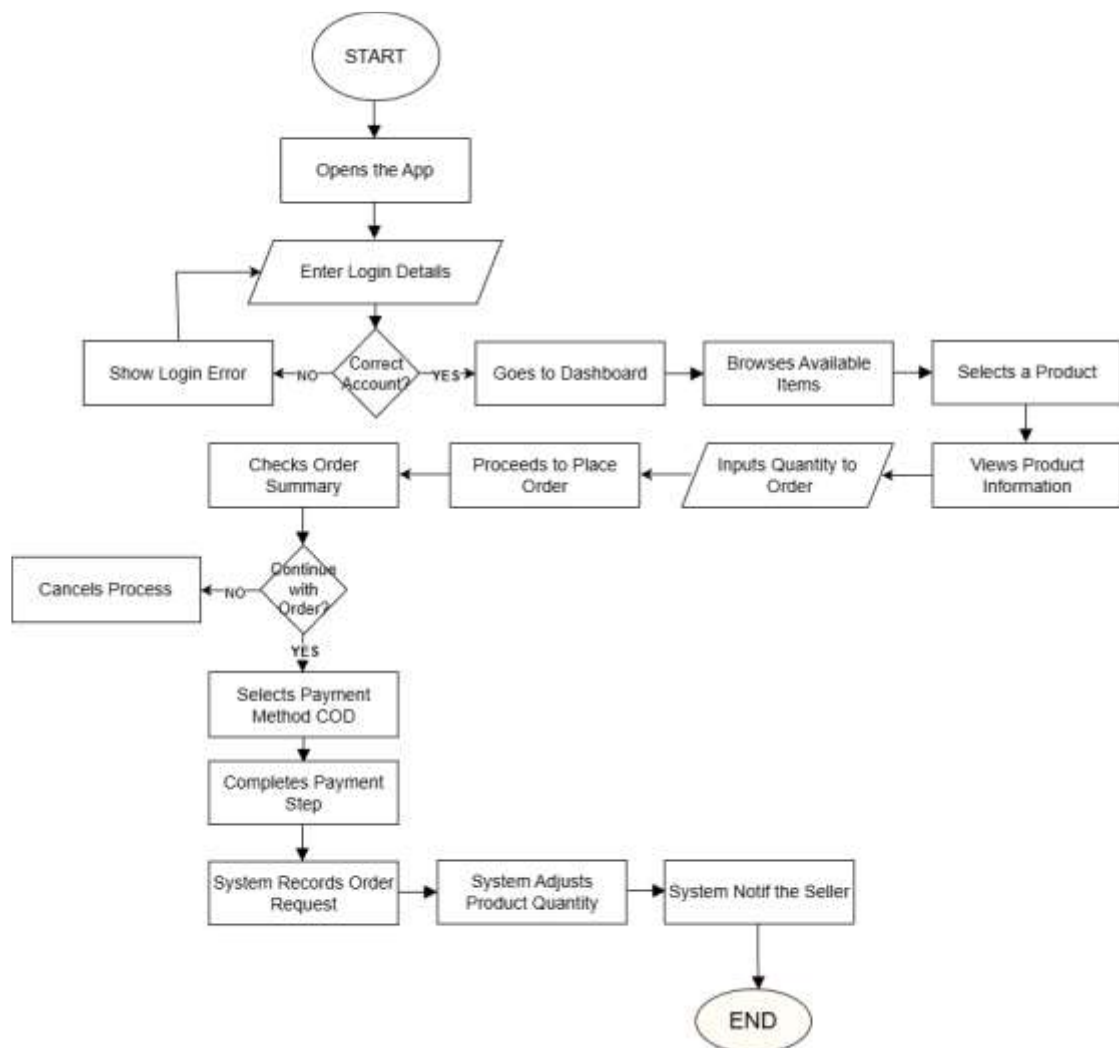


Figure 4.5 – PROGRAM FLOW : Buyer Places an Order

The Program Flow for Buyer Places an Order begins with the user logging in and entering their credentials. Once authenticated, the BUYER proceeds to browse products, select an item, and add it to their cart, reviewing the order summary. After selecting the payment method (COD), the system records the order, adjusts the product quantity, and notifies the SELLER of the new order.

Program Flow

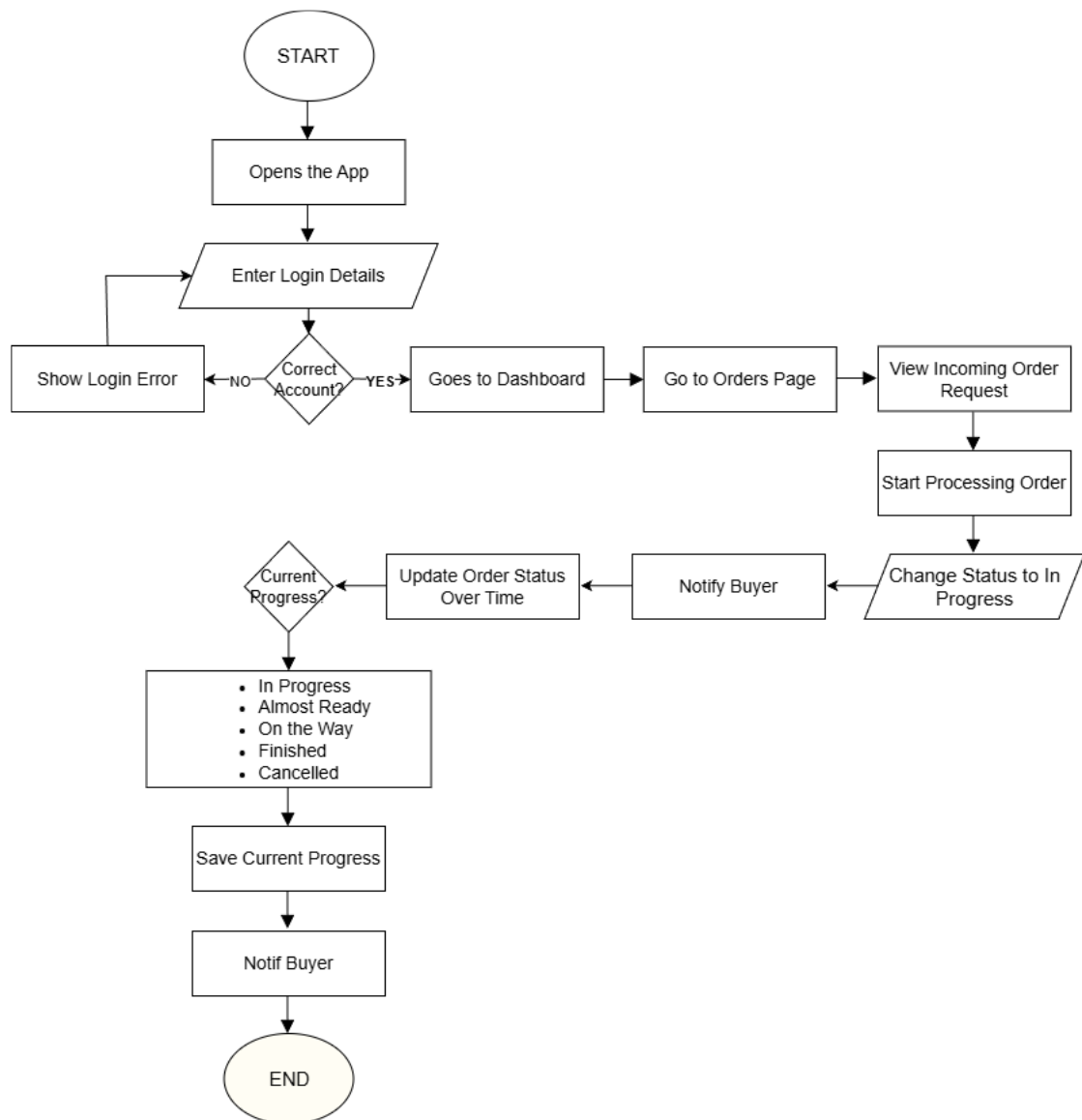


Figure 4.6 – PROGRAM FLOW : Seller Handles Order & Updates Progress

The Program Flow for Seller Handling Orders and Updating Progress begins with the seller logging in and viewing incoming order requests on the dashboard. The seller then processes the order by changing the status to "In Progress," updating it over time as necessary (e.g., "Almost Ready," "On the Way," "Finished," "Cancelled"). Finally, the system saves the updated status and notifies the buyer of the progress.

Program Flow

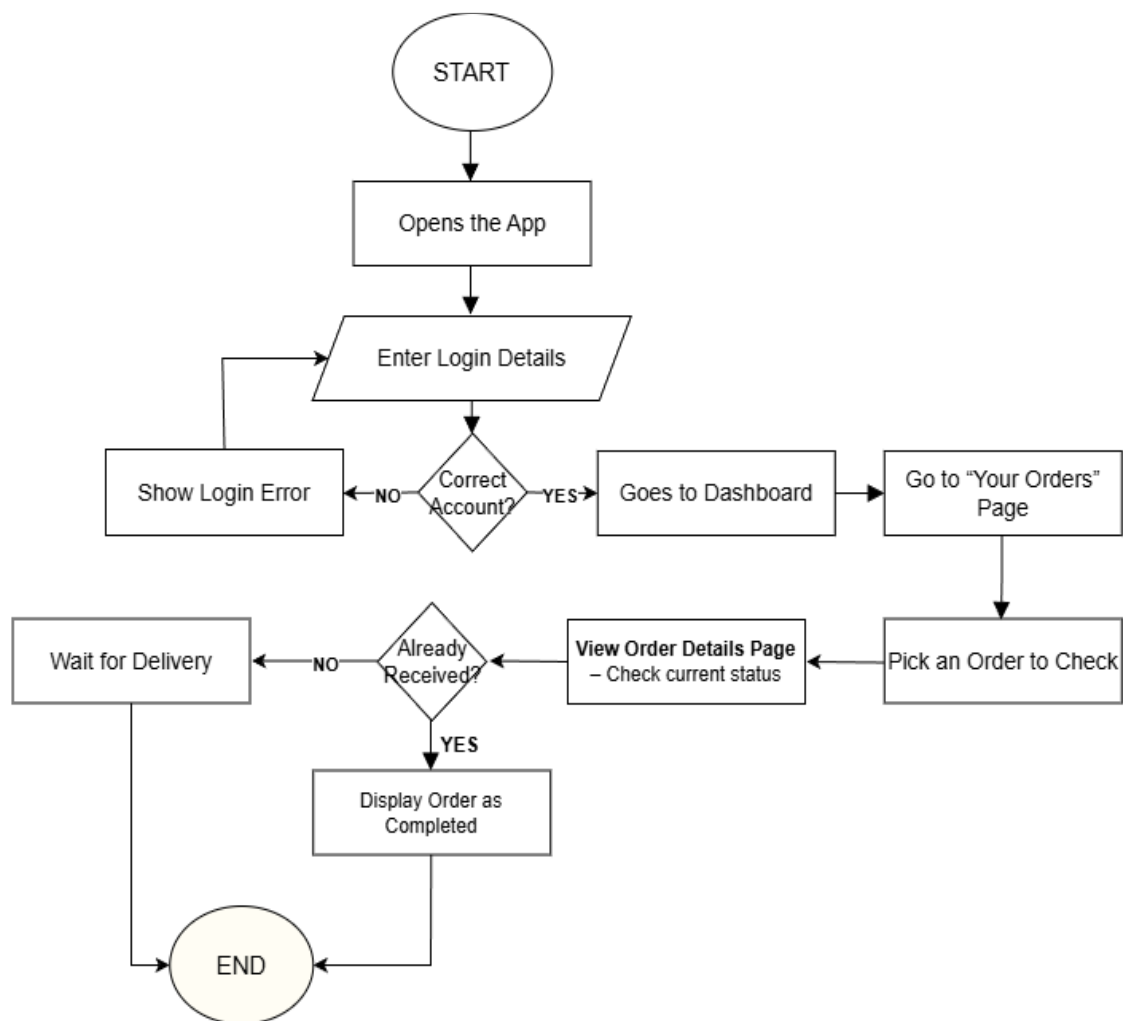


Figure 4.7 – PROGRAM FLOW : Buyer Tracks and Confirms Order

The Program Flow for Buyer Tracking and Confirming Orders begins with the buyer logging into the app and accessing their dashboard. The buyer can then view the order details by checking the status on the "Your Orders" page, where they can see if the order has already been received. Finally, once the order is received, the buyer confirms it by marking the order as completed.

Program Flow

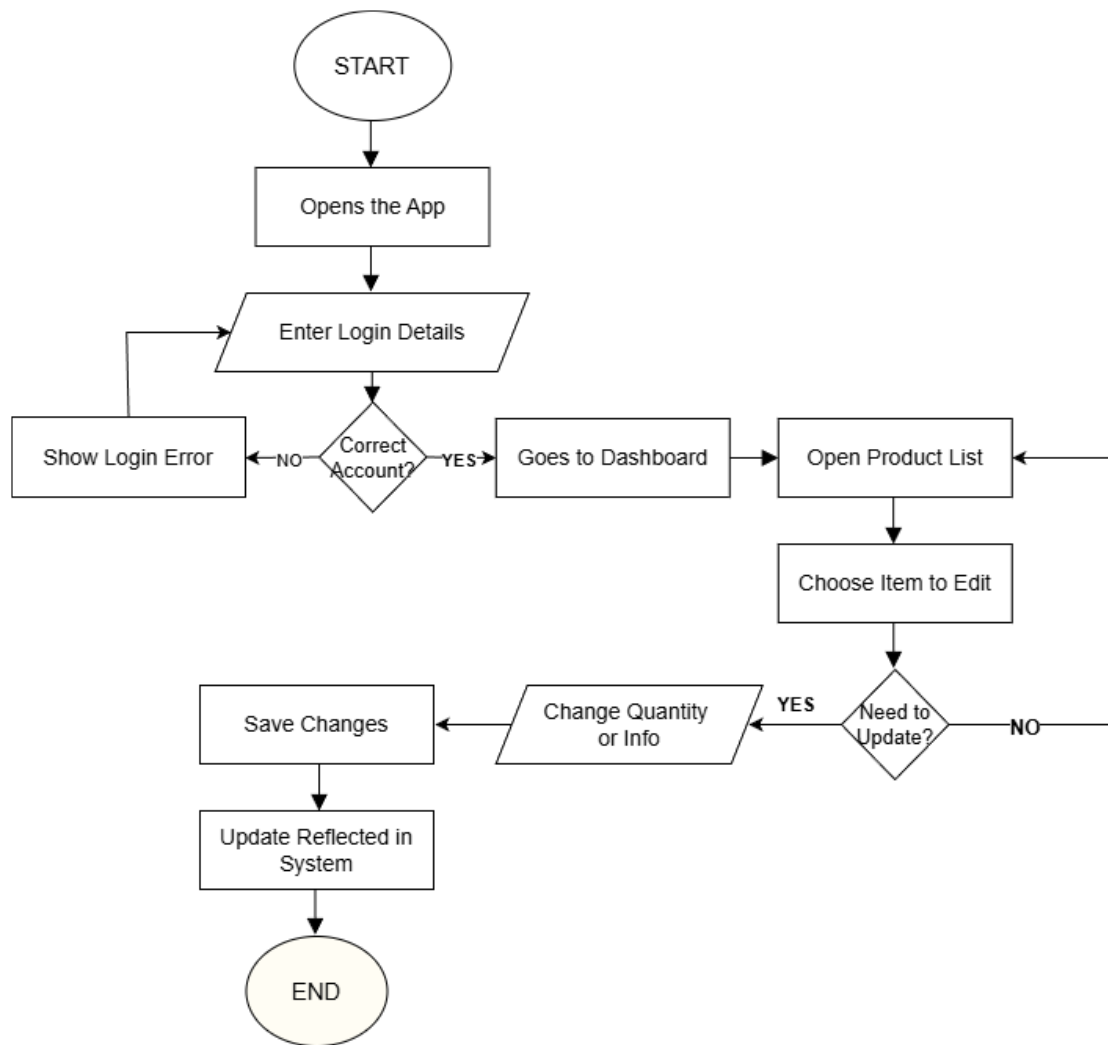


Figure 4.8 - PROGRAM FLOW: *Seller Manages Items*

The Program Flow for Seller Managing Items begins with the seller logging into the app and navigating to the dashboard. From there, they access the product list and select an item to edit, where they can change the quantity or information. Once changes are made, the seller saves the updates, and the system reflects the modifications.

Program Flow

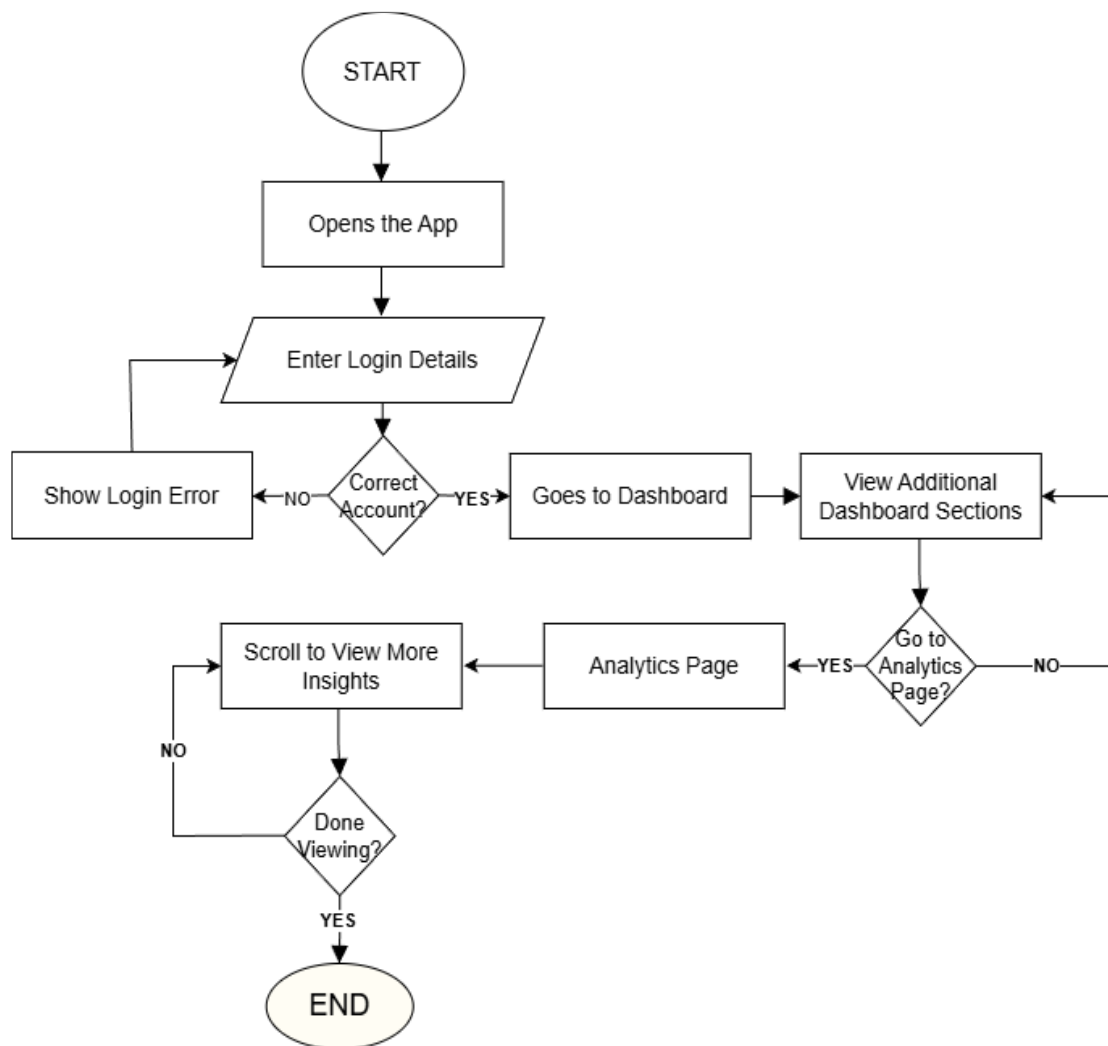


Figure 4.9 – PROGRAM FLOW : Admin Views Reports

The Program Flow for Admin Viewing Reports starts when the admin logs in and is authenticated, proceeding to the dashboard. From the dashboard, the admin can view additional sections, including the analytics page, by choosing to scroll and explore further insights. If the admin is done viewing, the process ends.

Program Flow

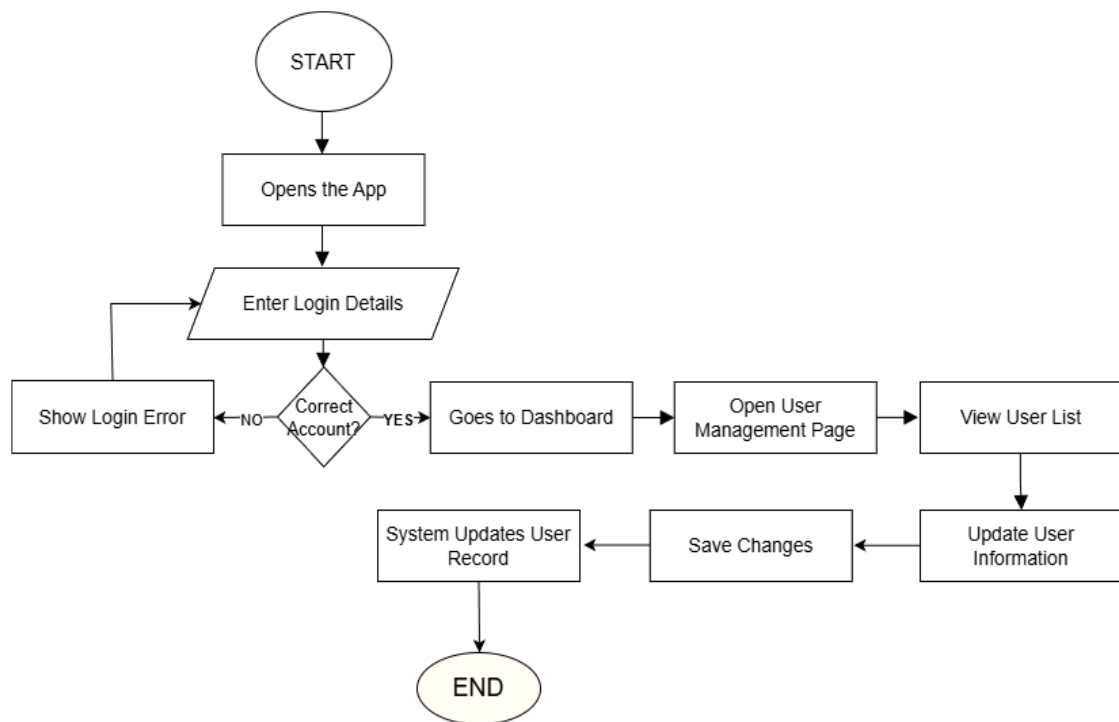


Figure 4.10 - PROGRAM FLOW : Admin Manages Users

The Program Flow for Admin Managing Users starts with the admin logging in and being directed to the dashboard. From there, the admin accesses the user management page, views the user list, and updates user information as needed. After saving the changes, the process ends.

Object Modeling

Modeling in the MaEase system defines key entities such as Buyer, Seller, Product, Order, and Admin, with their roles and behaviors. It ensures clear interactions, like Buyers placing orders, Sellers managing inventory, and Admins overseeing operations. This structure helps maintain system clarity and efficiency.

4.3.5 Use Case Diagram

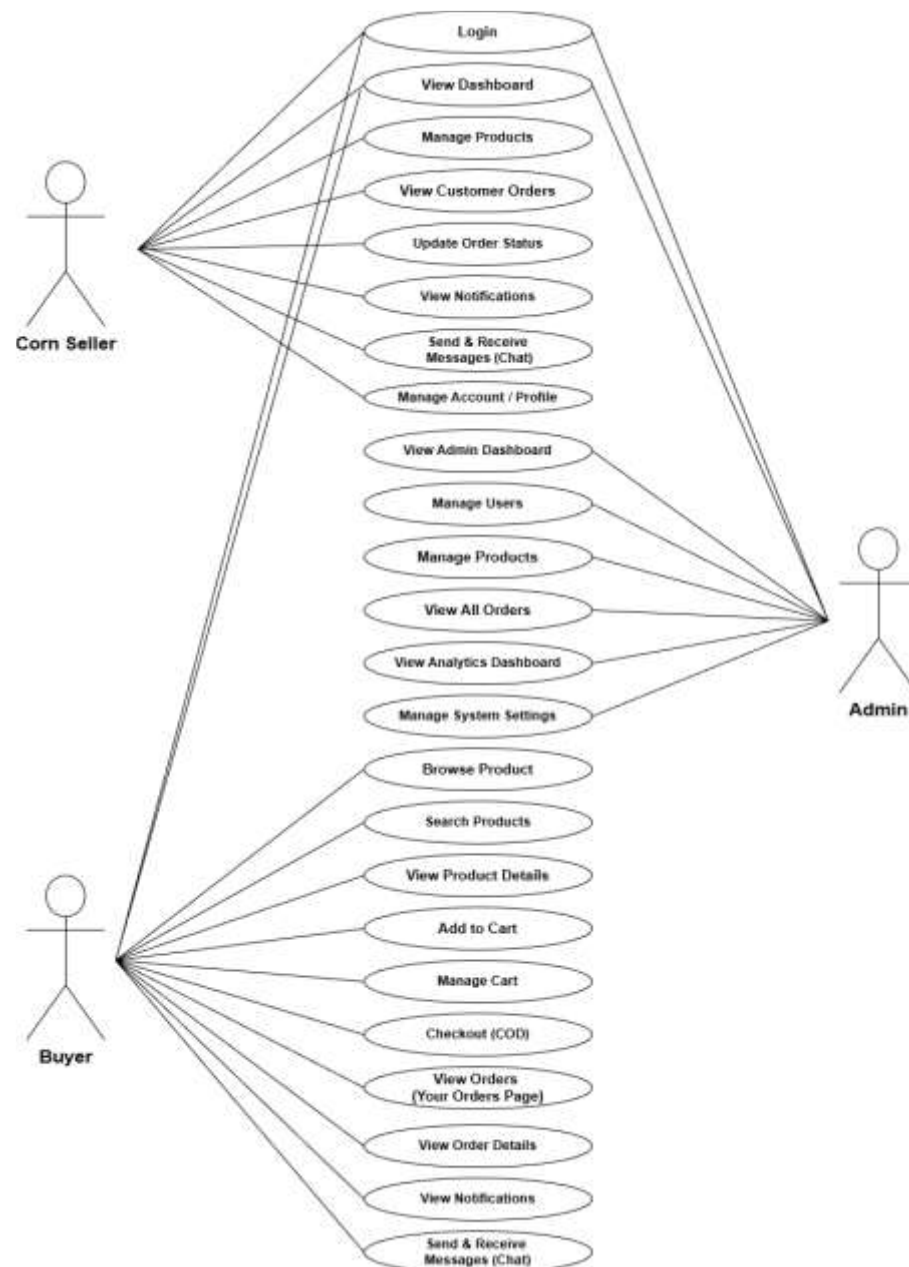


Figure 4.11 - Use Case Diagram

The Use Case Diagram for MaEase illustrates the various roles and interactions within the system, including the Buyer, Seller, and Admin. The diagram highlights key actions such as browsing products, managing orders, updating inventory, and viewing analytics, showing how each user type engages with the system. This visual representation helps in understanding the functional scope of the system, ensuring clear user roles and responsibilities.

4.3.6 Sequence Diagram

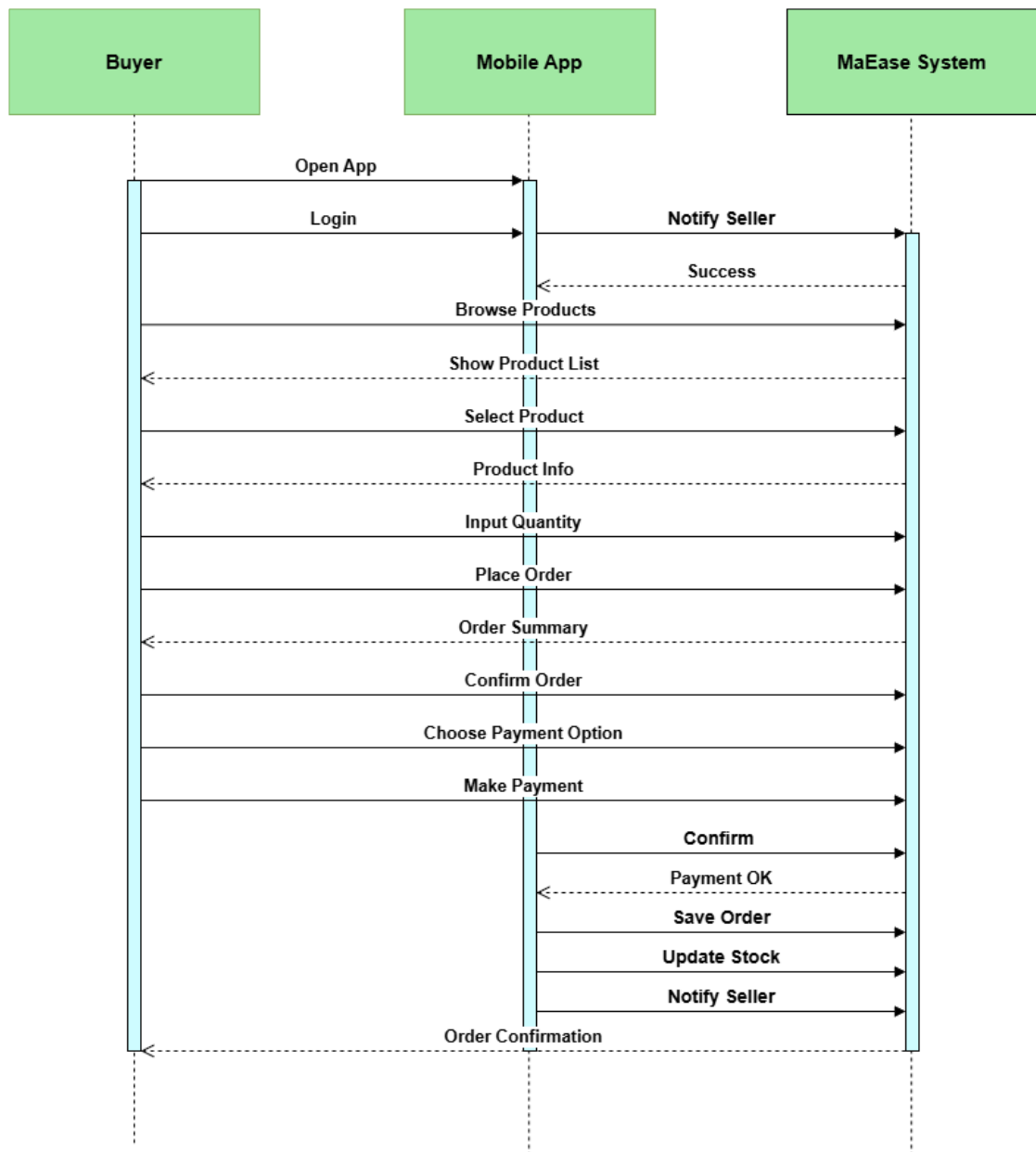


Figure 4.12 – *Sequence Diagram for Buyer Places an Order*

The sequence diagram for "Buyer Places an Order" shows the interactions between the Buyer, Mobile App, and MaEase System. The process begins with the Buyer logging into the mobile app, browsing products, and selecting an item. After confirming the order and payment details, the system updates the stock and notifies the Seller, completing the order confirmation process.

Sequence Diagram

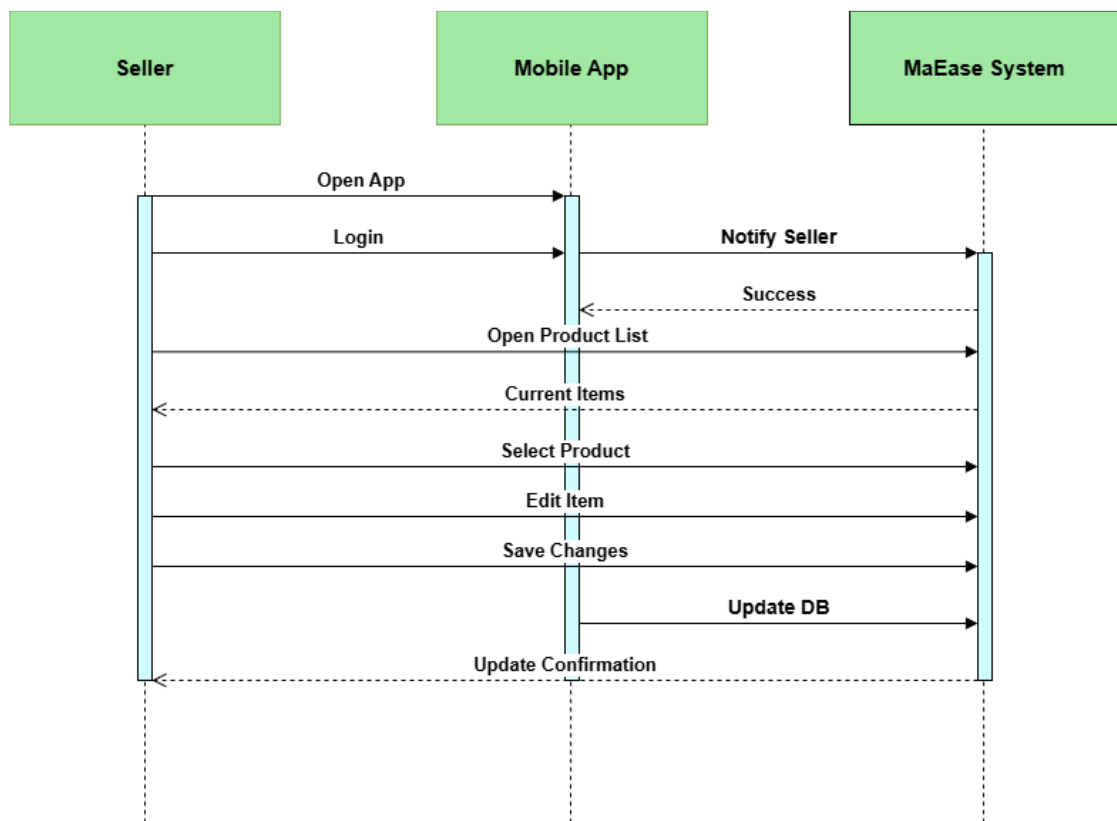


Figure 4.13 – *Sequence Diagram for Seller Manages*

The sequence diagram for "Seller Manages Items" shows the interaction between the Seller, Mobile App, and the MaEase System. The Seller logs into the app, browses and selects a product, and then edits it by updating its details. Once the changes are saved, the system updates the product details in the database and sends a confirmation to the Seller.

Sequence Diagram

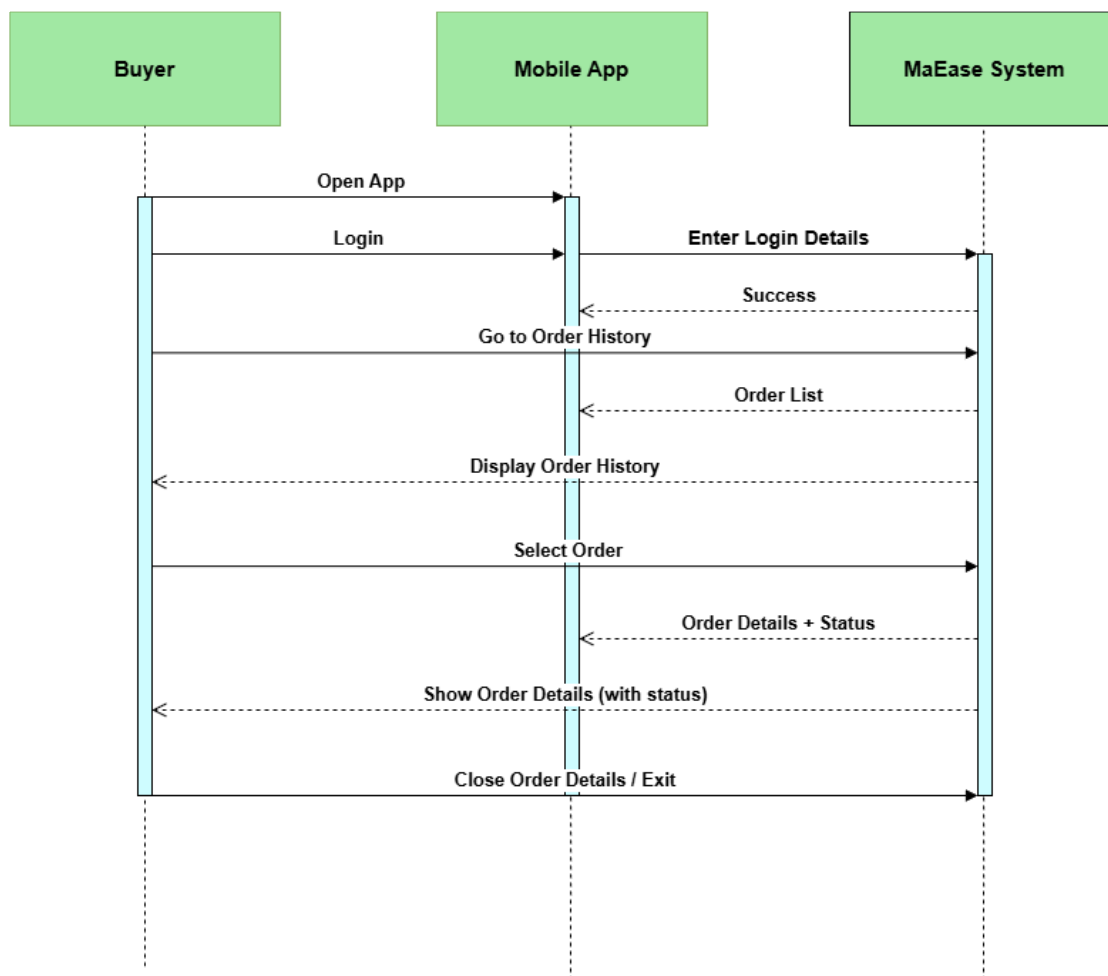


Figure 4.14 – *Sequence Diagram for Buyer Tracks Order Status*

The sequence diagram for "Buyer Tracks Order Status" illustrates how the Buyer interacts with the Mobile App and MaEase System. After logging in, the Buyer accesses their order history, selects an order, and views the order details, including its current status. The system retrieves and displays the order details with its latest status, allowing the Buyer to track progress.

Sequence Diagram

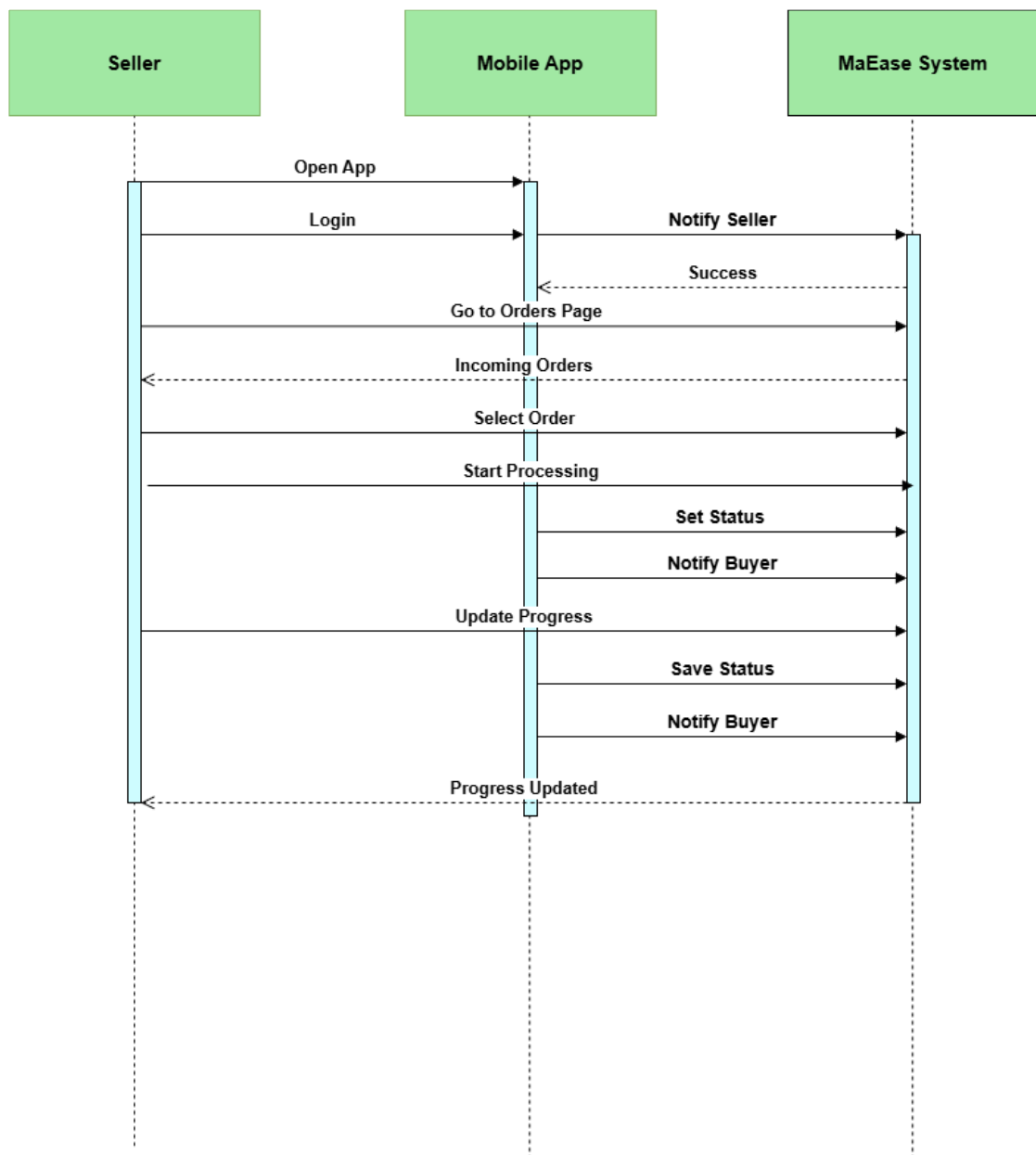


Figure 4.15 – *Sequence Diagram for Seller Handles Order & Updates Progress*

The sequence diagram for "Seller Handles Order & Updates Progress" outlines the process when the Seller logs in, checks incoming orders, and begins processing an order. The system allows the Seller to update the progress of the order, such as changing its status and notifying the Buyer about the update. The system then saves the status and sends a confirmation notification to the Buyer.

Sequence Diagram

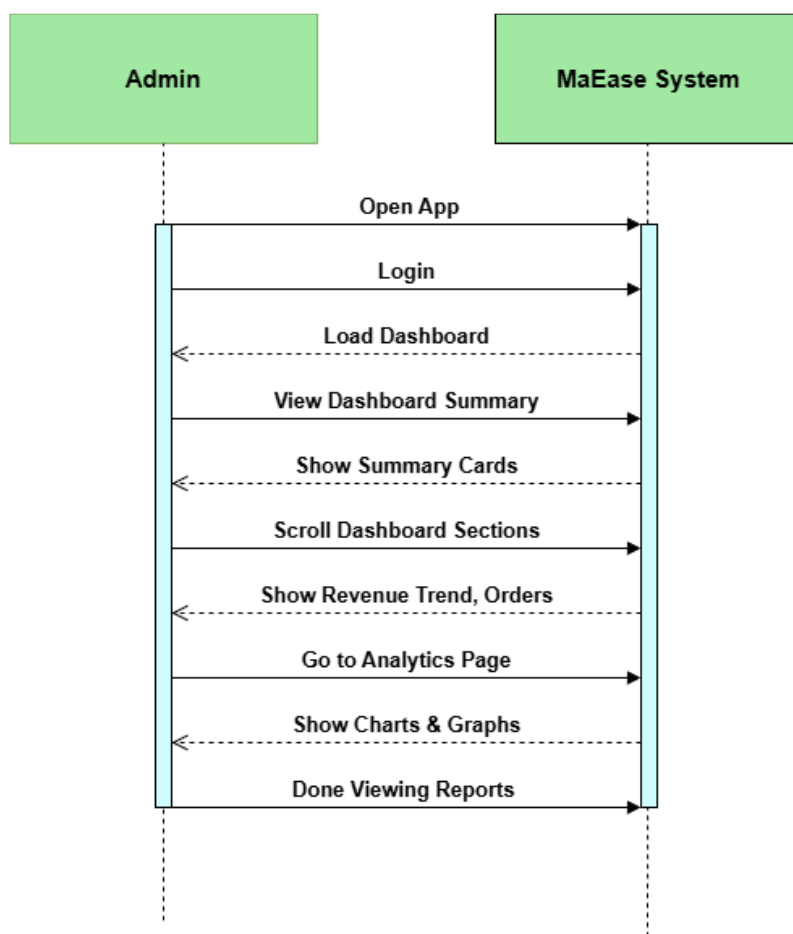


Figure 4.16 – *Sequence Diagram for Admin Views Reports*

The sequence diagram for "Admin Views Reports" illustrates the process where the Admin logs into the system and accesses the dashboard. The Admin views the dashboard summary, including various data points like revenue trends and order details, before navigating to the analytics page. The Admin is then able to view detailed charts and graphs related to these metrics before exiting the report viewing.

Sequence Diagram

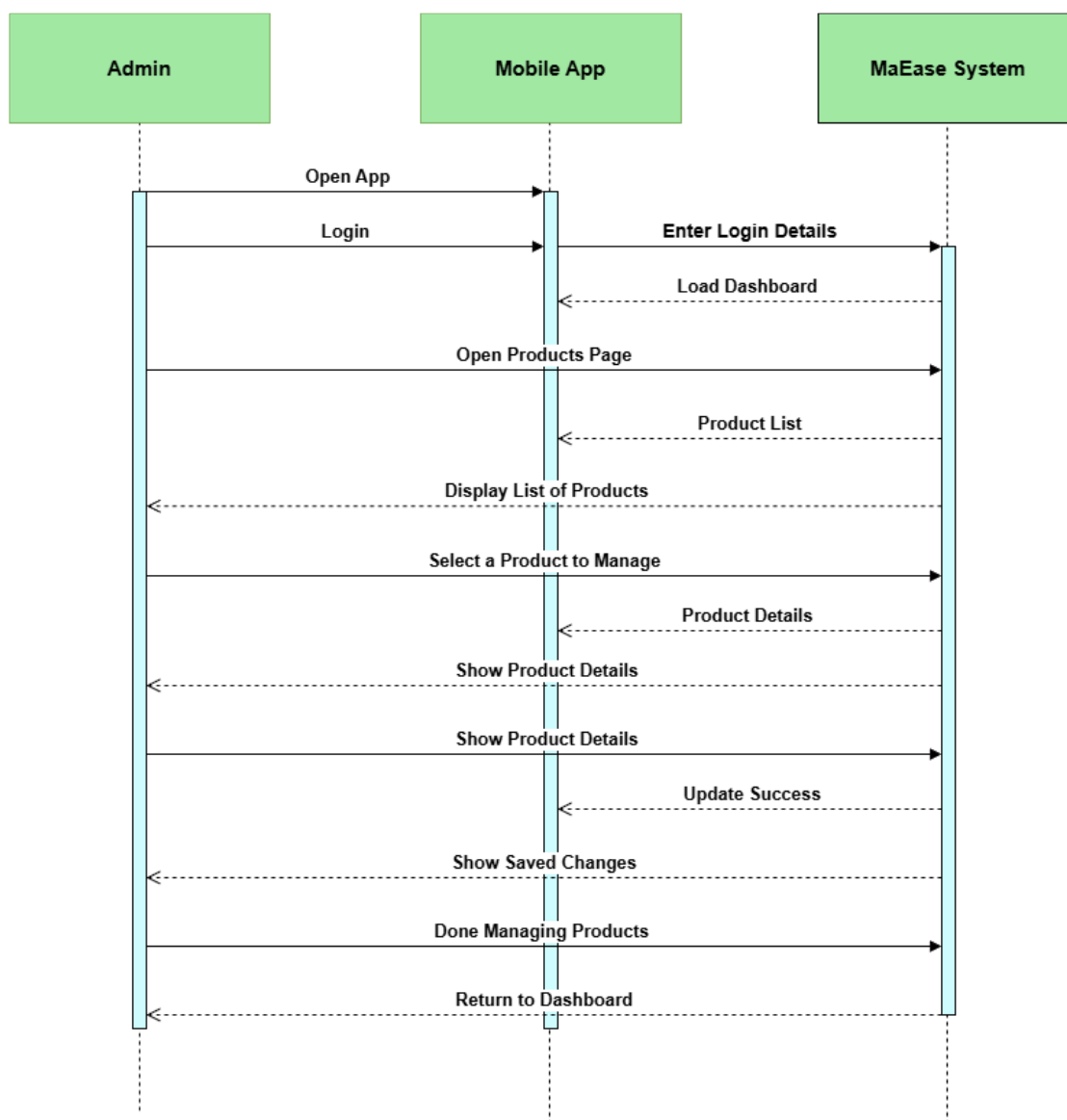


Figure 4.17 – *Sequence Diagram for Admin Manages Products*

The sequence diagram for "Admin Manages Products" depicts the process where the Admin logs into the system and accesses the products page. After selecting a product, the Admin views and updates its details, including saving the changes made. Finally, the Admin returns to the dashboard after completing the product management task.

4.3.7 Activity Diagram

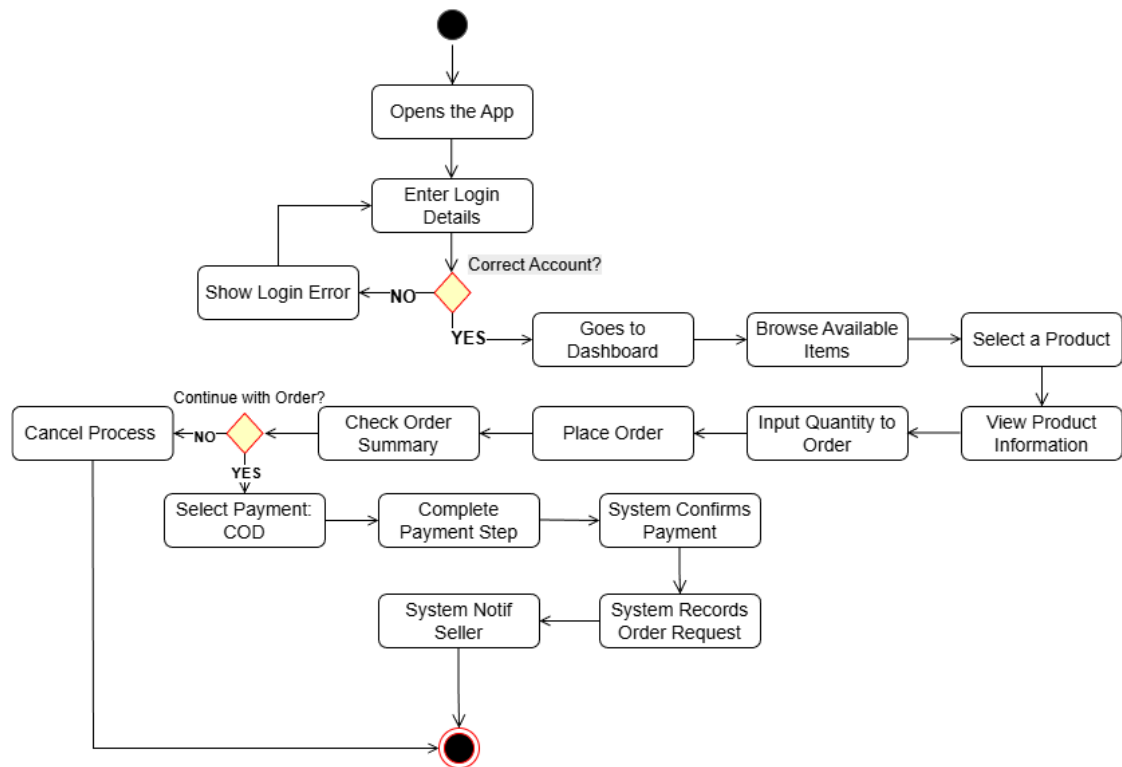


Figure 4.18 - Activity Diagram of Buyer Places an Order

The activity diagram for "Buyer Places an Order" outlines the process where a buyer logs into the system and browses available products. After selecting a product and inputting the quantity, the buyer proceeds to check the order summary, choose the payment method (COD), and confirms the payment. The system then records the order and notifies the seller about the purchase.

Activity Diagram

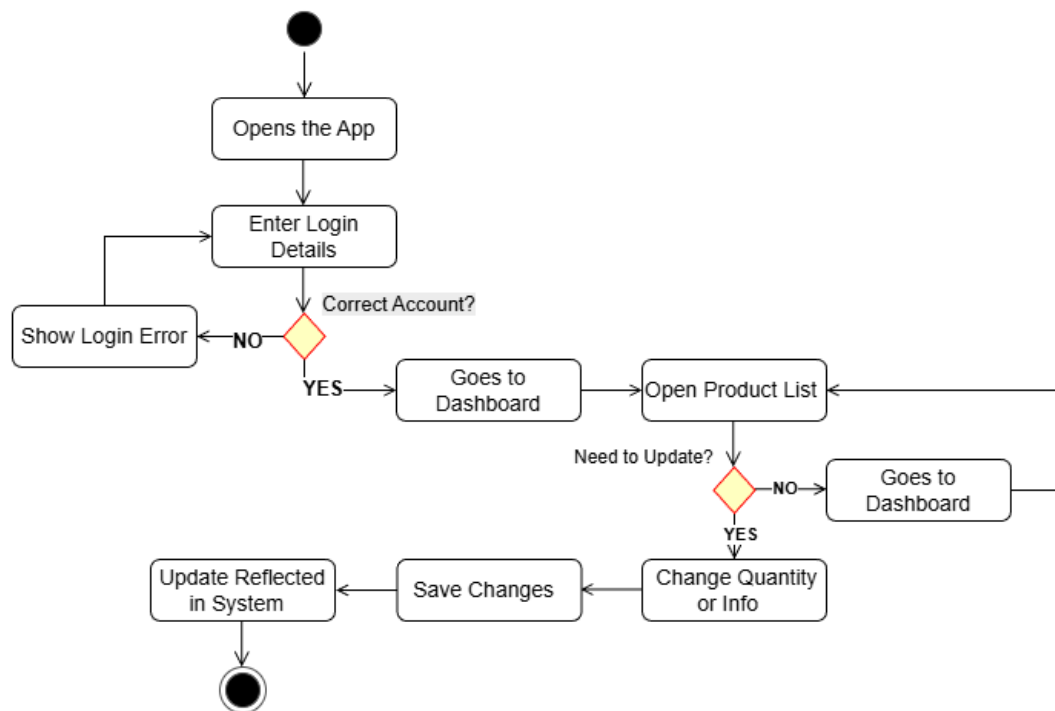


Figure 4.19 - Activity Diagram for Seller Manages Items

The activity diagram for "Seller Manages Items" starts when the seller logs into the app and accesses the product list from the dashboard. If the seller needs to update an item, they can change the quantity or product information, then save the changes. Once the updates are saved, they are reflected in the system.

Activity Diagram

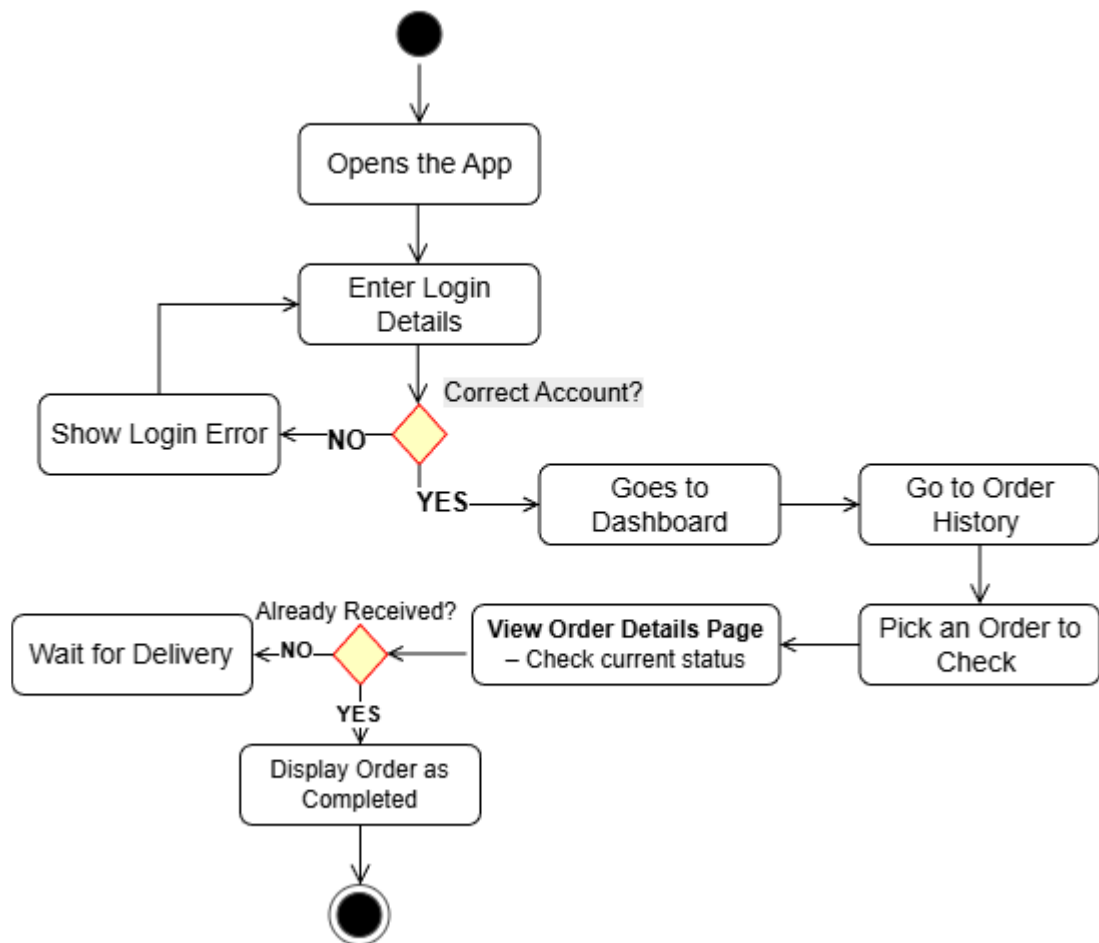


Figure 4.19 – Activity Diagram for Buyer Tracks and Confirms Order

The activity diagram for "Buyer Tracks Order Status" starts with the buyer logging into the app and checking their order history. The buyer can select an order to view its details and check its current status. If the order has already been received, the system will display it as completed; otherwise, the buyer waits for delivery updates.

Activity Diagram

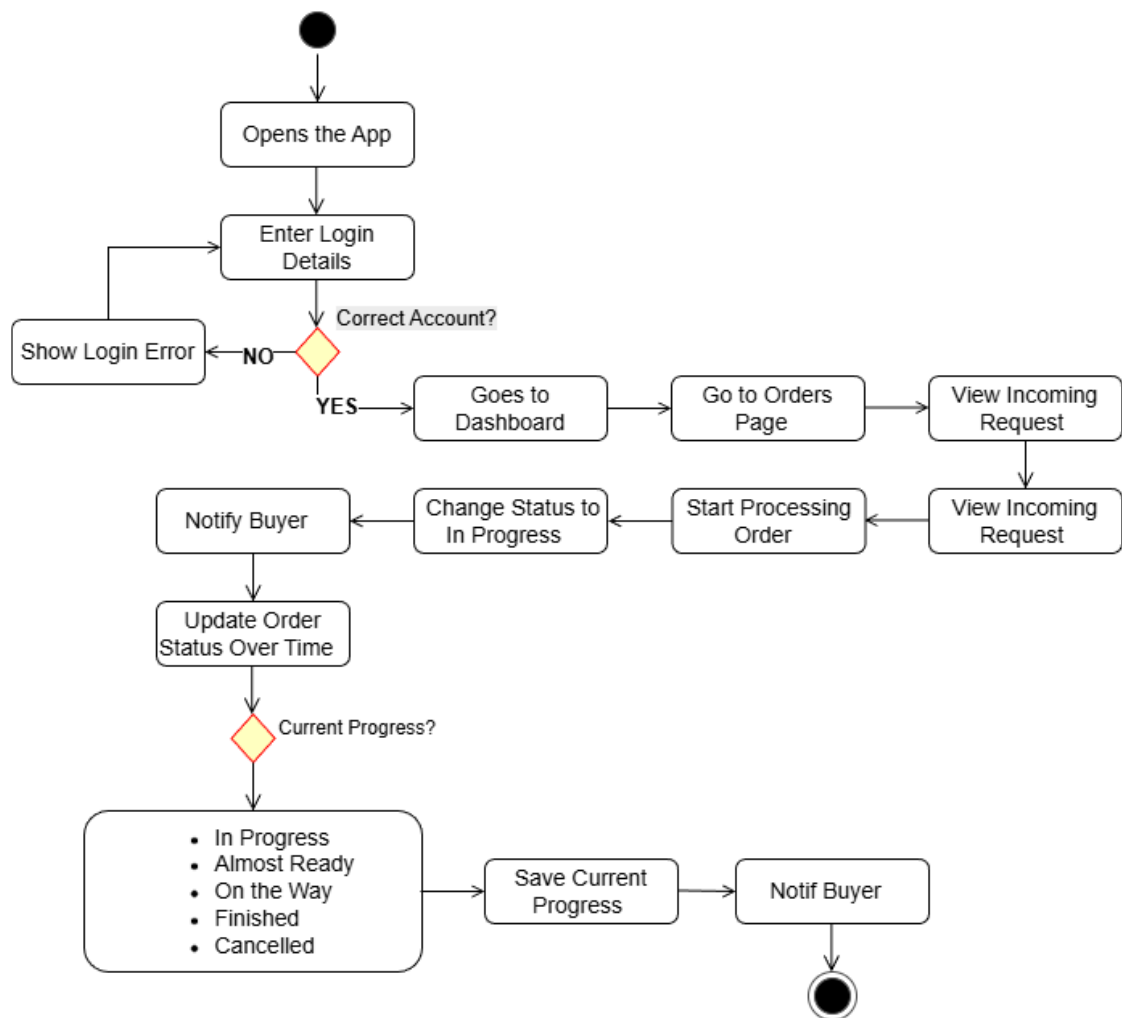


Figure 4.20 – Activity Diagram for Seller Handles Order & Updates Progress

The activity diagram for "Seller Handles Order & Updates Progress" begins with the seller logging in and viewing the incoming order request. The seller processes the order, updates the order status to "In Progress," and continuously updates the status over time, such as "Almost Ready," "On the Way," or "Finished." Finally, the seller saves the progress and notifies the buyer about the updated status.

Activity Diagram

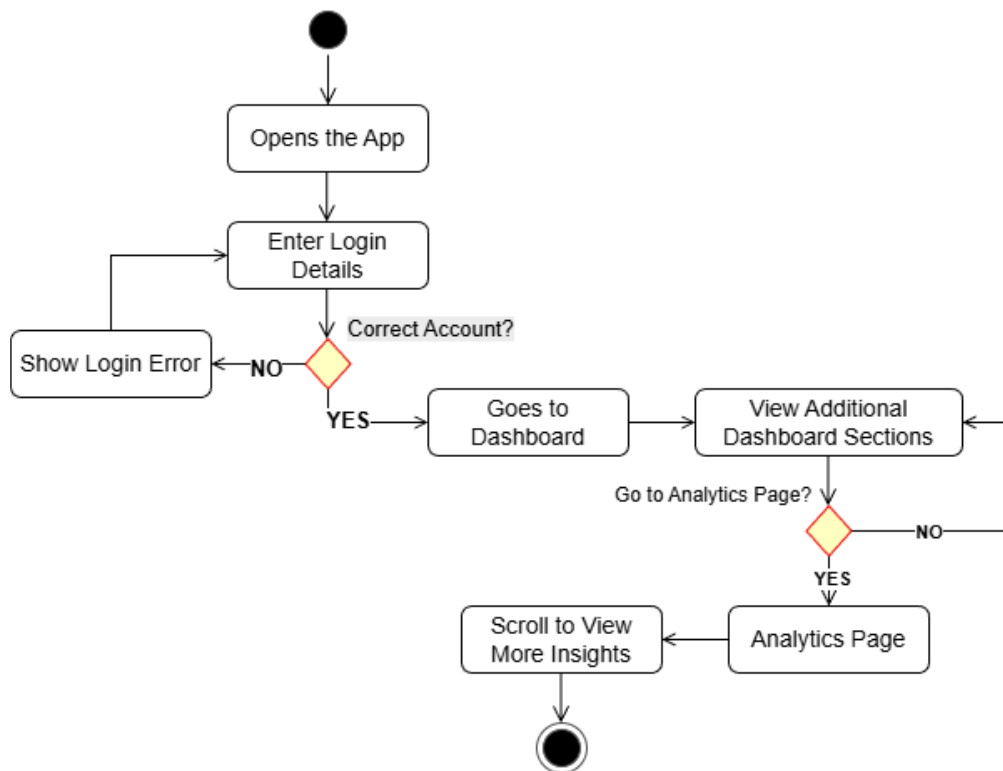


Figure 4.21 - Activity Diagram for Admin Manages Items

The activity diagram for "Admin Views Reports" begins with the admin logging in and accessing the dashboard. From there, the admin can view additional sections and scroll to see more insights or choose to go to the analytics page. Once on the analytics page, the admin can view detailed reports.

Activity Diagram

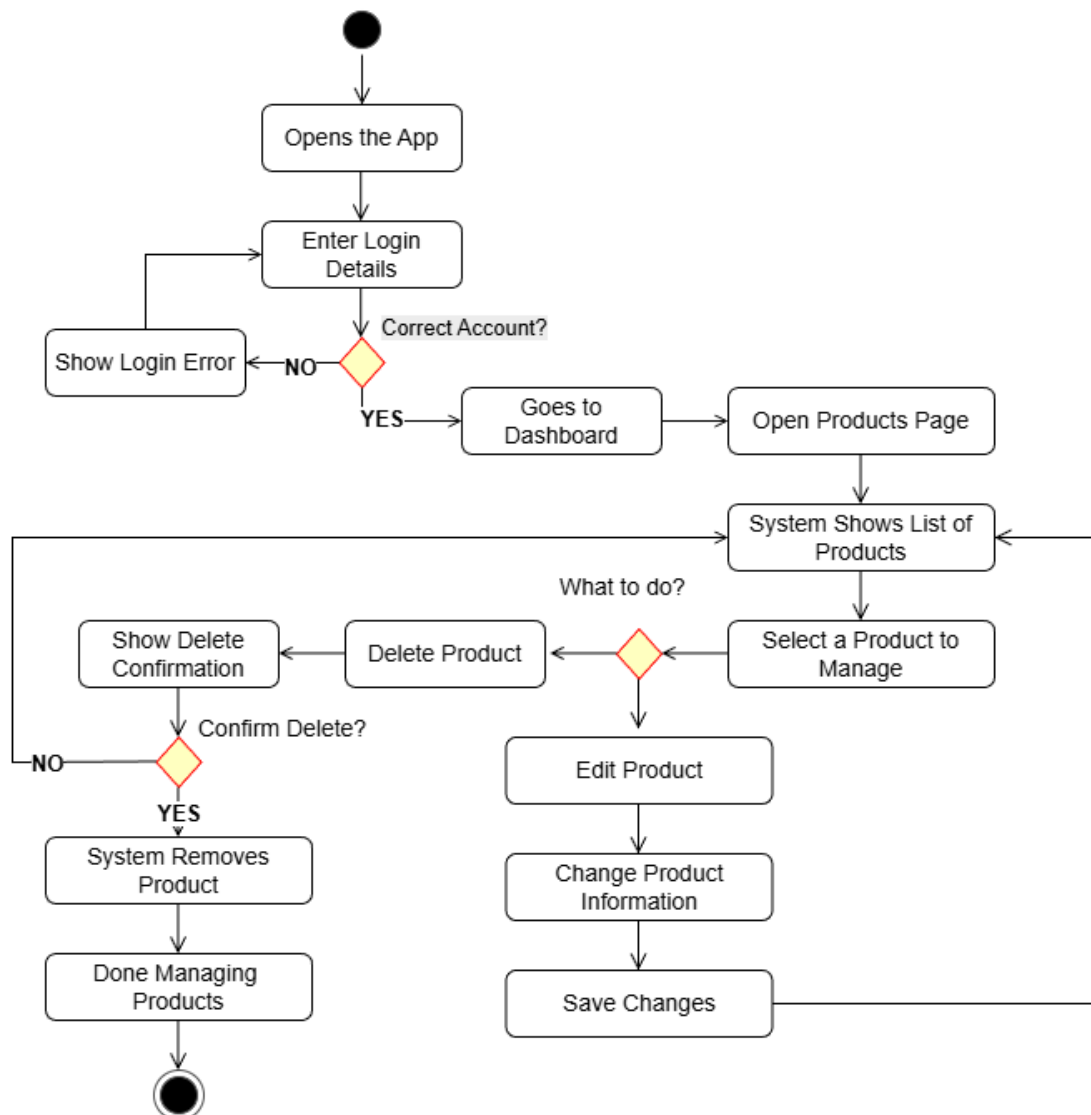


Figure 4.22 - Activity Diagram for Admin Manages Products

The activity diagram for "Admin Manages Products" starts with the admin logging in and accessing the dashboard. The admin can either delete or edit a product from the list displayed on the products page. If a product is deleted, the system removes it after confirmation, or if edited, the admin updates the product information and saves the changes.

4.4 System Design

The System Design section outlines the architectural and user interface (UI) aspects of the MaEase mobile and web application. It covers the design choices, including the layout of screens for different users (admin, seller, and buyer), as well as the forms and reports integrated into the system.

4.4.1 Output and User-Interface Design

1.Admin Interface

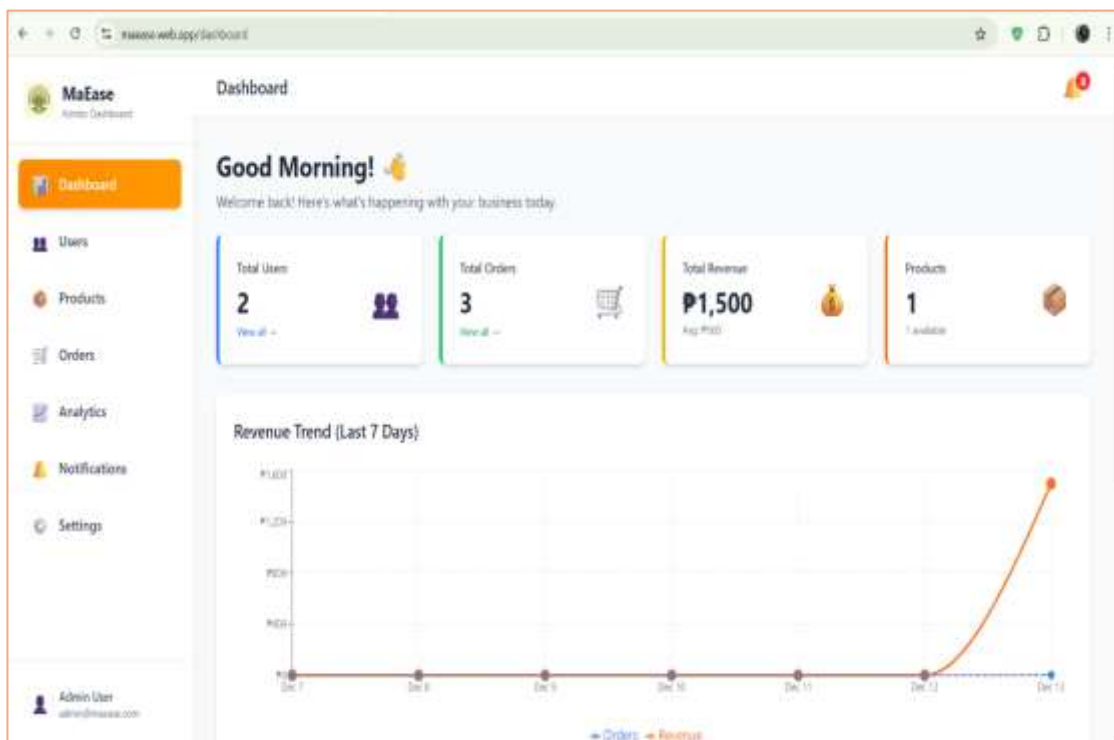


Figure 4.23 –Admin Dashboard Page

The admin dashboard, highlighting key metrics such as total users, total orders, and total revenue, along with the revenue trend (last 7 days) graph. Below, the recent orders section displays a list of the latest orders, including the order number, buyer's name, total amount, and payment method. This overview helps admins efficiently monitor system performance and manage incoming orders.

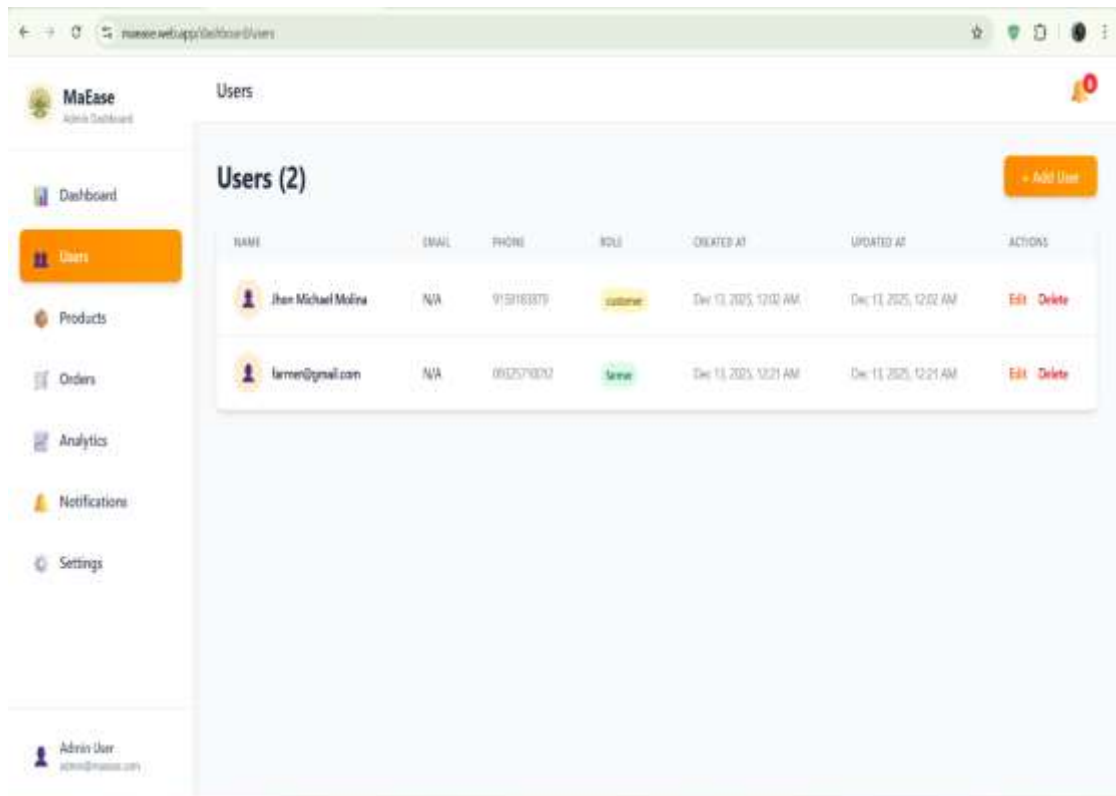


Figure 4.24 – Admin User Page

The user management page displays a list of registered users, showing details like name, email, role, and phone number. admins can easily manage users with options to edit or delete accounts, available under the actions column. to add new users, admins can simply click on the “+ add user” button at the top right.

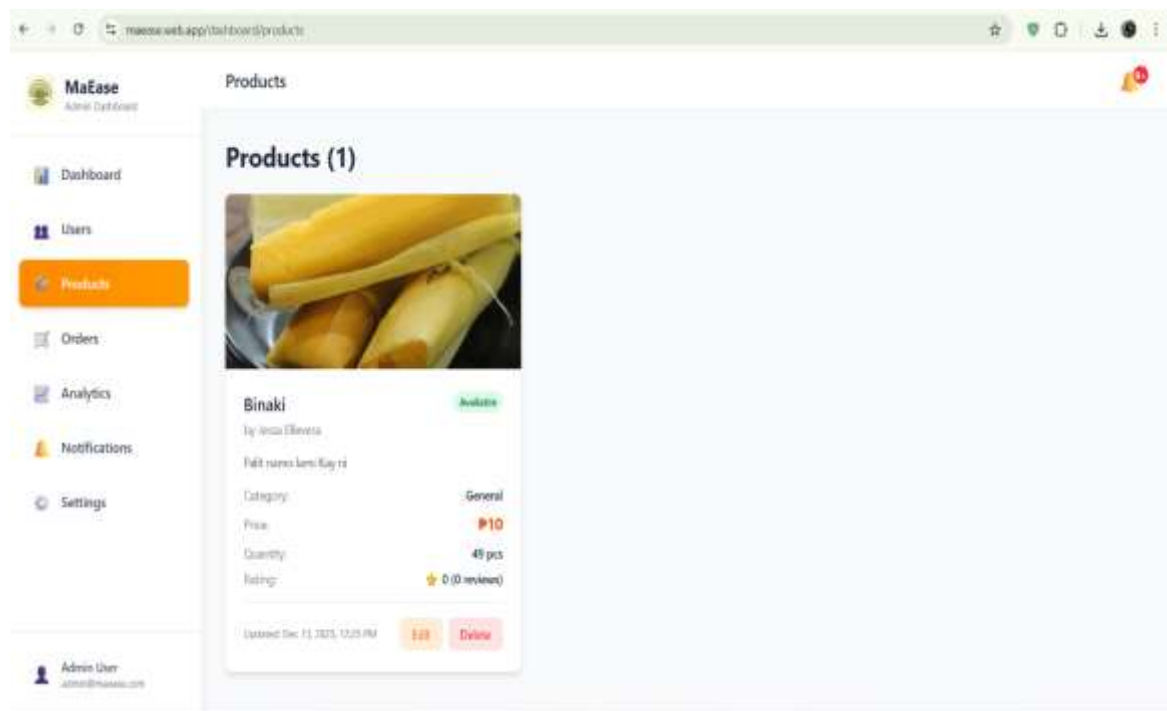
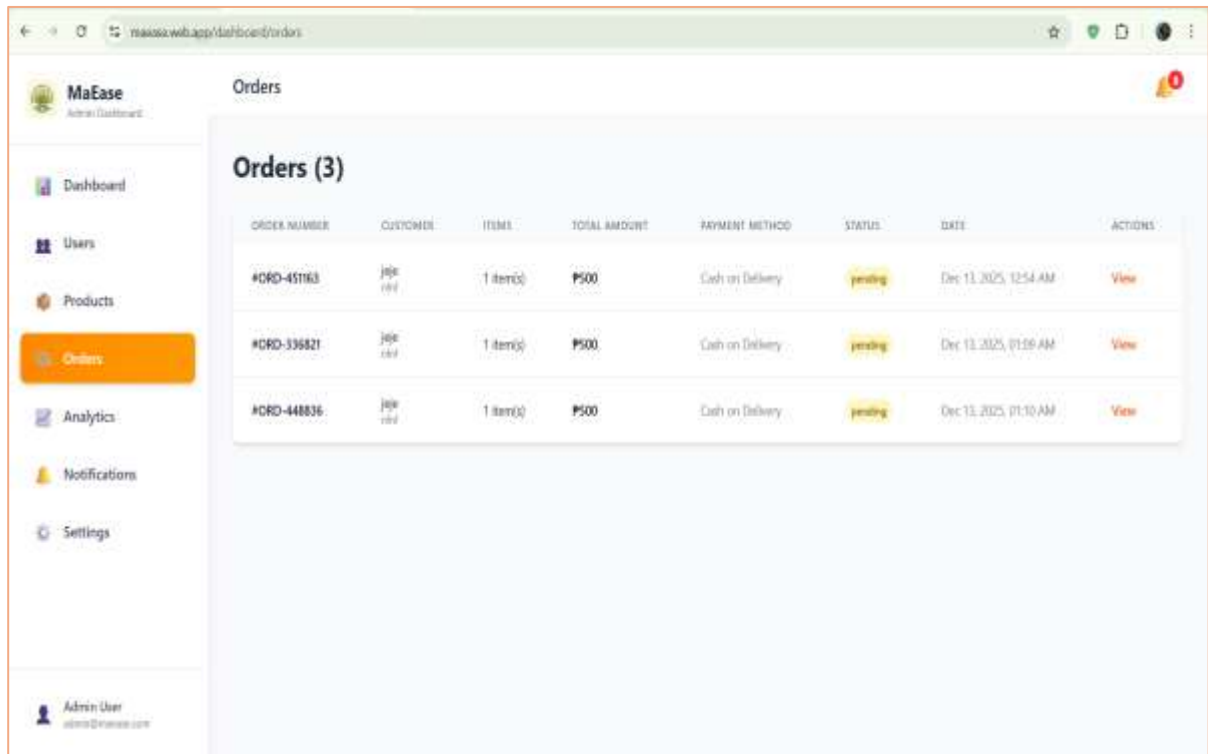


Figure 4.25 – Admin Products Page

This page displays detailed information about a product listed by a seller. The product page includes key details such as product name, price, quantity available, and a brief product description. Admins can manage the product by clicking edit or delete, allowing them to update product details or remove it from the system.



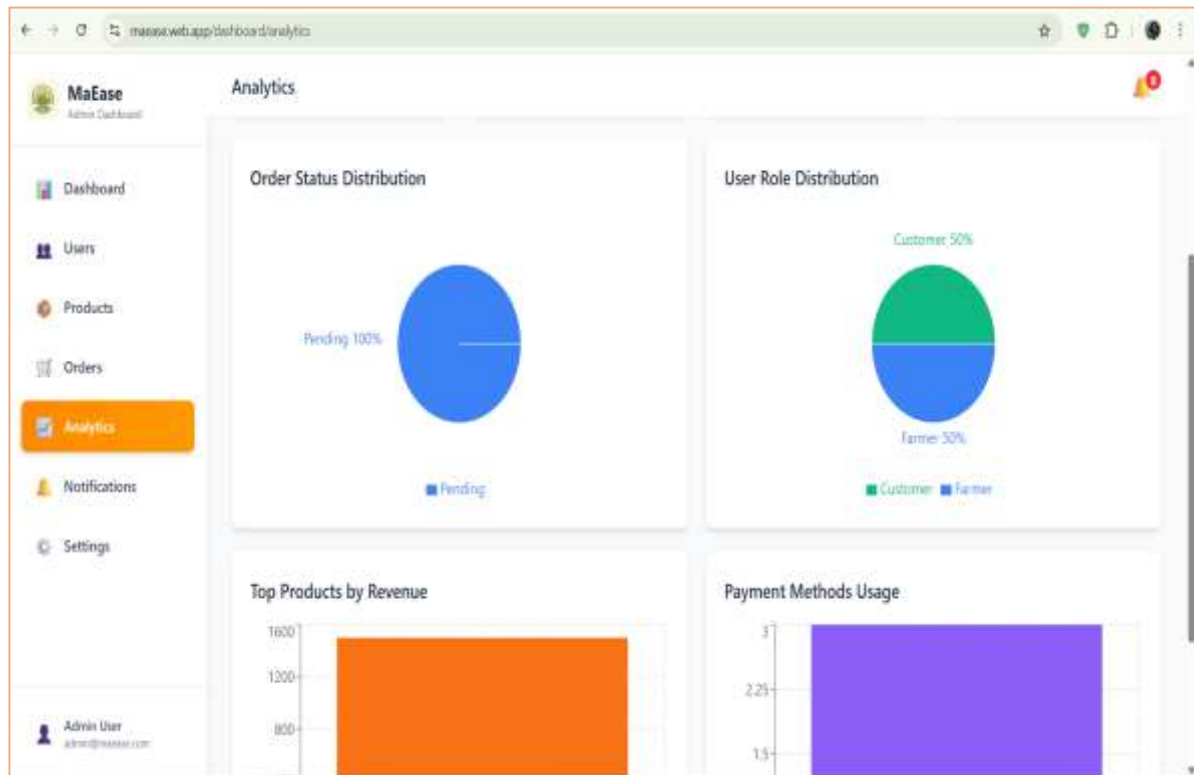
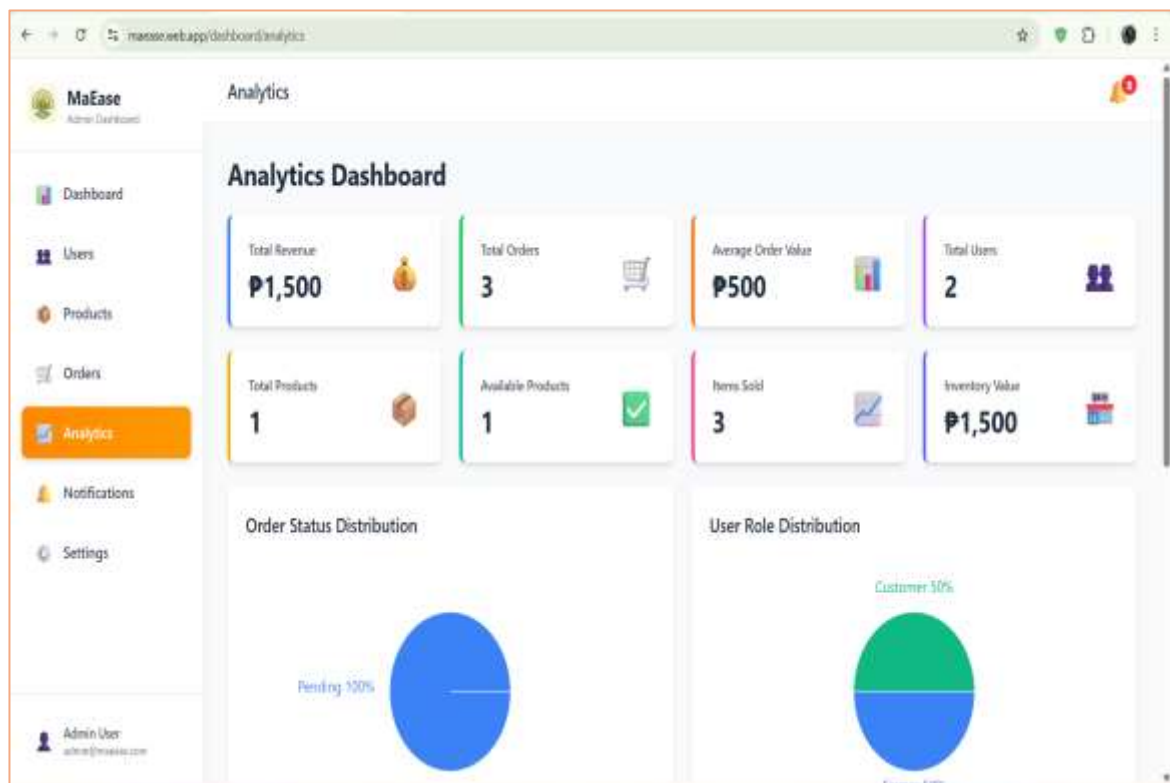
The screenshot displays the 'Orders' page in the MaEase Admin Dashboard. The sidebar on the left contains navigation links: Dashboard, Users, Products, Orders (highlighted), Analytics, Notifications, and Settings. The main content area is titled 'Orders (3)' and contains a table with the following data:

ORDER NUMBER	CUSTOMER	ITEMS	TOTAL AMOUNT	PAYMENT METHOD	STATUS	DATE	ACTIONS
#ORD-451163	jipe crl	1 item(s)	P500	Cash on Delivery	pending	Dec 13, 2025, 12:54 AM	View
#ORD-336821	jipe crl	1 item(s)	P500	Cash on Delivery	pending	Dec 13, 2025, 01:59 AM	View
#ORD-448836	jipe crl	1 item(s)	P500	Cash on Delivery	pending	Dec 13, 2025, 01:50 AM	View

The bottom of the sidebar shows the user 'Admin User' with the email 'admin@maease.com'.

Figure 4.26 – Admin Order Page

The order page in the admin dashboard, where admins can view a list of all customer orders. The page includes key details such as order number, customer name, total amount, payment method, and order status. Admins can click on the view button for more detailed order information.



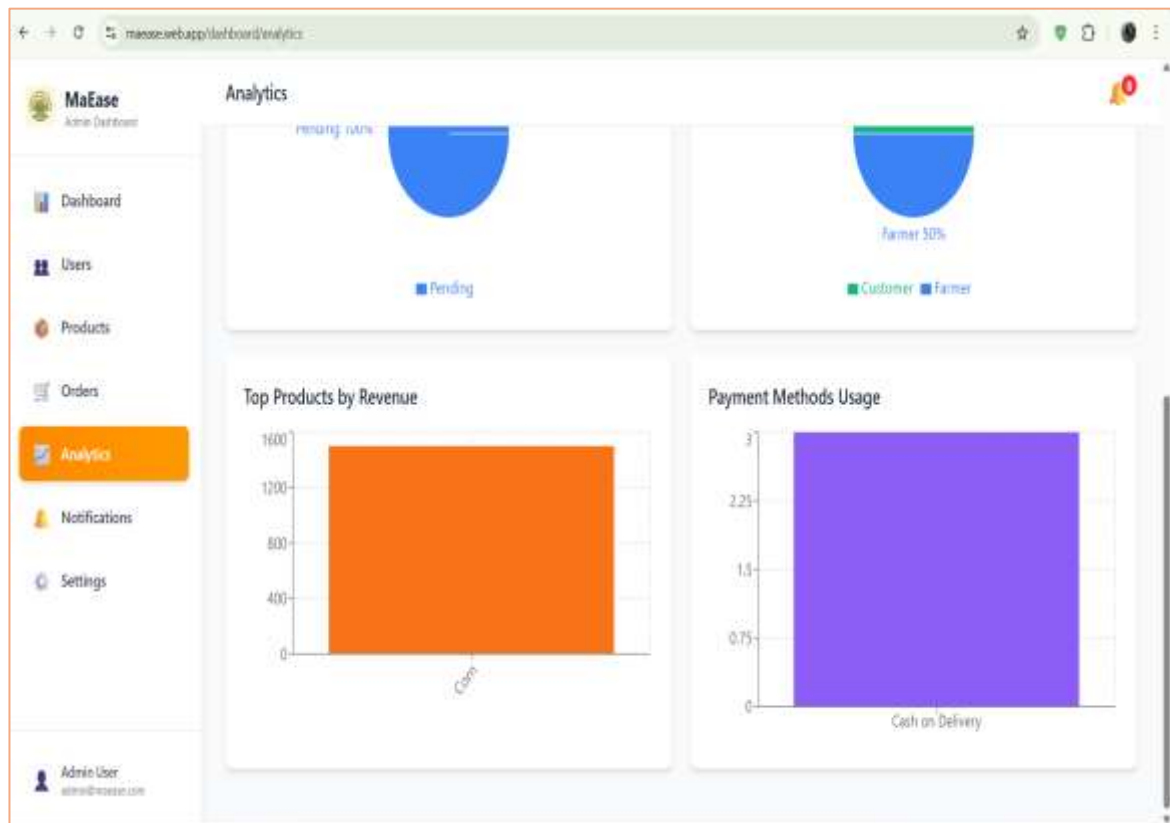


Figure 4.27 – Admin Analytics Dashboard Page

The analytics dashboard provides a comprehensive overview of the system's performance, including total revenue, total orders, inventory value, and user distribution. It features key metrics such as top products by revenue and payment methods usage, with visualizations like pie charts and bar graphs. Admins can easily analyze system data to monitor trends and make informed decisions.

2. Seller Interface

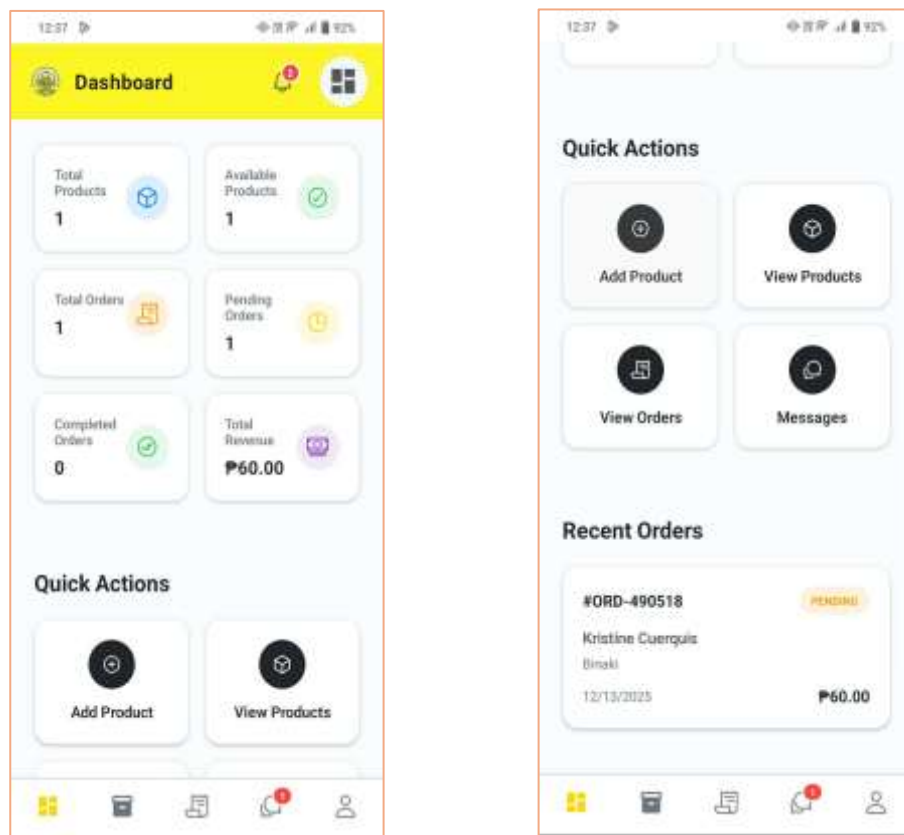


Figure 4.28 – Dashboard Page (Seller)

The home page for sellers in the maease mobile app, providing key metrics like total products, pending orders, and total revenue. Sellers can quickly navigate to quick actions such as adding a product, viewing products, and managing orders. The recent orders section displays the latest order details, helping sellers track their incoming orders and take timely action.

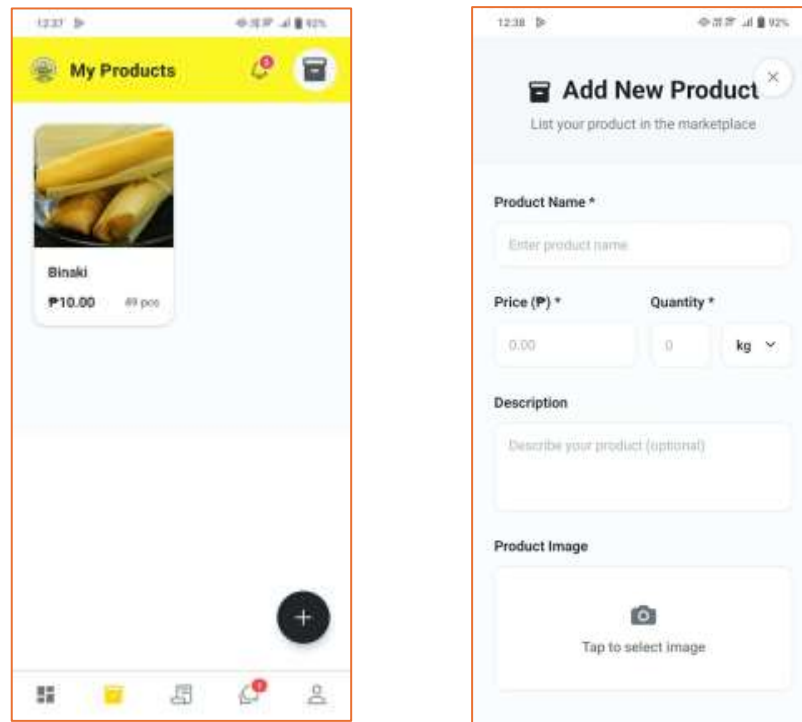


Figure 4.29 – My Products & Add Product Page (Seller)

The my products page displays a list of products available for sale, showing details like product name, price, and quantity. Sellers can easily add new products by tapping the “add product” button, which opens a form where they can input product name, price, quantity, description, and upload a product image. This intuitive interface allows sellers to manage their product listings quickly and efficiently.

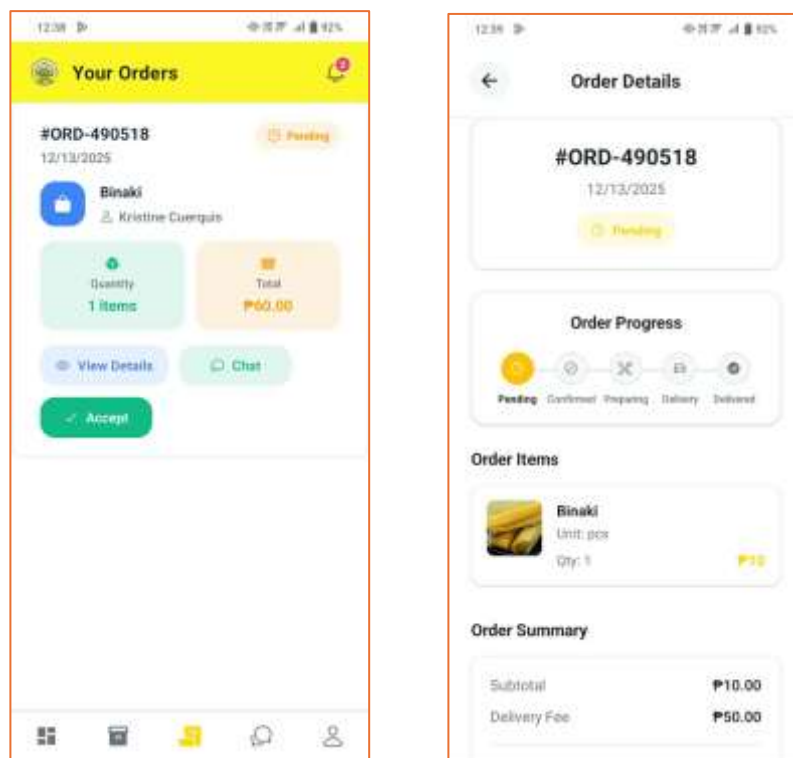


Figure 4.30 – Order Page & Order Details (Seller)

The order page shows a list of incoming orders, including order number, quantity, and total price, with options for the seller to accept, view details, or chat with the buyer. The order details page provides a comprehensive view of the order's status, including order progress (pending, confirmed, preparing, delivery, delivered) and itemized products. Sellers can also review order summary with details like subtotal, delivery fee, and total amount before processing the order.

4. Buyer Interface

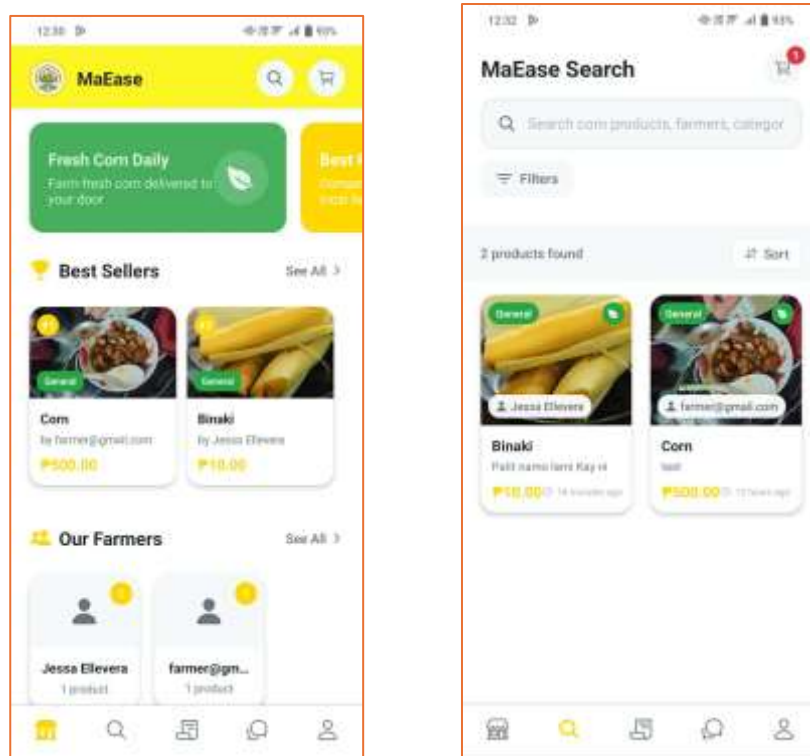


Figure 4.31 – Home Page & Search Bar (Buyer)

The home page for the buyer showcases best sellers and product categories, allowing users to quickly discover popular items like binaki and corn. The search bar enables buyers to search for specific products, farmers, or categories, making it easier to find desired items. This combination of features helps the buyer navigate the app efficiently and make informed purchasing decisions.

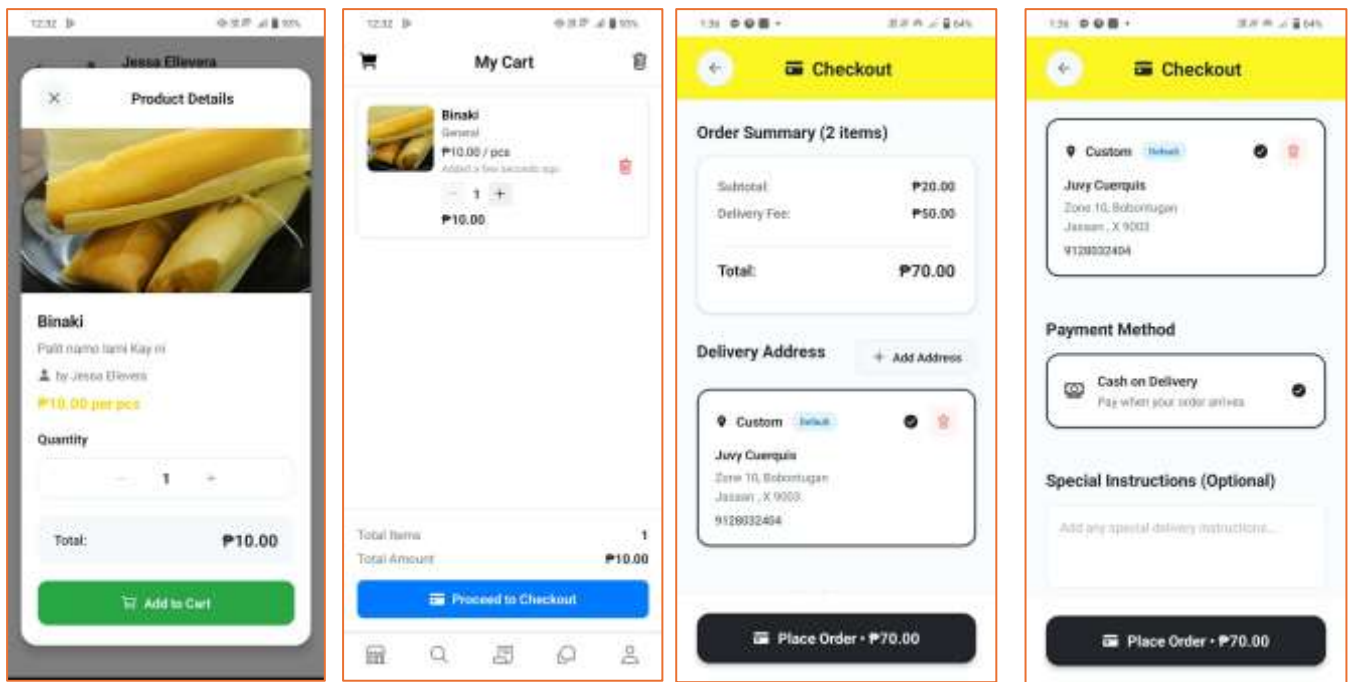


Figure 4.32 – Add to Cart, My Cart & Checkout Page (Buyer)

The add to cart page allows buyers to view detailed product information, select quantity, and add the product to their cart with the “add to cart” button. The my cart page displays the added products, quantity, and total price, giving buyers an overview before proceeding. On the checkout page, buyers can review their order summary, choose a payment method (e.g., cash on delivery), and enter their delivery address before confirming the purchase.

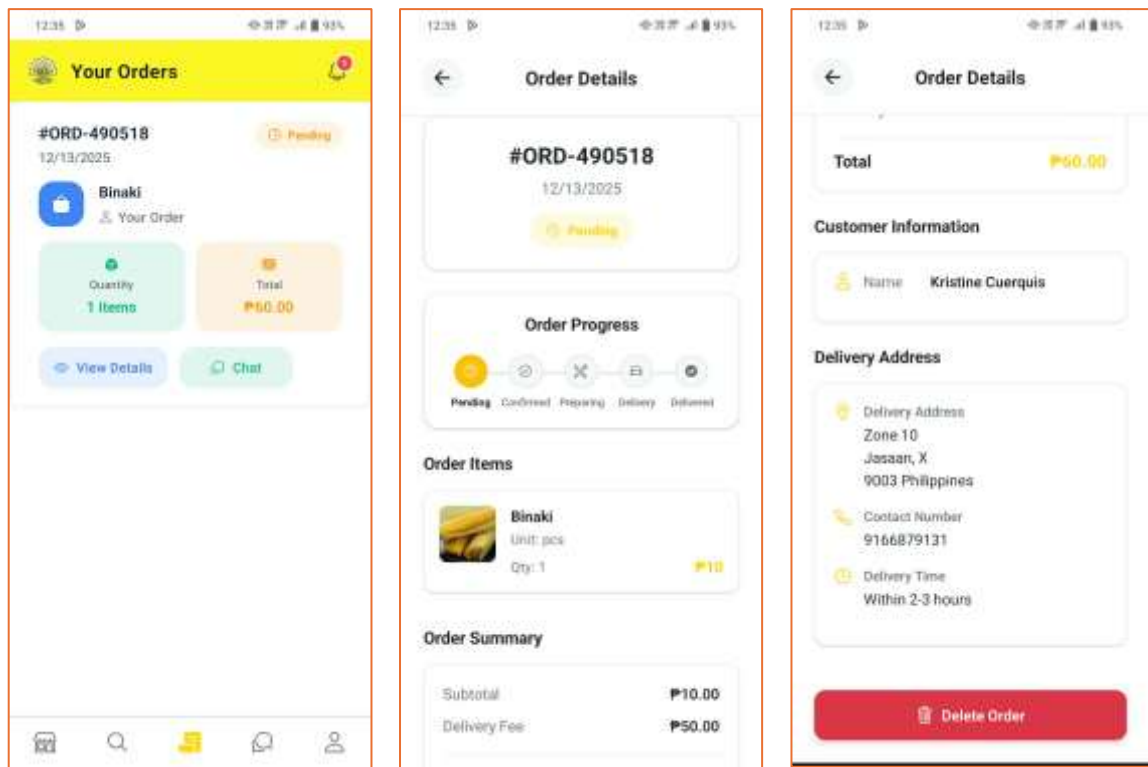


Figure 4.33 – Add to Cart, My Cart & Checkout Page (Buyer)

The "my orders" page allows buyers to view all their recent orders, including details such as the product name, order status, and total amount. It also provides options to view more details or chat with the seller for any inquiries. The "order details" page offers a deeper dive into each order, displaying the order progress, product items, delivery address, and the option to delete the order if needed.

Forms

The figure consists of two side-by-side screenshots of the Maease mobile app login interface. Both screens feature a circular logo at the top center with the text 'MAEASE' and 'AGRICULTURE' below it. The main heading is 'Welcome Back!' followed by a green leaf icon. Below this is the instruction: 'Sign in to your Maease account and continue your farming journey.'

Left Screenshot (Initial Login):

- Login Method:** Two buttons are shown: 'Password' (orange) and 'SMS' (white with orange border).
- Phone Number:** A text input field contains '09171251234'. Below it is a hint: 'Enter your Philippine mobile number (e.g., 09171251234)'.
- Password:** A text input field contains 'Enter your password'. To the right is an eye icon for toggling visibility.
- Sign In:** An orange button at the bottom with the text 'Sign In →'.

Right Screenshot (OTP Verification):

- Login Method:** The 'SMS' button is now orange, and the 'Password' button is white with an orange border.
- Phone Number:** The same number '09171251234' is in the input field, with the same hint below it.
- Send OTP:** An orange button with a green checkmark icon and the text 'Send OTP'.
- Enter OTP Code:** A text input field contains '000000'. Below it is a hint: 'Enter the 6-digit code sent to your phone'.

Figure 4.34 – Login Page (Buyer and Seller)

This screenshot shows the maease mobile app login page, where buyers and sellers can choose between logging in using password or sms (otp). Users enter their phone number to proceed with either login method and receive a one-time passcode (otp) for verification, or they can enter their password directly. The interface is designed to ensure quick access while maintaining user security, particularly for farmers with limited technical knowledge.

The image displays two sequential steps of the Maease mobile app registration process. Both screens feature a green circular logo at the top with a stylized plant icon, followed by the text 'Join Maease!' and a subtitle 'Create your account and connect with local farmers in your community'.

Left Screenshot (Step 1): This screen prompts the user to enter their full name, phone number, and OTP code. It includes a 'Send OTP' button and a 'Create a password' field at the bottom.

Right Screenshot (Step 2): This screen prompts the user to enter their password, confirm their password, and select their role as either 'Farmer' or 'Customer'. It includes a 'Create Account' button at the bottom.

Figure 4.35 – Register Page (Buyer and Seller)

The maease mobile app registration page, where buyers and sellers can create their accounts. Users enter their full name, phone number, and a one-time passcode (otp) sent to their mobile, followed by creating a password. They also choose their role (farmer or customer) and can then click “create account” to complete the registration process.

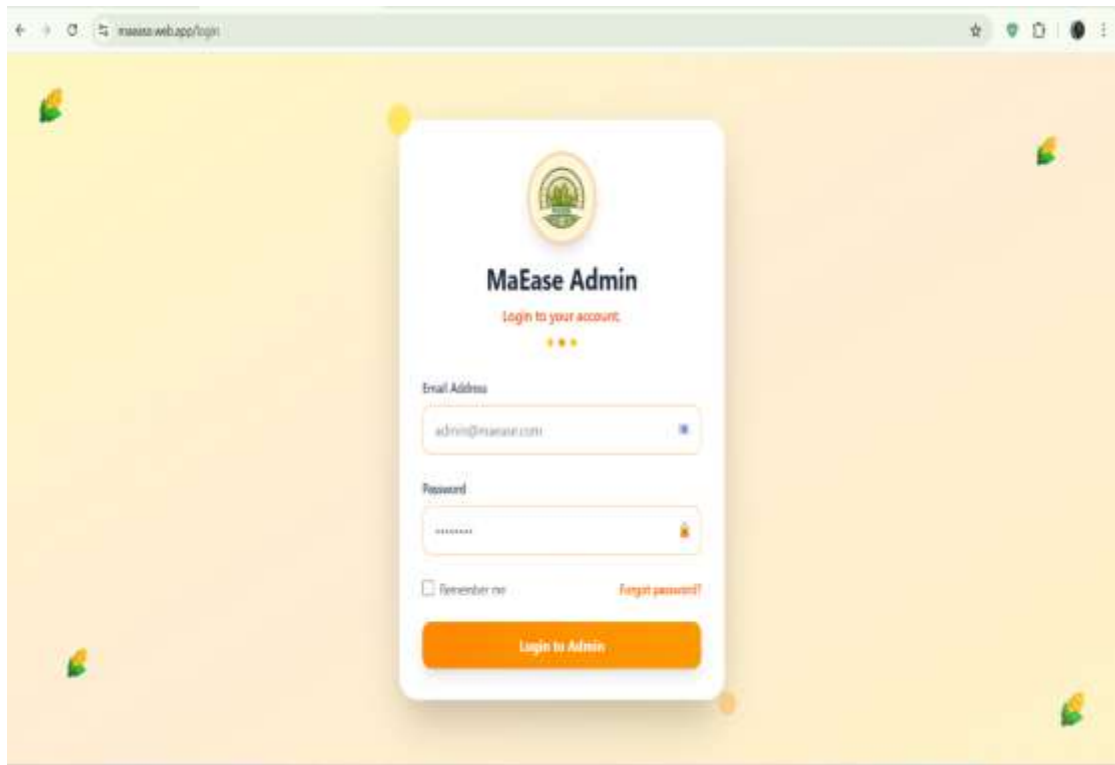


Figure 4.36 – Admin Login Page

This form shows the admin login page for the maease system, where the admin enters their email address and password to access the backend. the design is simple and user-friendly, featuring the maease logo at the top and a “login to admin” button. the page also offers options like “remember me” for saving login credentials and a “forgot password?” link for account recovery.

12:38

Add New Product ✕

List your product in the marketplace

Product Name *

Enter product name

Price (P) * **Quantity ***

0.00 0 kg

Description

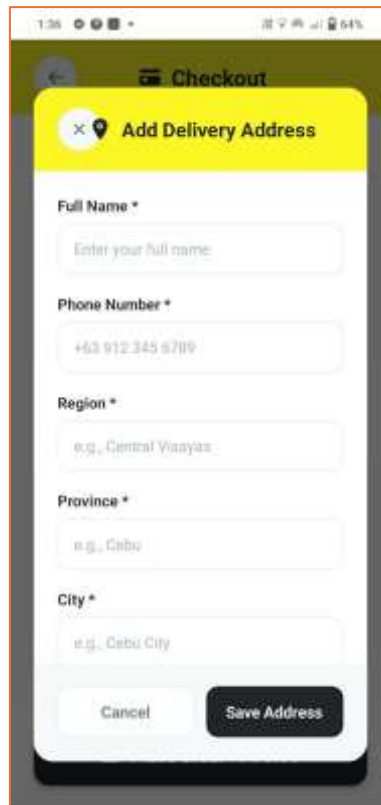
Describe your product (optional)

Product Image

Tap to select image

Figure 4.37 – Add Product Form (Seller)

The "add new product" form allows the seller to input details about their product, including the product name, price, quantity, and an optional description. The seller can also upload an image of the product, making it easier for buyers to view the item. This form is designed to streamline the process of listing new products for sale within the maease marketplace.



The screenshot shows a mobile application interface for adding a delivery address. At the top, there is a status bar with the time 1:36, signal strength, Wi-Fi, and battery level at 64%. Below the status bar is a dark green header with a back arrow and the word "Checkout". A yellow modal box with a close button and a location pin icon is titled "Add Delivery Address". The form contains five required fields: "Full Name *" with a placeholder "Enter your full name", "Phone Number *" with a placeholder "+63 912 345 6789", "Region *" with a placeholder "e.g., Central Visayas", "Province *" with a placeholder "e.g., Cebu", and "City *" with a placeholder "e.g., Cebu City". At the bottom of the form are two buttons: "Cancel" and "Save Address".

Figure 4.38 – *Add Delivery Address (Buyer)*

The "add delivery address" form allows the buyer to input their full name, phone number, and delivery address information, including the region, province, and city. This ensures that the seller has accurate shipping details to deliver the products efficiently. The form includes a "save address" button to store the address for future orders.

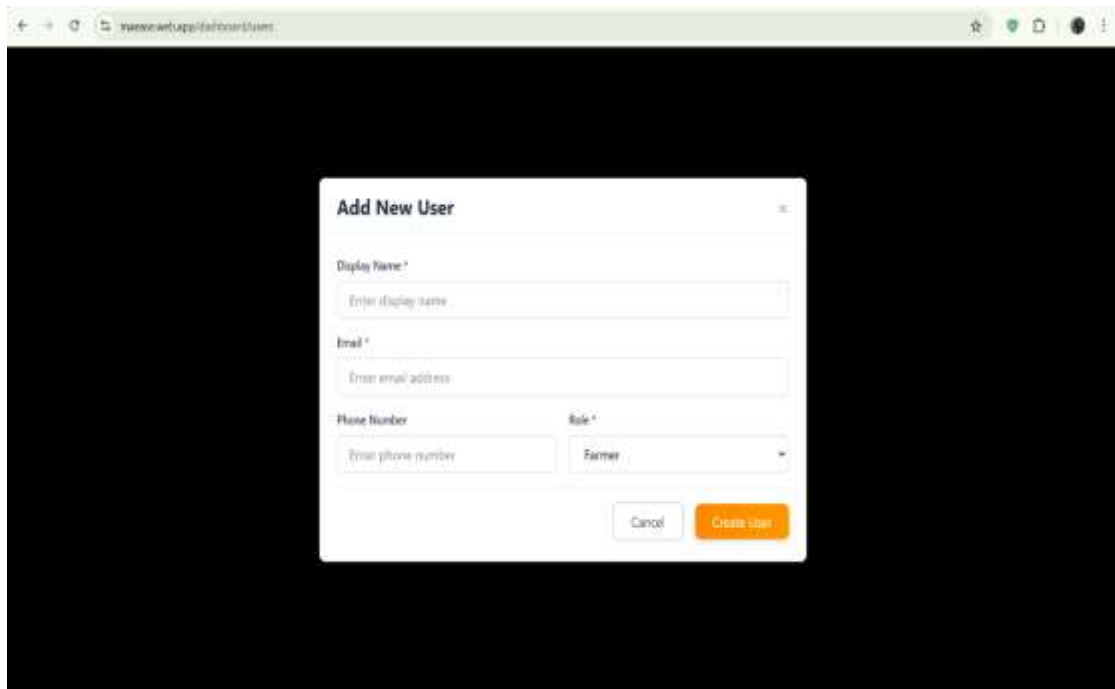
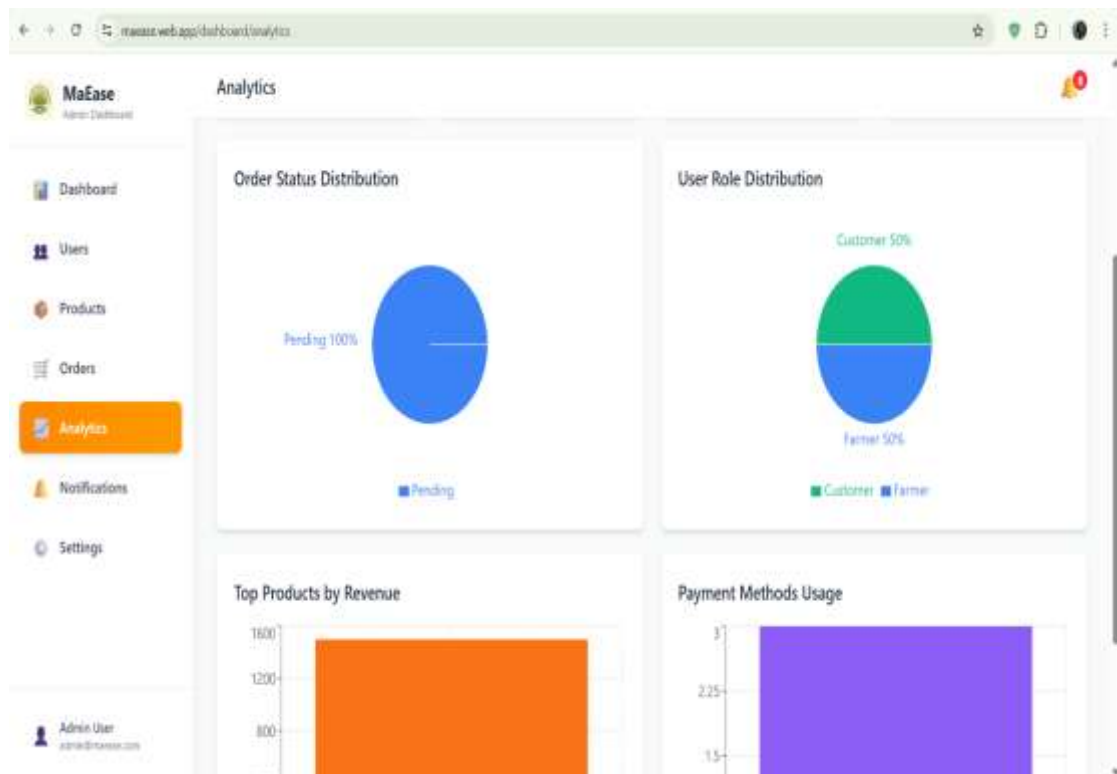
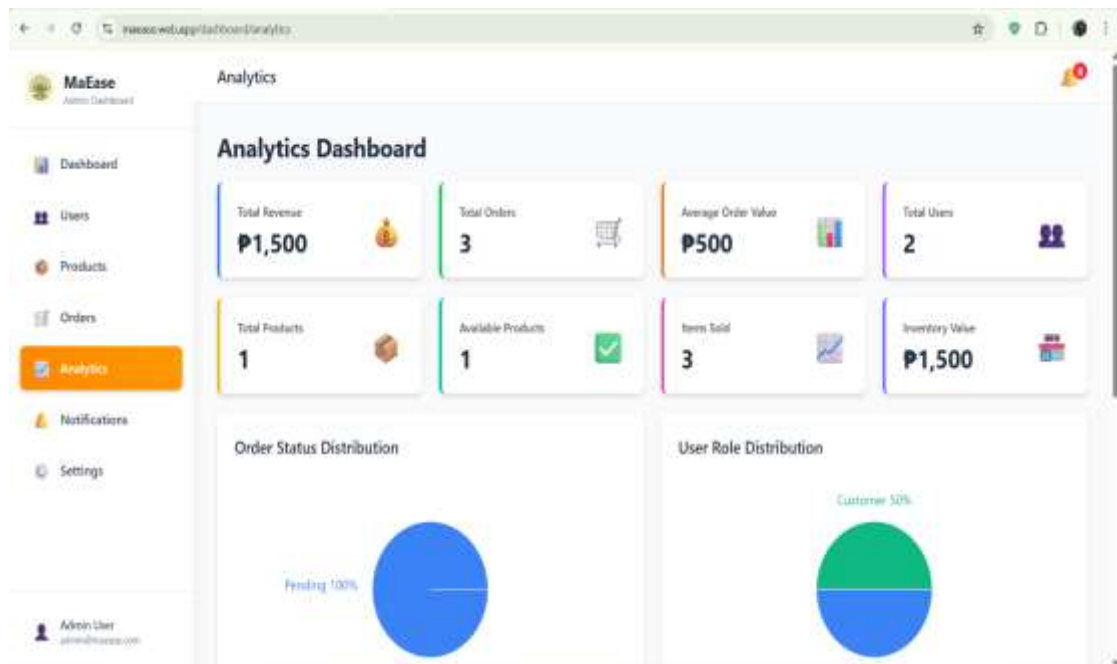
A screenshot of a web browser window showing a form titled "Add New User". The form is centered on a dark background. It contains four input fields: "Display Name" with placeholder text "Enter display name", "Email" with placeholder text "Enter email address", "Phone Number" with placeholder text "Enter phone number", and "Role" with a dropdown menu currently showing "Farmer". At the bottom right of the form are two buttons: a grey "Cancel" button and an orange "Create User" button. The browser's address bar at the top shows the URL "http://www.wetup.com/#!/users".

Figure 4.39 – *Add New User Form (Admin)*

The add new user form enables admins to create user accounts by entering required details, such as display name, email, phone number, and role. Once the form is completed, the admin can click “create user” to finalize the registration. The form is simple and requires all fields to be filled for successful submission.

Reports



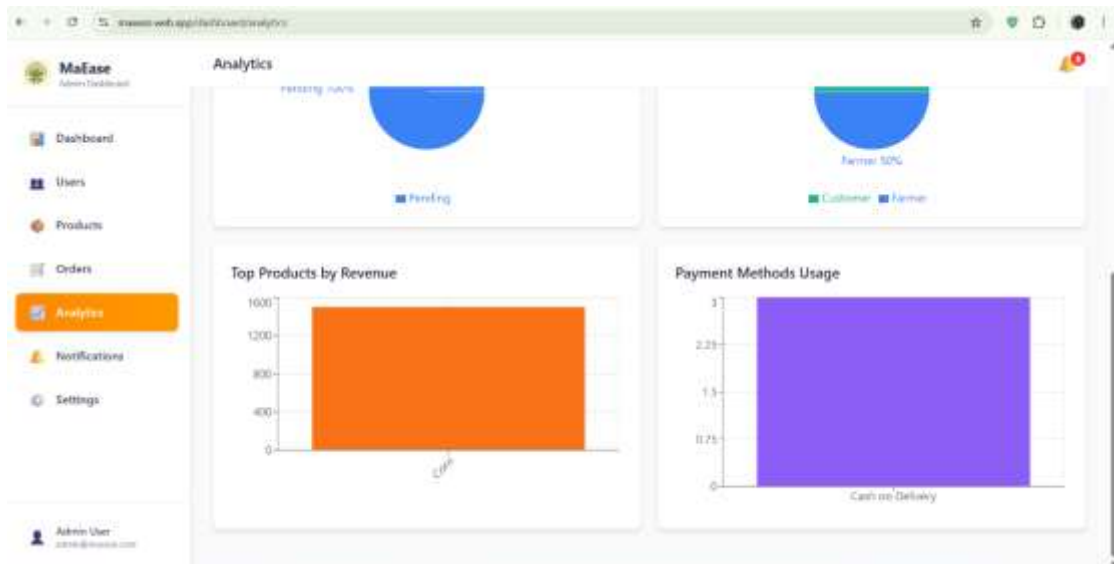


Figure 4.40 – Analytics reports (Admin)

This screenshot shows the generated reports section of the maease admin dashboard, presenting key performance metrics such as total revenue, total orders, and total products. The analytics include visual data, such as order status distribution and user role distribution, enabling admins to monitor sales performance and user trends effectively. These reports assist in business decision-making by providing insights into inventory, order fulfillment, and customer activity patterns.

4.5 Data Design

4.5.1 Entity Relationship Diagram

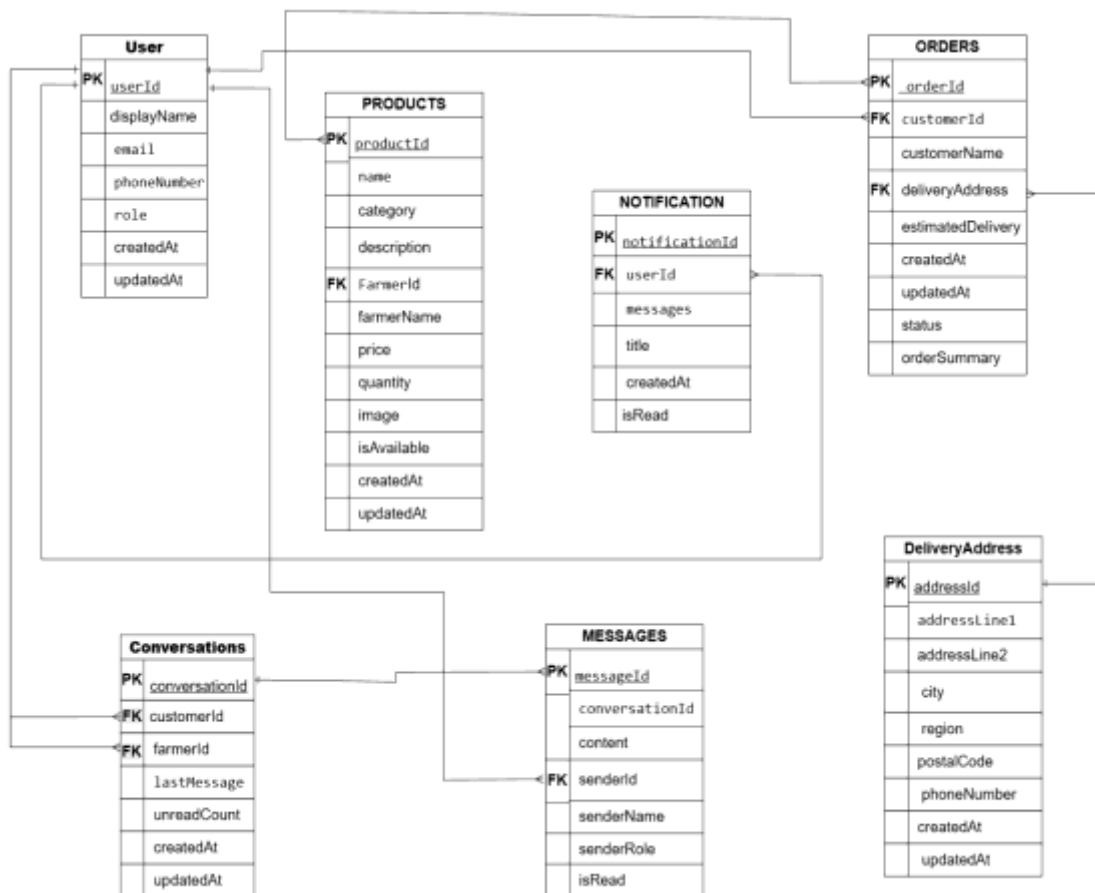


Figure 4.41 - Entity Relationship Diagram

This entity relationship diagram (erd) represents the data model for the maease system, showing how entities like users, products, orders, notifications, and messages are related. It illustrates key relationships, such as users placing orders, receiving notifications, and engaging in conversations with sellers. The diagram highlights primary keys (pk) and foreign keys (fk) to ensure proper data linkage and relational integrity.

4.5.2 Data Dictionary

Field Name	Data Type	Description
userId	string	Unique user identifier
displayName	string	User full name
email	string	Email address
phoneNumber	string	Contact number
role	string	User role (admin, customer, farmer)
createdAt	timestamp	Account creation date
updatedAt	timestamp	Last profile update
userId	string	Unique user identifier
displayName	string	User full name
email	string	Email address

Table 4.1: Data Dictionary of Users

Field Name	Data Type	Description
productId	string	Unique product identifier
name	string	Product name
description	string	Product description
farmerId	string	Farmer user ID
farmerName	string	Farmer name
price	number	Product price
quantity	number	Available stock
image	string	Product image URL
isAvailable	boolean	Availability status
createdAt	timestamp	Creation date
updatedAt	timestamp	Last update date

Table 4.1 - Data Dictionary of Products

Field Name	Data Type	Description
orderId	string	Unique order identifier
customerId	string	Customer user ID
customerName	string	Customer name
deliveryAddress	map	Embedded delivery address
estimatedDelivery	timestamp	Estimated delivery date
status	string	Order status
orderSummary	map	Cost breakdown summary
createdAt	timestamp	Order creation date
updatedAt	timestamp	Last order update

Table 4.2 - Data Dictionary of Orders

Field Name	Data Type	Description
addressLine1	string	Street / Zone
addressLine2	string	Barangay / Area
city	string	City / Municipality
region	string	Region
postalCode	string	Postal code
label	string	Address label
phoneNumber	string	Contact number

Table 4.3 - Data Dictionary of Delivery Address

Field Name	Data Type	Description
messageId	string	Unique message identifier
conversationId	string	Conversation ID
content	string	Message content
senderId	string	Sender user ID
senderName	string	Sender name
senderRole	string	Sender role
messageType	string	Message type

isRead	boolean	Read status
timestamp	timestamp	Time sent

Table 4.4 - Data Dictionary of Delivery Messages

Field Name	Data Type	Description
notificationId	string	Unique notification identifier
userId	string	Receiver user ID
title	string	Notification title
message	string	Notification message
priority	string	Priority level
type	string	Notification type
isRead	boolean	Read status
createdAt	timestamp	Creation time

Table 4.5 - Data Dictionary of Delivery Notifications

Field Name	Data Type	Description
conversationId	string	Unique conversation identifier
customerId	string	Customer user ID
farmerId	string	Farmer user ID
lastMessage	map	Latest message snapshot
unreadCount	map	Unread message counters
createdAt	timestamp	Creation time
updatedAt	timestamp	Last update time

Table 4.6 - Data Dictionary of Delivery Conversations

This data dictionary defines the structure, fields, and data types used in the MAEASE system based on the approved Entity Relationship Diagram (ERD). It serves as a reference for understanding how data is stored, organized, and related across different collections in the NoSQL database. Each table describes the attributes required to support user management, product listings, order processing, messaging, notifications, and delivery information. The data dictionary ensures consistency, clarity, and proper implementation during system development and maintenance.

4.6 System Architecture

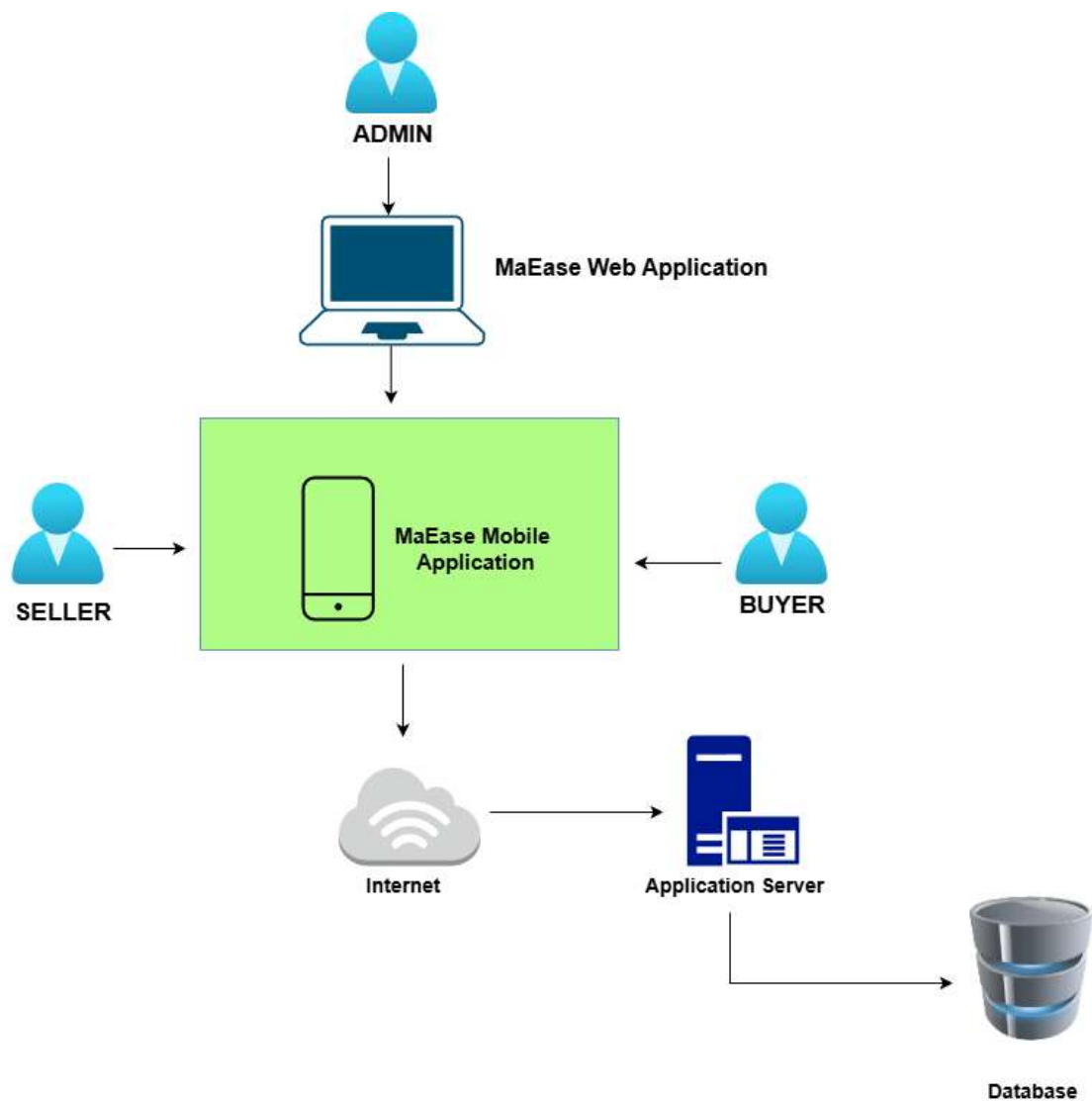


Figure 4.42 – System Architecture

The system architecture diagram for the maease system illustrates the interaction between various components: the admin, seller, and buyer, each accessing different interfaces. The admin accesses the system via the maease web application, while the seller and buyer interact through the maease mobile application. Communication between the mobile app, web app, and database is routed through the internet and managed by the application server. The database stores all the necessary data, such as user profiles, orders, and product details, ensuring data consistency and accessibility across the system.

4.7 Development

4.7.1 Software Specification

- **Frontend (Web Application):**
 - React JS – used to build a dynamic and interactive user interface for the web-based system.
 - JavaScript (ES6+) – the primary programming language used for application logic.
 - HTML5 and CSS3 – used for structuring and styling the web application.
- **Mobile Application:**
 - React Native (Expo Framework) – used to develop a cross-platform mobile application compatible with both Android and iOS devices.
 - Expo SDK – provides access to native device features such as camera, location services, and notifications.
- **Backend and Database:**
 - Firebase Authentication – used for secure user authentication and account management.
 - Firebase Firestore – a cloud-based NoSQL database used for real-time data storage and synchronization.
 - Firebase Storage – used to store images and other media files.
- **Other Development Tools:**
 - Node.js and npm – used for package management and dependency handling.
 - Git and GitHub – used for version control and collaborative development.

4.7.2 Hardware Specification

- **Development Computer:**
 - Processor: Intel Core i5 / AMD Ryzen 5 or higher
 - RAM: Minimum of 8 GB
 - Storage: 256 GB SSD or higher
 - Operating System: Windows 10 or Windows 11
- **Mobile Devices (for Testing):**
 - Android smartphone running Android 9.0 or higher
 - iOS device running iOS 13 or higher (optional)
 - USB cable or wireless connection via Expo Go
- **Internet Connectivity:**
 - Stable internet connection required for Firebase cloud services and real-time data synchronization.

4.7.3 Program Specification

- **Core Functionalities:**
 - User registration and authentication
 - Real-time data management using Firebase Firestore
 - Create, Read, Update, and Delete (CRUD) operations
 - Secure storage and handling of user information
 - Cross-platform accessibility (Web and Mobile)
- **Application Architecture:**
 - The application follows a component-based architecture, allowing features to be divided into reusable components for improved maintainability and scalability.

4.7.4 Programming Environment

- **Code Editor:** Visual Studio Code
- **Frameworks and Libraries:**
 - React JS
 - React Native with Expo
- **Backend Platform:** Firebase
- **Package Manager:** npm
- **Testing Tools:** Expo Go for mobile application testing
- **Deployment Environment:**
 - Firebase Hosting for the web application
 - Expo Application Services (EAS) for building and deploying the mobile application

4.7.5 Test Plan

The test plan for the MAEASE system outlines the structured procedures and strategies used to validate the system's functionality, reliability, and usability. It ensures that all findings are robust, reproducible, and meet the specified objectives. The plan includes the following components:

Objectives

- To verify that all system features function correctly according to specifications.
- To ensure data integrity, real-time synchronization, and secure user authentication.
- To assess usability, performance, and reliability across web and mobile platforms.

Scope

- Included: User authentication, product management, order processing, messaging, notifications, and delivery address handling.
- Excluded: External third-party system failures and network outages beyond the control of the development environment.

Testing Strategy

- Unit Testing: Validate individual components such as user login, CRUD operations, and data input forms.
- Integration Testing: Check interactions between web and mobile applications with Firebase backend.
- User Acceptance Testing (UAT): Conduct testing with real users to evaluate usability and user experience.
- Performance Testing: Assess response times and system behavior under normal and peak loads.

Environment

- Hardware: Development computers (Intel Core i5 / AMD Ryzen 5, 8 GB RAM), Android and iOS mobile devices.
- Software: Visual Studio Code, Node.js, npm, React JS, React Native with Expo, Firebase services.

Schedule

- Testing is conducted iteratively during development, with final validation performed prior to deployment.

Deliverables

- Test reports documenting passed and failed cases, issues found, and corrective actions taken.
-

Risks and Mitigation

- Possible network or device issues may affect testing; mitigated by using controlled test environments and multiple devices.

Tools

- Bug tracking via GitHub Issues; mobile testing with Expo Go; web testing via standard browsers.

4.8 Testing

This section outlines the different testing phases conducted for the MaEase system to ensure it functions as intended and meets the specified requirements.

4.8.1 Unit Testing

Unit testing focuses on verifying individual system components or functions to ensure they work correctly in isolation. This testing was conducted during the development phase to validate the correctness of each major feature of the MaEase system. It helped identify and resolve errors early before integrating the components into the complete system.

Test ID	Major Function	Input/ Action	Expected Output	Result
1	User Login (Buyer/Seller)	Enter valid phone number and password	User successfully logs in and is redirected to the correct dashboard	Passed
2	User Registration	Enter required user details and submit registration form	Account is successfully created and saved in the database	Passed
3	Product Listing (Seller)	Add product name, price, quantity, and image	Product is added and displayed in seller product list	Passed
4	Update Product (Seller)	Edit product details and save changes	Product information is updated successfully	Passed

5	Delete Product (Seller)	Select product and click delete	Product is removed from the product list	Passed
6	Browse Products (Buyer)	View product listings on home page	Products are displayed correctly	Passed
7	Add to Cart (Buyer)	Select product and add to cart	Product is added to cart with correct quantity	Passed
8	Place Order (Buyer)	Confirm order and submit	Order is recorded and seller is notified	Passed
9	Order Status Update (Seller)	Update order status (Preparing, Delivered)	Order status is updated and buyer is notified	Passed
10	Order Tracking (Buyer)	View order status in My Orders	Updated order status is displayed correctly	Passed
11	Messaging (Buyer/Seller)	Send message through chat feature	Message is delivered and displayed in conversation	Passed
12	Notification Alert	Trigger new order or status update	Notification is sent to relevant user	Passed
13	Admin User Management	Add or edit user details	User information is updated successfully	Passed
14	Admin Reports	View analytics dashboard	System displays correct sales and user data	Passed

Table 4.7 - Unit Testing table Results

4.8.2 Integration Testing

Integration testing focuses on verifying the interaction between different system modules to ensure they work together correctly as a unified system. This testing was conducted after unit testing to check data flow and communication between the mobile application, backend services, and database. It ensured that integrated features such as order processing, notifications, and user management functioned properly without errors.

4.8.3 Performance Testing

Performance testing evaluates the system's responsiveness, stability, and reliability under normal and peak usage conditions. It ensures that the MaEase application can handle multiple users and transactions without delays or system failures.

4.8.4 System Testing

System testing evaluates the fully integrated MaEase system to ensure that all features work correctly together and meet the specified requirements in an end-to-end, production-like environment.

4.8.5 Performance of the Evaluation Result

MAEASE SUS CALCULATION											
Participant	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	SUS Score
P1	5	1	5	1	5	2	5	1	5	1	95.0
P2	5	2	5	1	5	1	5	1	5	2	90.0
P3	5	1	5	2	5	1	5	2	5	1	92.5
P4	5	1	5	1	5	1	5	1	5	1	100.0
P5	4	1	5	1	5	1	5	1	5	1	97.5
P6	5	2	5	1	4	2	5	1	5	2	85.0
P7	5	1	5	2	5	1	4	2	4	2	82.5
P8	5	1	5	1	5	1	5	1	4	1	97.5
P9	4	2	5	1	5	2	5	1	5	2	87.5
P10	5	1	4	2	5	1	5	1	5	1	92.5

P1	5	1	5	1	5	2	5	1	5	1	95.0
										AVERAGE: 92.0%	

Table 4.8 - System Usability Scale Result

Figure 4.36 presents the System Usability Scale (SUS) results, indicating a high level of user satisfaction and overall usability of the MaEase system based on the evaluation scores.

Chapter V

Summary, Conclusions and Recommendations

5.1 Summary of Findings

This study aimed to design and develop MaEase, a mobile-based application intended to support small-scale corn sellers and buyers in Bobontugan, Jasaan, Misamis Oriental. The primary objective was to address common problems in corn-related businesses, including manual transaction handling, lack of organized order tracking, limited communication between buyers and sellers, and absence of digital sales records.

Using the Design and Development Research (DDR) approach and the Waterfall System Development Life Cycle, the study involved requirements analysis, system design, development, testing, and evaluation. The MaEase system was developed with key features such as product listing, order management, order tracking, messaging, notifications, and basic reporting, accessible through mobile applications for buyers and sellers and a web interface for administrators.

Results from system testing and usability evaluation using the System Usability Scale (SUS) and ISO/IEC 25010-based criteria showed that the system is functional, user-friendly, and suitable for users with limited technical skills. The findings indicate that MaEase effectively improves transaction organization, communication efficiency, and overall management of corn selling activities.

5.2 Conclusions

Based on the results and findings of the study, the following conclusions are drawn:

- The absence of a digital platform significantly contributes to inefficiencies in small-scale corn businesses, particularly in order handling, communication, and transaction tracking.
- The MaEase application effectively addresses these issues by providing a centralized and structured system for managing products, orders, and buyer-seller interactions.
- The role-based access design ensures that sellers, buyers, and administrators can perform their tasks efficiently while maintaining proper system control.
- Usability evaluation results indicate that the system is easy to use and acceptable to its intended users, confirming that MaEase is appropriate for rural communities with limited digital exposure.

- The implementation of MaEase demonstrates that mobile-based solutions can positively support digital transformation and operational efficiency in small-scale agricultural enterprises.

5.3 Recommendations

Based on the conclusions drawn from the study, the following recommendations are proposed:

- **For System Improvement**
Future enhancements may include online payment integration, GPS-based delivery tracking, and more detailed sales analytics to further improve system functionality.
- **For Users and Stakeholders**
Local corn sellers and buyers are encouraged to adopt MaEase to improve transaction efficiency, reduce manual errors, and maintain organized sales records.
- **For Implementation and Deployment**
The system may be deployed to other barangays or agricultural communities to assess its scalability and adaptability to different rural settings.
- **For Training and Support**
Providing basic user training or instructional materials is recommended to ensure effective system adoption, especially for users with limited technical knowledge.

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APPENDICES



UNIVERSITY OF SCIENCE AND TECHNOLOGY OF SOUTHERN PHILIPPINES

Alubijid | Balubal | Cagayan de Oro | Claveria | Jasaan | Oroquieta | Panaon | Villanueva

MaEase System Evaluation Questionnaire

Instructions:

Please read and understand each statement below. Choose the number that best matches your experience after using the MaEase system.

Use the following scale:

- 1 – Dili Gyud Uyon (*Strongly Disagree*)
- 2 – Dili Uyon (*Disagree*)
- 3 – Walay Komento / Neutral (*Neutral*)
- 4 – Uyon (*Agree*)
- 5 – Uyon Kaayo (*Strongly Agree*)

System Usability Scale – SUS	Strongly Disagree				Strongly Agree
Statement	1	2	3	4	5
Ang paggamit sa MaEase system makatabang nako nga mapadali ang akong mga buluhaton. <i>(Using the MaEase system helps me complete my tasks more efficiently.)</i>					
Lisod para nako ang pagkat-on unsaon paggamit sa MaEase system. <i>(I find the MaEase system difficult to learn.)</i>					
Sayun ug klaro ang pag-navigate sa MaEase system. <i>(The MaEase system is easy and clear to navigate.)</i>					
Kinahanglan pa nako og tabang sa uban aron magamit ang MaEase system. <i>(I need the support of others to be able to use the MaEase system.)</i>					

Maayo ang pagkahiusa ug pagtinabangay sa mga feature sa MaEase system. <i>(The features of the MaEase system are well integrated.)</i>					
Daghan kaayo og komplikasyon ang MaEase system. <i>(The MaEase system is unnecessarily complex.)</i>					
Dali ra matun-an ang paggamit sa MaEase system. <i>(Most people would learn to use the MaEase system very quickly.)</i>					
Medyo bug-at ug dili komportable gamiton ang MaEase system. <i>(I find the MaEase system cumbersome to use.)</i>					
Komportable ug confident ko sa paggamit sa MaEase system. <i>(I feel confident using the MaEase system.)</i>					
Kinahanglan pa og dugang nga pagkat-on aron magamit og maayo ang MaEase system. <i>(I need to learn a lot of things before I can use the MaEase system effectively.)</i>					

Comments / Suggestions:

(Please provide any additional feedback or suggestions to improve the MaEase system.)

Informed Consent Form for Research Participation

Title of the Study:

MaEase: Developing a Mobile Application to Empower Corn Farmers with Comprehensive Service Solutions

School Name: University of Science and Technology of Southern Philippines

Course/Strand Name: BS – Information Technology

Researchers:

Kristine G. Cuerquis

Jessa P. Ellevera

Jean Mae S. Pabecca

Purpose of the Study

We are conducting a research study to develop a mobile application called MaEase, which aims to support corn product handlers, sellers, and buyers in our local community. The purpose of this interview is to gather insights from individuals like you who are directly involved in corn-related activities. Your responses will help us understand your needs, challenges, and suggestions, which will serve as the basis for designing the system.

Procedures

If you agree to participate, we will conduct a short face-to-face interview with you. The interview will last approximately 15-30 minutes, and your answers will be written down or recorded with your permission. The interview questions will focus on your experience in handling, selling, or buying corn products and the services you think could be improved through a digital platform.

Voluntary Participation

Your participation in this study is entirely voluntary. You may refuse to answer any question or stop the interview at any time without penalty. There are no risks involved, and you will not be required to pay or receive any form of compensation.

Confidentiality and Data Privacy

All information you share during this interview will be kept strictly confidential. Your name or any identifying information will not appear in our final report. The data will be stored securely and used only for academic purposes related to this research project. Only the researchers and our adviser will have access to your answers.

Consent Statement

I have read and understood the details above. I was given the chance to ask questions and all were answered clearly. I voluntarily agree to participate in the interview for this research.

Name of Respondent: _____

Signature: _____

Date: _____

Mrs. Cristina D. Nuñez
Corn Owner
Bobontugan, Jasaan, Misamis Oriental

Subject: Request for Permission to Conduct Interview and Gather Non-Confidential Data

Dear Ma'am,

Good day!

We, the undersigned BSIT-4 students of the University of Science and Technology of Southern Philippines (USTP) – Jasaan Campus, respectfully request permission to conduct an interview and gather non-confidential information for our capstone project entitled:

“MAEASE: Developing a Mobile Application to Empower Corn Farmers with Comprehensive Service Solutions.”

In support of the data-gathering phase of our study, we kindly request your permission to collect non-confidential information related to current corn selling and buying practices, such as order handling, product management, and transaction processes. All information gathered will be used strictly for academic purposes and will be treated with confidentiality and respect.

We hope for your favorable approval and are willing to adjust to any schedule that is convenient for you. Thank you very much for your time and cooperation.

Respectfully yours,

CUERQUIS, KRISTINE G.
ELLEVERA, JESSA P.
PABECCA, JEAN MAE S.
BS in Information Technology
USTP Jasaan Campus

Noted by:

Ma'am Nevel Taurine C. Auxilio
Capstone Adviser

Letter of Consent

Kristine G. Cuerquis

Project Manager

BS – Information Technology

University Of Science And Technology Of Southern Philippines

August 04, 2023

To:

Ma'am Nevel Taurine C. Auxilio

Capstone Adviser

University Of Science And Technology Of Southern Philippines

Subject: Request for Consent to Conduct Client Interviews

Dear Ma'am Nevel Taurine ,

Good day!

We, the undersigned researchers of the study entitled **“MaEase: Developing a Mobile Application to Empower Corn Farmers with Comprehensive Service Solutions,”** respectfully seek your consent to conduct interviews with selected clients involved in corn-related operations.

These interviews are essential to gather relevant data that will guide the design and development of our mobile application. We assure you that all interviews will be conducted ethically and responsibly, with full respect for participants' privacy and consent. Interviewees will be asked to sign an informed consent form before participating.

We kindly ask for your approval before we proceed. Your guidance is highly valued throughout our research process.

Thank you very much for your support.

Respectfully yours,

Research Team:

Kristine G. Cuerquis

Jessa P. Ellevera

Jean Mae S. Pabecca

Noted and Approved:

Nevel Taurine C. Auxilio

Capstone Adviser

August 04, 2025

Client Consent and Agreement Letter

Date: _____

We, the researchers of the study entitled: “MaEase: Developing a Mobile Application to Empower Corn Farmers with Comprehensive Service Solutions,” are respectfully seeking your permission and agreement to involve you in our capstone research activities.

This includes participation in interviews or surveys, feedback for development, and possible testing of the system within your area.

All information shared will be kept confidential, and your participation is voluntary. You may withdraw at any time without penalty.

We appreciate your support and cooperation in helping us complete this academic study.

Thank you very much.

Agreed and Signed by:

Name: _____

Signature: _____

Date: _____

Curriculum vitae



JESSA P. ELLEVERA

Programmer

Career Objective

To enhance my foundational skills in programming by participating in real-world academic projects, and to build confidence and experience through collaboration with fellow students.

Contact Information

☎ 0916-687-9131

✉ jessaellevera661@gmail.com

📍 Zone 7, Bobontugan, Jasaan, Misamis Oriental

Personal Information

Date of Birth: January 24, 1998

Nationality: Filipino

Languages: Bisaya, Tagalog, English

Education

Bobontugan Elementary School
2016-2017

Bobontugan National High School
2019-2020

Bobontugan Senior High School
2021-2022

Bachelor of Science in Information Technology
University of Science and Technology of Southern Philippines – Present

Project Expertise

Capstone Project Title:

MaEase: Developing a Mobile Application to Empower Corn Farmers with Comprehensive Service Solutions (Ongoing)

Project Description:

- A simple digital platform designed to assist local corn businesses in buying and selling their products.
- Allows corn merchants to post product details and monitor customer orders.
- Helps improve income opportunities for corn-based families through technology.
- Aims to simplify the selling process for those in rural areas.
- Encourages the use of mobile and web apps in agricultural business transactions.

Skills

- Basic use of HTML and JavaScript
- Knows how to create simple forms and buttons
- Familiar with PowerPoint presentations
- Able to search and research effectively online
- Willing to work and learn with a team



JEAN MAE S. PABECCA

Programmer

Career Objective

To gain hands-on experience in software development by applying the basic IT skills I have learned, and to grow professionally by working on meaningful school-based projects that help the community.

Contact Information

☎ 09386607087

✉ pabeccajeanmae@gmail.com

📍 Zone 9 San Antonio Jasaan,
Misamis Oriental

Personal Information

Date of Birth: December 31, 2003

Nationality: Filipino

Languages: Bisaya, Tagalog, English

Education

San Antonio Elementary School
2016–2017

Bobontugan National High School
2019–2020

Bobontugan Senior High School
2021–2022

Bachelor of Science in Information Technology
University of Science and Technology of Southern Philippines – Present

Project Expertise

Capstone Project Title:

MaEase: Developing a Mobile Application to Empower Corn Farmers with Comprehensive Service Solutions (Ongoing)

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- Allows corn merchants to post product details and monitor customer orders.
- Helps improve income opportunities for corn-based families through technology.
- Aims to simplify the selling process for those in rural areas.
- Encourages the use of mobile and web apps in agricultural business transactions.

Skills

- Basic knowledge in HTML and CSS
- Simple website layout design
- Typing and encoding using Microsoft Word
- Basic file organization and storage
- Able to follow team instructions



KRISTINE G. CUERQUIS

Project Manager

Career Objective

To develop strong leadership and coordination skills by managing academic projects, while contributing to technology-based solutions that benefit the local community.

Contact Information

☎ 0910-187-2779

✉ kcuerquis5@gmail.com

📍 Zone 10, Bobontugan, Jasaan, Misamis Oriental

Personal Information

Date of Birth: August 28, 2003

Nationality: Filipino

Languages: Bisaya, Tagalog, English

Education

Bobontugan Elementary School
2016–2017

Bobontugan National High School
2019–2020

Bobontugan Senior High School
2021–2022

Bachelor of Science in Information Technology
University of Science and Technology of Southern Philippines – Present

Project Expertise

Capstone Project Title:

MaEase: Developing a Mobile Application to Empower Corn Farmers with Comprehensive Service Solutions (Ongoing)

Project Description:

- A simple digital platform designed to assist local corn businesses in buying and selling their products.
- Allows corn merchants to post product details and monitor customer orders.
- Helps improve income opportunities for corn-based families through technology.
- Aims to simplify the selling process for those in rural areas.
- Encourages the use of mobile and web apps in agricultural business transactions.

Skills

- Basic communication and leadership
- Simple documentation writing
- Familiar with Microsoft Word and Excel
- Can manage group schedules and assignments
- Able to guide group decision-making

